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# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment for oesophageal cancer: a systematic review and meta-analysis



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# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment for oesophageal cancer: a systematic review and meta-analysis

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LJZ, YLS, and NJ contributed equally to this work.

**Trial registration:** The study protocol has been registered in the International Prospective Register of Systematic Reviews (PROSPERO, CRD42020169230, 28/04/2020).

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## Abstract

**Objectives:** Despite remarkable advances in the treatment of oesophageal cancer (OC), the role of anti-epidermal growth factor receptor (anti-EGFR) agents in treating OC remains controversial. A systematic review and meta-analysis were performed to elucidate the efficacy and safety of anti-EGFR agents.

**Design:** A meta-analysis of randomised controlled trials (RCTs).

**Setting:** Hospitals.

**Participants:** Patients with OC were randomly assigned to conventional therapy with or without anti-EGFR agents.

**Interventions:** We searched PubMed, EMBASE, Web of Science, ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine, China National Knowledge Infrastructure, and Wanfang Data Knowledge Service Platform for RCTs of anti-EGFR treatment versus conventional treatment in patients with OC, from inception to December 2019.

**Primary and secondary outcome measures:** Overall survival (OS) and progression-free survival (PFS) were primary outcome measures, while objective response rate (ORR), disease control rate (DCR), and toxicities were secondary outcome measures.

**Results:** In total, 25 RCTs with 3406 OC patients were included. Overall, anti-EGFR treatment significantly improved the OS (HR 0.81, 95% CI 0.74–0.89, $P < 0.00001$), ORR (RR 1.33, 95% CI 1.16–1.52, $P < 0.0001$), and DCR (RR 1.22, 95% CI 1.11–1.34, $P < 0.0001$), but not PFS (HR 0.91, 95% CI 0.76–1.08, $P = 0.26$). Anti-EGFR treatment was significantly associated with higher incidences of myelosuppression (RR 0.42, 95% CI 0.21–0.86, $P = 0.02$), diarrhoea




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(RR 2.26, 95% CI 1.42–3.60, $P = 0.0005$), acne-like rash (RR 9.83, 95% CI 3.54–27.28, $P < 0.0001$), and hypomagnesaemia (RR 2.61, 95% CI 1.02–6.68, $P = 0.04$).

**Conclusions:** Overall, anti-EGFR treatment has positive effects on the OS, ORR, and DCR in OC. However, considering the high incidence of adverse effects such as myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia following anti-EGFR treatment, careful monitoring is recommended.

**Strengths and limitations of this study**

- This systematic review and meta-analysis utilised a comprehensive study design to evaluate the efficacy and safety of anti-EGFR agents combined with conventional treatments versus conventional treatments alone for oesophageal cancer (OC).
- No clear evidence is available whether the combination of anti-EGFR agents and conventional treatments is superior to the conventional treatments alone for OC.
- This study has been registered a priori and reported in line with Preferred Reporting Items for Systematic Reviews and Meta-Analysis.
- Since most of the eligible studies were from Asia, which might result in the overestimation of this meta-analysis, physicians should interpret our findings with caution when applying anti-EGFR agents in their clinical practices.



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## Introduction

Oesophageal cancer (OC) is the seventh most common cancer type and the sixth leading cause of cancer mortality, accounting for approximately one in every twenty cancer-related deaths worldwide in 2018.[1] However, its incidence and mortality rates vary considerably, as demonstrated by a two- to three-fold difference in these statistics between regions all over the world.[2] OC is common in Eastern and Southern Africa, and the highest incidence rates have been reported in East Asia, with rates in Mongolia and China being among the top five worldwide.[1 2]

OC is diagnosed in the later stages in 80% of the patients. Typically, few patients benefit from single-modality therapies, such as surgery, chemotherapy (CT), and radiotherapy (RT).[3] Therefore, chemoradiotherapy (CRT) has become a standard of care for OC. However, controlling locoregional recurrence or persistent disease is still a major challenge.[4] Attempts such as the use of higher radiation doses, combination of various CT regimens and RT or other combination therapies (e.g. surgery, neoadjuvant CT) have failed to achieve any improvement.[5-8] Therefore, the addition of novel therapies to conventional treatments is urgently needed.

The epidermal growth factor receptor (EGFR) signalling pathway has been implicated in the proliferation, invasion, angiogenesis, and survival of cancer cells.[9] Moreover, overexpression of EGFR in OC has been associated with poor prognosis.[10] Previous studies have shown the synergistic effects of EGFR inhibitors and conventional treatments.[11 12] Therefore, there exists a strong rationale for testing drugs targeting EGFR as an effective treatment strategy against



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OC. Tyrosine kinase inhibitors (TKIs) and monoclonal antibodies (mAbs) are two promising anti-EGFR drugs. Various anti-EGFR agents, such as cetuximab,[13 14] nimotuzumab,[15 16] panitumumab,[17] erlotinib,[18 19] and gefitinib,[20 21] have been administered for treating OC. However, studies examining the role of anti-EGFR agents in combination with conventional treatments have produced controversial results, and the toxicities associated with these agents are variable and uncertain.[13-16 18-21] Moreover, there is no clear evidence whether the combination of anti-EGFR agents and conventional treatments is superior to conventional treatments alone for OC. To address this research gap, we conducted a systematic review and meta-analysis of randomised controlled trials (RCTs) to evaluate the efficacy and safety of anti-EGFR agents combined with conventional treatments versus conventional treatments alone in patients with OC.

## Methods

### Literature Search Strategy

A literature search of the following databases: PubMed, EMBASE, Web of Science, ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine (CBM), China National Knowledge Infrastructure (CNKI), and Wanfang Data Knowledge Service Platform was performed by two authors to identify studies from inception to December 2019. Furthermore, a manual search using the reference lists of all the retrieved articles and relevant journals was conducted to identify additional eligible studies. Unpublished studies were retrieved from the System for Information on Grey Literature database to reduce publication bias. The



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combinations of Medical Subject Headings (MeSH)/Emtree terms and free-text terms were used to search the above databases. The main search terms used were as follows: oesophageal neoplasms, epidermal growth factor receptor inhibitor, cetuximab, panitumumab, nimotuzumab, erlotinib, gefitinib, randomised controlled trial, and clinical trial. The full search strategies utilised are presented in **Supplementary Data S1**.

Two authors independently screened the studies based on the selection criteria. An initial screening of each title and abstract was performed, and eligible studies were retrieved as full text. Any disagreements between the two authors during the selection process were settled by discussion and consensus with a third author.

**Inclusion and Exclusion Criteria**

Eligible studies were included if they met the following criteria: (i) patients with OC were randomly assigned to conventional therapy with or without anti-EGFR agents; (ii) hazard ratios (HRs) with 95% confidence intervals (CIs) for overall survival (OS) and/or progression-free survival (PFS) were reported or sufficient data were provided to calculate these values.

Studies were excluded if they were repeated studies, reviews, comments, letters, case reports, meeting abstracts, study protocols, or animal studies. Duplicate and incomplete data were also excluded.

**Data Extraction and Quality Assessment**

Data extraction and quality assessment were carried out by two independent authors, and discrepancies were resolved by discussion and consensus with the third author.

The information extracted included the first author's name, publication year, country, study



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design, enrolment period, sample size, mean/median age (range, years), treatment protocol, and major clinical outcomes (OS, PFS, objective response rate (ORR), disease control rate (DCR), and grade 3-5 toxicities) of each eligible study.

The quality of each eligible study was assessed by the Cochrane Handbook for Systematic Reviews of Interventions Version 6.0 (www.training.cochrane.org/handbook). Random sequence generation, allocation concealment, blinding, incomplete outcome data, and selective outcome reporting were examined to categorise each eligible study as low risk of bias (A), moderate risk of bias (B), or high risk of bias (C).

**Statistical Analysis**

The primary outcomes were OS and PFS, expressed as HR with 95% CI that could be calculated from the number of observed deaths minus the number of estimated deaths and its variance.[22] OS was defined as the time from diagnosis to death or last follow-up and PFS was defined as the time from diagnosis to the first documented recurrence. Kaplan–Meier curves were read using Engauge Digitizer version 10.8 (free software downloaded from http://sourceforge.net/projects/digitizer/). Secondary outcomes, including ORR, DCR, and haematological and non-haematological toxicities, were expressed as relative risks (RRs) ratio with 95% CIs using the Mantel-Haenszel method. ORR was defined as the rate of partial response (PR) and complete response (CR), and DCR was defined as the rate of PR, CR, and stabilisation.

Statistical heterogeneity across studies was evaluated by the chi-squared test, with a significant level at $P < 0.1$, and quantified with the $I^2$ statistic. The fixed-effects model was



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used if the heterogeneity test showed no statistical significance ($I^2 < 50\%$; $P > 0.1$); otherwise, the random-effects model was adopted. Publication bias was assessed by inspecting the funnel plots for asymmetries and outliers. Sensitivity analysis was conducted to explore the robustness of the results by excluding each study to recalculate the pooled HR and RR estimates. Five studies were excluded because of small sample sizes or treatments with surgery and/or induction chemotherapy.[13 19 23-25] Subgroup analysis was performed according to the different treatment regimens.

All analyses were conducted in Review Manager version 5.3 (Cochrane Collaboration, Oxford, UK). A two-sided *P*-value < 0.05 was considered significant for all analyses except heterogeneity tests. This study was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.[26]

**Patient and public involvement**

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

## Results

### Study Selection

A total of 1374 records were identified by searching the databases, and no additional records were found through other sources. After removing duplicates, 1298 records remained. Based on title screening, 952 records were excluded as they did not meet the eligibility criteria. Next, 346 abstract records were screened for further evaluation, of which 301 records were excluded



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as they were reviews (n = 114), non-RCTs (n = 108, including case reports, meeting abstracts, letters, editorials, and news), animal studies (n = 3), or studies on other cancers and drugs (n = 76) unrelated to our study. After screening 45 full-text articles, 20 records were excluded as they were non-RCTs (n = 5), ongoing studies (n = 2), studies on other cancers and drugs (n = 8), and studies without outcomes to be combined (n = 5). Finally, 25 RCTs were included in the quantitative synthesis for this meta-analysis. The flow diagram of the selection process is shown in **Figure 1**.

**Study Characteristics**

The eligible studies were published between 2009 and 2018, and a total of 3406 patients were included in this meta-analysis. The sample size of each study ranged from 16 to 225. In total, 1699 patients were in the anti-EGFR arm and 1707 patients were in the non-anti-EGFR arm. The age of the included patients ranged from 26 to 87 years. Seventeen studies were conducted in China,[18-20 23-25 27-37] three in the United Kingdom of Great Britain and Northern Ireland (UK)[21 38 39], and one each in the United States of America (USA),[14] Germany,[11] Switzerland,[13] Cuba,[16] and Brazil.[15] Two studies from the UK were performed on the same group of study patients and had partly overlapping study periods.[38 39] However, different outcomes were analysed in each study. For the anti-EGFR arms, patients were given anti-EGFR agents, including cetuximab, nimotuzumab, erlotinib, or gefitinib in combination with surgery, CT, RT, or CRT. In case of the non-anti-EGFR arms, patients received only surgery, CT, RT, or CRT. The basic characteristics of the 25 selected studies are presented in **Table S1**.

**Quality Assessment**



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The quality assessment of each study is shown in **Table 1**. Briefly, among the 25 RCTs, one study was rated as A level,[21] 17 studies as B level,[14 19 20 23-25 27-37] and seven studies as C level.[1 13 15 16 18 38 39] Eleven of the included studies were assessed as being at low risk of random sequence generation,[13-15 21 27 29 35-39] and six studies used allocation concealment.[11 13 15 21 38 39] However, almost all the studies did not use blinding, except for one study which reported the blinding of participants and outcome assessment.[21] All the studies were assessed as having low risk of bias because they addressed incomplete outcome data and selective reporting. No other source of bias was identified from these studies.

**Table 1** Quality assessment of each study

| Study (year) | Random sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Other bias | Rank |
|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Unclear | Yes | No | No | Yes | Yes | Yes | C |
| Crosby et al. 2013[39] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014a[23] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Crosby et al. 2017[38] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Suntharalingam et al. 2017[14] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Lu et al. 2017[35] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ruhstaller et al. 2018[13] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Ramos-Suzarte et al. 2012[16] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Xu et al. 2013[34] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Shi et al. 2014[33] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Sun et al. 2014[24] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2014[25] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Luo et al. 2015[31] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016b[28] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |



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**Table 1** Continued

| Study (year) | Random sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Other bias | Rank |
|---|---|---|---|---|---|---|---|---|
| Fu et al. 2017[27] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Castro et al. 2018[15] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014b[19] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhang et al. 2014[32] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ye et al. 2015[30] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhou et al. 2015[36] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016a[37] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhao et al. 2016[29] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2018[18] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Zeng et al. 2014[20] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Dutton et al. 2014[21] | Yes | Yes | Yes | Yes | Yes | Yes | Yes | A |

## Efficacy Analysis

Fifteen studies[11 13-15 18 21 24 28 30 31 33 35-38] were included in the meta-analysis for analysing OS, which showed significant improvement in the anti-EGFR arm (HR 0.81, 95% CI 0.74–0.89, $P < 0.00001$; $I^2 = 40\%$, $P = 0.05$) (**Figure 2**). Subgroup analysis indicated that OS benefit was observed when anti-EGFR agents were administered in combination with CT, CRT, or RT for the treatment of OC. Both anti-EGFR mAbs and TKIs, especially erlotinib and nimotuzumab, significantly improved the OS (**Figure S1**).

Six studies[11 13 21 24 34 38] were combined for evaluating PFS, which demonstrated no significant difference between the anti-EGFR arm and the control arm (HR 0.91, 95% CI 0.76–1.08, $P = 0.26$; $I^2 = 53\%$, $P = 0.06$) (**Figure 3**). Although longer PFS was identified in patients treated with



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nimotuzumab, no significant difference was found in other treatments (**Figure S2**).

Analysis of the 16 studies,[11 16 19 20 24 25 27-29 31-37] which evaluated ORR, indicated that ORR significantly increased in the anti-EGFR arm (RR 1.33, 95% CI 1.16–1.52, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S3**). Higher ORR was detected in patients with OC who received anti-EGFR agents in combination with CT, CRT, surgery, or RT. Both the categories of anti-EGFR drugs, mAbs and TKIs, especially nimotuzumab and erlotinib, significantly increased ORR (**Figure S4**).

Fifteen studies,[11 16 19 20 24 27-36] which analysed DCR, showed that DCR was significantly improved in the anti-EGFR arm (RR 1.22, 95% CI 1.11–1.34, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S5**). Treatment with the combination of anti-EGFR agents and CT or RT resulted in higher DCR. Anti-EGFR mAbs or TKIs, particularly cetuximab or erlotinib, decreased the risk of DCR (**Figure S6**).

**Safety Analysis**

Six different types of haematological toxicities were analysed in this meta-analysis, including anaemia, leucopoenia, neutropenia, thrombocytopenia, lymphopenia, and myelosuppression. Results from the pooled analysis demonstrated that there was no significant association between anti-EGFR agents and the increased risk of haematological toxicities, except for myelosuppression (RR 0.42, 95% CI 0.21-0.86, $P = 0.02$) (**Figure S7**). Subgroup analysis revealed that the addition of anti-EGFR agents to CT was associated with an increased risk of myelosuppression compared with CT alone, while no significant difference was found with CRT. Additionally, all patients who received anti-EGFR mAbs (e.g., cetuximab and



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nimotuzumab), and only nimotuzumab displayed a significant difference in terms of the incidence of myelosuppression (RR 0.39, 95% CI 0.18–0.83, $P = 0.02$) (**Figure S8**).

Eight non-haematological toxicities, including cardiac toxicity, gastrointestinal toxicity, dermatologic toxicity, pulmonary toxicity, renal toxicity, constitutional symptoms, infection, and laboratory findings, were reported. The results indicated statistically significant differences for dermatologic toxicity (RR 9.83, 95% CI 3.54–27.28, $P < 0.0001$) and laboratory findings (RR 1.59, 95% CI 1.05–2.42, $P = 0.03$) in the anti-EGFR arm, while no significant differences were observed for other non-haematological toxicities (**Table S2**). Subgroup analysis confirmed that patients receiving anti-EGFR treatment had higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia than those in the control group (**Table S2**). Higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia were observed in patients receiving anti-EGFR agents in combination with CRT than in those receiving CRT alone (**Figure S9, S10, S11**). Increased risk of diarrhoea, acne-like rashes, and hypomagnesaemia were found among patients who received anti-EGFR mAbs (**Figure S9, S10, S11**), especially cetuximab (**Figure S9, S10, S11**). Additionally, anti-EGFR TKIs, particularly erlotinib, increased the incidence of acne-like rash (**Figure S10**). Gefitinib was also associated with an increased risk of diarrhoea (**Figure S9**).

**Sensitivity Analysis**

Sensitivity analysis was executed by excluding the studies by Ruhstaller et al. 2018[13], Chen et al. 2014a,[23] Chen et al. 2014b,[19] Sun et al. 2014,[24] and Wu et al. 2014.[25] Any significant changes in the pooled estimate in terms of OS, PFS, ORR, and DCR indicated that the results were



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statistically stable and reliable (**Table S3**). No other sensitivity analysis had any effect on the results.

**Publication Bias**

A funnel plot of the included studies, plotted to assess the possibility of publication bias, showed asymmetry, thereby indicating that publication bias existed (**Figure 4**).

**Discussion**

OC, a malignant digestive system neoplasm with poor prognosis, is regarded as a threat to public health. Currently, anti-EGFR agents are commonly used as valuable therapeutic options for OC. However, various studies have published contradictory results regarding these agents. Thus, a meta-analysis of RCTs was conducted to evaluate the efficacy and safety of anti-EGFR agents in combination with conventional treatment when compared with conventional treatment alone. To the best of our knowledge, this is the first meta-analysis to evaluate the benefits and toxicities of anti-EGFR and non-anti-EGFR treatment strategies for OC. The findings of this meta-analysis indicated that the anti-EGFR therapies might be more effective in terms of OS, ORR, and DCR than non-anti-EGFR therapies. However, anti-EGFR agents might be associated with higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia.

Adding anti-EGFR agents to CT, CRT, or RT significantly improved the OS. However, considering the fact that CRT is considered as the standard treatment approach for OC, the combination of anti-EGFR agents with CT or RT was not considered owing to limited



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availability of studies for this meta-analysis. According to this meta-analysis, the addition of anti-EGFR agents to CRT may be an important strategy to improve clinical outcomes in patients with OC. Moreover, anti-EGFR mAbs and TKIs, especially nimotuzumab and erlotinib, significantly improved the OS. The clinical benefits of nimotuzumab may be explained by the direct inhibition of the EGFR signalling pathway, thereby exerting an antiproliferative effect on tumour cells.[40] Furthermore, nimotuzumab enhanced the sensitivity of tumour cells to RT and CT.[41] These results might provide the potential explanation for the positive results obtained in this study. The combination of erlotinib with conventional treatment for OC is still controversial because of the limited number of RCTs on this TKI. G1/S cell cycle arrest induced by TKIs has already been described in various cancers. Studies have reported that TKIs inhibit the growth of OC cells,[42-45] which might be the reason for the significant improvement of OS in our study. Cetuximab did not significantly contribute to increased OS, which was consistent with a previous report.[46] It has been proposed that the reduced efficacy of cetuximab in combination with CRT is due to proinflammatory and antitumour proliferative effects.[47] Moreover, cetuximab might limit the free-radical damage by platinum drugs, which could again explain the non-significant outcomes in this meta-analysis.[48] For gefitinib, no improvement in OS was observed, which was in accordance with the results of a previous meta-analysis.[49] Currently, there is no evidence that gefitinib could suppress EGFR, thus indicating its limited clinical efficacy.

Considering the outcome of the PFS reported in this meta-analysis, there was no significant improvement in the anti-EGFR arm, which was consistent with the results of a previous meta-



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analysis.[50] Longer PFS was identified with nimotuzumab combination therapy, which was also reported by Jing et al.[51] The biological characteristics of nimotuzumab differed from those of other anti-EGFR agents, which might be the reason for its different clinical effects.

With regards to ORR and DCR, significant differences were noted between the anti-EGFR arm and non-anti-EGFR arm. Both anti-EGFR mAbs and TKIs, especially nimotuzumab, erlotinib, and cetuximab, had significant differences in ORR and DCR for OC. However, previous studies have reported different ORRs and DCRs.[52]

In terms of ORR, our results were in line with those of some studies that indicated a significant advantage of combining anti-EGFR agents with CT, CRT, or RT.[53 54] Furthermore, an ORR of 47.8% in the nimotuzumab group compared to 15.4% in the control group was noted.[16] Additionally, anti-EGFR TKIs were superior to conventional therapy in ORR.[55] Our meta-analysis indicated that erlotinib has a significant benefit in terms of ORR. However, combined data of ORR have not been reported in OC. Whether erlotinib therapy is better than other treatment strategies is still unknown owing to insufficient and conflicting homogeneity studies.

DCR improved greatly when anti-EGFR agents were combined with CT or RT; however, no confirmed conclusions could support our result as few studies reported this outcome. Erlotinib and cetuximab demonstrated a notable beneficial effect on DCR for treating OC, which was in agreement with previous studies.[46 55] Most meta-analysis studies have reported only the results of mAb and TKI therapy. However, as no specific drugs were reported, it was impossible to verify the effectiveness of each drug.



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This meta-analysis confirmed that patients with OC in the anti-EGFR arm had higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia than the non-anti-EGFR arm, which was consistent with previous studies.[56 57] An increased risk of myelosuppression was found in anti-EGFR with CT. Moreover, this meta-analysis showed that nimotuzumab increased the risk of myelosuppression. These results might be related to the finding that nimotuzumab could enhance the sensitivity of tumour cells to CT.[41] However, these findings remain controversial due to the lack of sufficient high-quality, multi-centre, and large-sample RCTs, and thus should be interpreted cautiously.

Non-haematological toxicities, such as diarrhoea, acne-like rashes, and hypomagnesaemia occurred more often in patients with OC receiving anti-EGFR agents with CRT than in those receiving CRT alone. Furthermore, the combination of cetuximab and gefitinib and conventional treatment resulted in a higher risk of diarrhoea than conventional treatment alone. The underlying mechanism of diarrhoea caused by either anti-EGFR mAb or TKI remains poorly understood and might be induced by multiple factors, including altered gut motility, colonic crypt damage, changes to intestinal microflora, and altered transport in the colon.[58] A plausible explanation could be the inhibition of the EGFR pathway in enterocytes.[59] Thus, a better understanding of its mechanism may help in the identification of new approaches to prophylaxis, reduce risk, and develop more effective management.

Acne-like rash is the most common side effect of anti-EGFR agents,[60] and our meta-analysis confirmed that anti-EGFR mAbs and TKIs, such as cetuximab and erlotinib, could induce acne-like rashes. The mechanism leading to skin toxicity is not well known, but it is undoubtedly



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the result of modifications of the signals associated with its activation, particularly the RAS/RAF/MEK/ERK pathway, which affects cell cycle regulation, including the proliferation and differentiation of epidermis cells.[61] Skin toxicities often lead to physical and psychological discomforts, which might result in treatment interruptions, dose modifications, or even non-adherence to anti-cancer therapies.[60] Therefore, it is imperative to understand skin toxicities and develop proper approaches to control or relieve their adverse effects on patients.

An increased incidence of hypomagnesaemia was found in cetuximab, which was in accordance with another report.[46] Hypomagnesaemia, a frequent side effect of cetuximab, is a serious electrolyte abnormality associated with an increased risk of seizures and cardiac events.[62] The exact mechanism of cetuximab-induced hypomagnesaemia has not been well defined. Some researchers discovered that EGFR was strongly expressed in the kidney, particularly in the ascending limb of the loop of Henle, where 70% of the filtered magnesium was reabsorbed, and cetuximab might interfere with the magnesium transport as it blocks EGFR.[62 63] However, hypomagnesaemia is often ignored because of its non-specific symptoms, which are easily attributed to underlying tumours or previous CT regimens.[64] Consequently, serum magnesium should be monitored when cetuximab-based therapy is performed.

Several limitations should be considered while interpreting the results of this meta-analysis. First, we lacked access to individual patient data, which might have caused publication, reporting, or selection bias. Second, because of the insufficient number of studies conducted, we included different anti-EGFR agents and conventional treatments in meta-analysis, which increased clinical heterogeneity among the included studies. Subsequently, subgroup and



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sensitivity analyses were performed to increase the reliability of our results. Third, most of the studies were conducted in Asia, and nimotuzumab is widely used in China, which might result in the overestimation of this meta-analysis. The geographical limitation is due to the higher incidence and mortality rates of OC in East Asia. Fourth, the methods were unclear, especially in the domain of 'allocation concealment and blinding', which might affect the overall methodological quality of the included studies. High methodology quality of RCTs is needed for further research. Finally, considering all these limitations, the results of this meta-analysis should be interpreted cautiously, and further research is necessary for verifying the effect of anti-EGFR therapy in OC.

In conclusion, this meta-analysis indicated that adding anti-EGFR agents to multimodal therapy benefited patients with OC in terms of the OS, ORR, and DCR. Therefore, anti-EGFR agents might be recommended as a treatment option for OC to improve treatment outcomes. The results from this study may help in establishing the optimum therapeutic regimen for OC. However, considering the higher incidence of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia when using the combination of anti-EGFR agents and multimodal therapy, careful monitoring and effective management is recommended. Further high quality large-scale and multi-centre studies are needed to confirm the efficacy and safety of anti-EGFR agents in OC.

**Contributors** LJZ and RY designed the study; LJZ, YLS, and NJ performed the literature search, extracted the data, and evaluated the quality of included studies; LJZ, YQH, BD, WL,



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YZH, YC, and HB- L conducted the statistical analysis; LJZ, YLS, and NJ were involved in the interpretation of the results; LJZ, YLS, NJ, and RY were responsible for the writing and critical revision of the manuscript. All authors read and approved the final manuscript.

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## Figure legends

**Figure 1** Flow diagram of the selection process of relevant studies

**Figure 2** Forest plot of the overall survival (OS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure 3** Forest plot of the progression-free survival (PFS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure 4** Funnel plot of publication bias

**Figure S1** Subgroup analysis of the overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S2** Subgroup analysis of the progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S3** Forest plot of the objective response rate (ORR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S5** Forest plot of the disease control rate (DCR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment



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regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S7** Forest plot of grade 3–5 haematological toxicities in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

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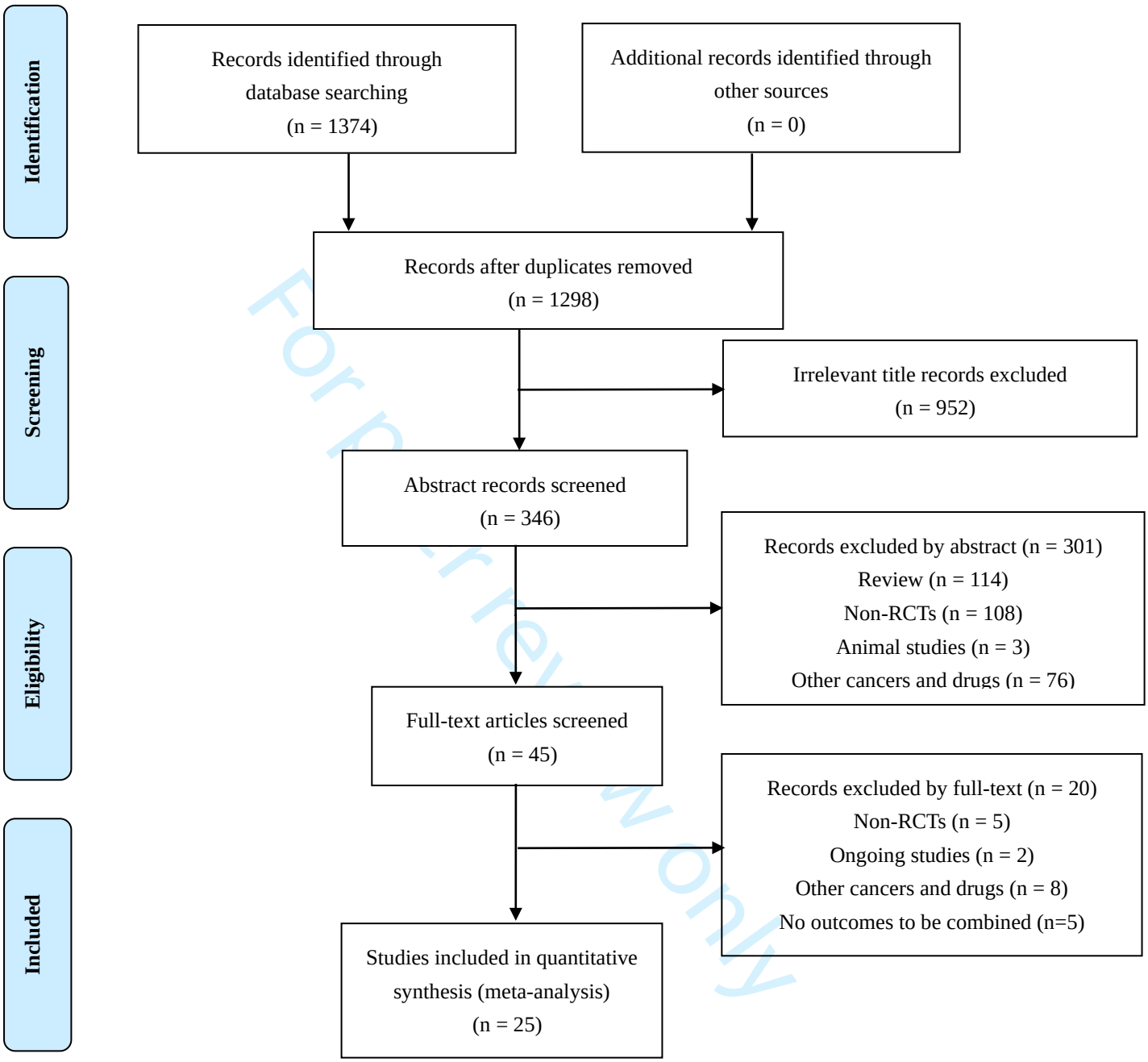



**Figure 1** Flow diagram of the selection process for relevant studies

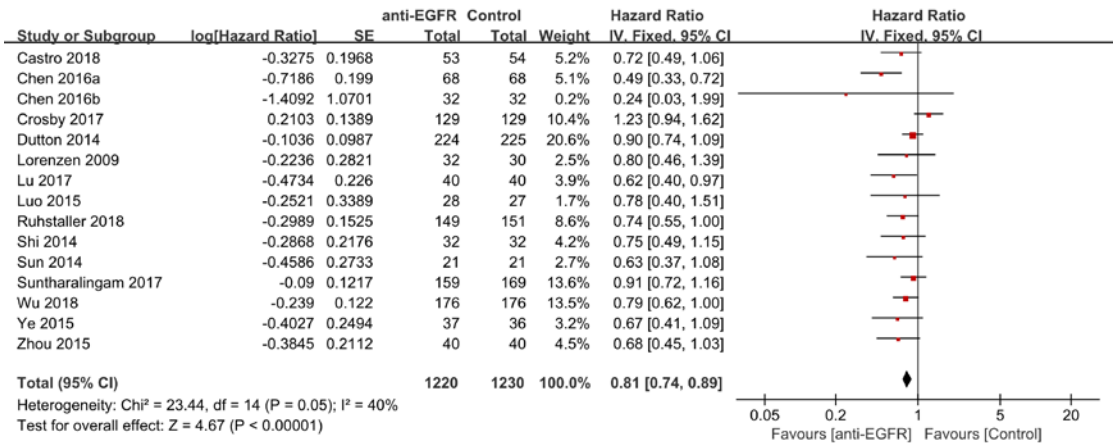



**Figure 2** Forest plot of the overall survival (OS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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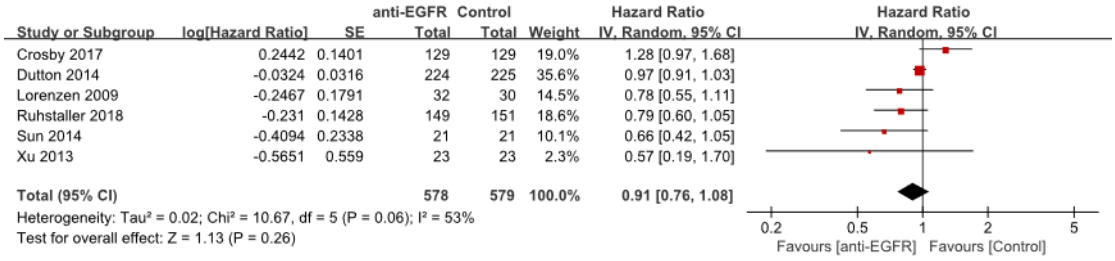



**Figure 3** Forest plot of the progression-free survival (PFS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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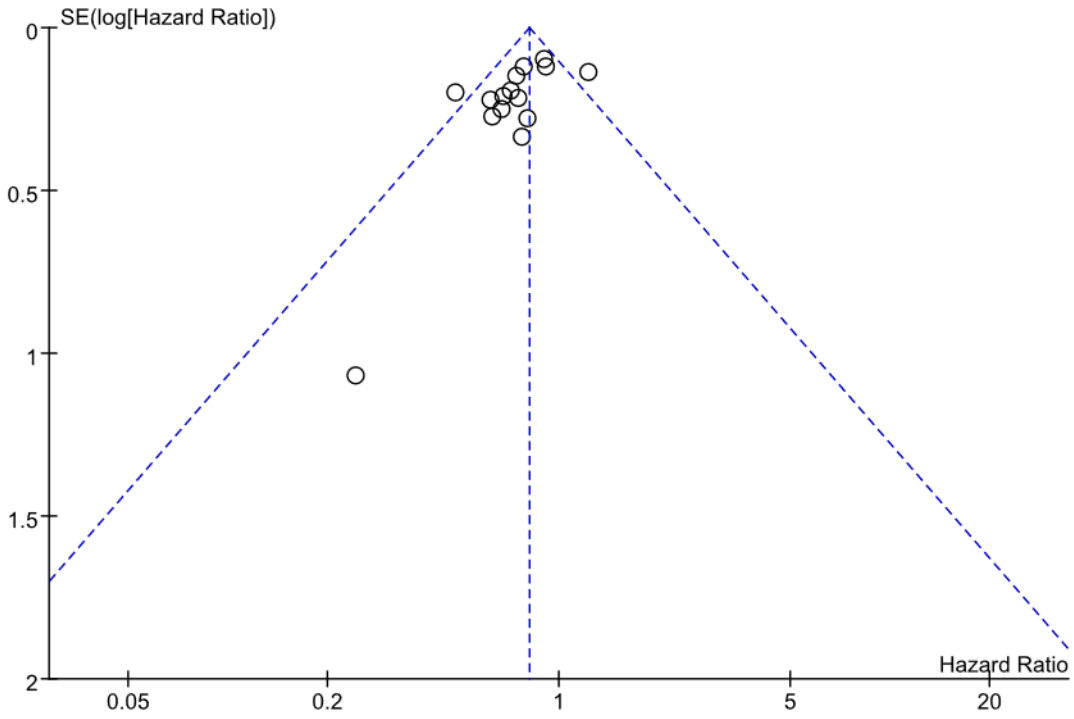



**Figure 4** Funnel plot of publication bias

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**Table S1** Basic characteristics of included studies in the meta-analysis

| Study (year) | Country | Study design | Enrolment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Germany | RCT Multicentre | 2004.12–2006.12 | 32 | 30 | 61 (40–76) | 62 (40–74) | Cetuximab+CT (Cisplatin+5-Fluorouracil) | CT (Cisplatin+5-Fluorouracil) | OS, PFS, ORR, DCR, Toxicities |
| Crosby et al. 2013[39] | UK | RCT Multicentre | 2008.02–2012.02 | 129 | 129 | 66.9 (44.9–84.1) | 66.6 (35.7–81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, Toxicities |
| Chen et al. 2014a[23] | China | RCT Single-institution | 2011.07–2012.01 | 20 | 20 | 57.3 (44–73) | | Surgery+Cetuximab+CT (Cisplatin+5-Fluorouracil） | Surgery+CT (Cisplatin+5-Fluorouracil) | OS, PFS, Toxicities |
| Crosby et al. 2017[38] | UK | RCT Multicentre | 2008.02–2012.02 | 129 | 129 | 66.9 (44.9–84.1) | 66.6 (35.7–81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, PFS, Toxicities |
| Suntharalingam et al. 2017[14] | USA | RCT Multicentre | 2008.06–2013.02 | 159 | 169 | 65 (39–87) | 63 (32–85) | Cetuximab+CRT (Cisplatin+Paclitaxel+RT) | CRT (Cisplatin+Paclitaxel+RT) | OS, Toxicities |
| Lu et al. 2017[35] | China | RCT Single-institution | 2013.01–2015.06 | 40 | 40 | 61.2 (45–80) | 61.3 (45–80) | Cetuximab+CT (Cisplatin+5-Fluorouracil) | CT (Cisplatin+5-Fluorouracil) | OS, ORR, DCR, Toxicities |
| Ruhstaller et al. 2018[13] | Switzerland | RCT Multicentre | 2010.05–2016.10 | 149 | 151 | 61 (56–68) | 61 (53–68) | Cetuximab+IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+ RT) +Surgery | IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+RT) + Surgery | OS, PFS, Toxicities |
| Ramos-Suzarte et al. 2012[16] | Cuba | RCT Multicentre | 2005.12–2008.12 | 33 | 30 | 58.5 | 58.9 | Nimotuzumab+CRT (Cisplatin+5-Fluorouracil+RT) | CRT (Cisplatin+5-Fluorouracil+RT) | ORR, DCR, Toxicities |
| Xu et al. 2013[34] | China | RCT Single-institution | 2009.01–2012.12 | 23 | 23 | 63 (40–74) | 62 (41–75) | Nimotuzumab+CT (Cisplatin+5-Fluorouracil) | CT (Cisplatin+5-Fluorouracil) | PFS, ORR, DCR, Toxicities |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrolment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Shi et al. 2014[33] | China | RCT Single-institution | 2009.10–2010.10 | 32 | 32 | 57 (42–68) | | Nimotuzumab+RT | RT | OS, ORR, DCR |
| Sun et al. 2014[24] | China | RCT Single-institution | 2008.01–2009.07 | 21 | 21 | 54 (36–78) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, PFS, ORR, DCR, Toxicities |
| Wu et al. 2014[25] | China | RCT Single-institution | 2009.10–2011.09 | 16 | 16 | 55 (38–68) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | ORR |
| Luo et al. 2015[31] | China | RCT Single-institution | NR | 28 | 27 | 62 (32–68) | 63 (31–69) | Nimotuzumab+CRT (Docetaxel+RT] | CRT (Docetaxel+RT] | OS, ORR, DCR, Toxicities |
| Chen et al. 2016b[28] | China | RCT Single-institution | 2012.01–2013.01 | 32 | 32 | 53.43 (39–68) | 54.52 (41–68) | Nimotuzumab+CRT (Nadaplatin+RT] | CRT (Nadaplatin+RT] | OS, ORR, DCR |
| Fu et al. 2017[27] | China | RCT Single-institution | 2013.12–2016.12 | 40 | 40 | 60.55 (31–76) | 61.38 (31–75) | Nimotuzumab+CRT (Docetaxel+Cisplatin++RT) | CRT (Docetaxel+Cisplatin++RT) | ORR, DCR |
| Castro et al. 2018[15] | Brazil | RCT Multicentre | 2009.02–2011.08 | 53 | 54 | 60.0 (40–75) | 58.5 (36–82) | Nimotuzumab+CRT (Cisplatin+Fluorouracil+RT) | CRT (Cisplatin+Fluorouracil+RT) | OS, Toxicities |
| Chen et al. 2014b[19] | China | RCT Single-institution | 2010.01–2012.01 | 60 | 60 | 58.7 (49–75) | | Surgery+Erlotinib+CT (Cisplatin+5-Fluorouracil) | Surgery+CT (Cisplatin+5-Fluorouracil) | CR, ORR, DCR |
| Zhang et al. 2014[32] | China | RCT Single-institution | 2009.10–2012.10 | 42 | 42 | 59.1 (49–74) | 58.4 (47–72) | Erlotinib+RT | RT | CR, ORR, DCR |
| Ye et al. 2015[30] | China | RCT Single-institution | 2011.02–2012.04 | 37 | 36 | 53.8 (NR) | 55.2 (NR) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, DCR |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrolment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Zhou et al. 2015[36] | China | RCT Single-institution | 2009.05–2011.11 | 40 | 40 | 55 (46–70) | 54 (45–68) | Erlotinib+RT | CT+RT (Docetaxel+Cisplatin)+RT | OS, ORR, DCR |
| Chen et al. 2016a[37] | China | RCT Single-institution | 2012.11–2014.11 | 68 | 68 | 47.3 (28–76) | 47.9 (26–78) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, ORR |
| Zhao et al. 2016[29] | China | RCT Single-institution | 2013.08–2014.04 | 38 | 38 | 64.5 (61–83) | 63.4 (59–81) | Erlotinib+RT | RT | CR, ORR, DCR |
| Wu et al. 2018[18] | China | RCT Multicentre | 2007.12–2015.06 | 176 | 176 | 61 (35–70) | 61.5 (40–70) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, Toxicities |
| Zeng et al. 2014[20] | China | RCT Single-institution | 2018.03–2010.08 | 26 | 26 | NR (40–80) | | Gefitinib+CT (Oxaliplatin+5-Fluorouracil) | CT (Oxaliplatin+5-Fluorouracil) | ORR, DCR |
| Dutton et al. 2014[21] | UK | RCT Multicentre | 2009.03–2011.11 | 224 | 225 | 64.7 (58.0–70.1) | 64.9 (58.2–70.7) | Gefitinib+CRT (NR) | Placebo+CRT (NR) | OS, PFS, Toxicities |

*RCT randomized controlled trial, NR Not reported, OS Overall survival, PFS Progression-free survival, ORR Objective response rate, DCR Disease control rate, RT Radiotherapy, CRT Chemoradiotherapy, IC Induction chemotherapy, CT Chemotherapy*

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**Table S2** Non-haematological toxicities of grade 3–5

| Toxicities | Subgroups | No. of studies | Sample Size | | RR [95% CI] | *P* | Heterogeneity | |
|---|---|---|---|---|---|---|---|---|
| | | | anti-EGFR (events/total) | Control (events/total) | | | $I^2$ | *P* |
| Cardiac toxicity** | ALL | 4 | 19/664 | 9/676 | 2.13 [0.97, 4.67] | 0.06 | 0% | 0.72 |
| Gastrointestinal toxicity | ALL | 28 | 359/2927 | 330/2990 | 1.11 [0.97, 1.28] | 0.13 | 16% | 0.23 |
| | Nausea and Vomiting | 6 | 74/558 | 81/565 | 0.94 [0.70, 1.26] | 0.67 | 43% | 0.12 |
| | Diarrhoea | 7 | 55/782 | 24/790 | 2.26 [1.42, 3.60] | 0.0005* | 30% | 0.21 |
| | Dehydration | 2 | 37/206 | 28/220 | 1.42 [0.91, 2.23] | 0.12 | 40% | 0.20 |
| | Dysphagia | 4 | 100/493 | 104/505 | 0.97 [0.76, 1.24] | 0.83 | 0% | 0.79 |
| | Oesophagitis | 5 | 64/520 | 63/531 | 1.03 [0.75, 1.42] | 0.84 | 0% | 0.70 |
| | Anorexia | 4 | 29/368 | 30/379 | 1.00 [0.61, 1.63] | 1.00 | 0% | 0.99 |
| Dermatologic toxicity | Acne-like rash | 6 | 37/749 | 2/760 | 9.83 [3.54, 27.28] | <0.0001* | 0% | 0.90 |
| Pulmonary toxicity | Dyspnoea | 3 | 10/335 | 6/349 | 1.72 [0.63, 4.68] | 0.29 | 26% | 0.25 |
| Renal toxicity | Renal insufficiency | 4 | 7/588 | 6/601 | 1.17 [0.41, 3.34] | 0.77 | 0% | 0.54 |
| Constitutional symptoms | ALL | 9 | 120/1084 | 104/1109 | 1.17 [0.92, 1.49] | 0.20 | 0% | 0.67 |
| | Fatigue | 6 | 109/749 | 93/760 | 1.19 [0.92, 1.53] | 0.19 | 0% | 0.51 |
| | Weight loss | 3 | 11/335 | 11/349 | 1.05 [0.47, 2.35] | 0.90 | 0% | 0.47 |
| Infection | ALL | 9 | 15/817 | 7/839 | 2.21 [0.91, 5.36] | 0.08 | 0% | 0.97 |
| | Fever | 4 | 2/271 | 1/280 | 2.17 [0.20, 23.69] | 0.53 | Not applicable | Not applicable |
| | Pneumonitis | 3 | 6/364 | 2/376 | 3.14 [0.64, 15.34] | 0.16 | 0% | 0.70 |
| | Stomatitis | 2 | 7/182 | 4/183 | 1.77 [0.53, 5.92] | 0.36 | 0% | 0.83 |
| Laboratory findings | ALL | 9 | 52/1005 | 34/1047 | 1.59 [1.05, 2.42] | 0.03* | 0% | 0.70 |
| | Hypomagnesaemia | 3 | 15/335 | 6/349 | 2.61 [1.02, 6.68] | 0.04* | 5% | 0.31 |
| | Hypokalaemia | 3 | 20/335 | 19/349 | 1.10 [0.60, 2.01] | 0.75 | 0% | 0.70 |
| | Hyponatraemia | 3 | 17/335 | 9/349 | 1.97 [0.91, 4.25] | 0.08 | 0% | 0.80 |

*$P < 0.05$, **including cardiac disorder/ischaemia/infarction*



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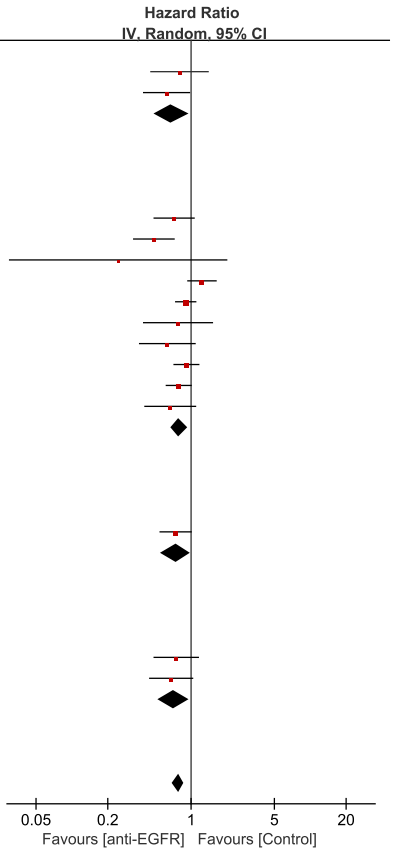
**Table S3** Sensitivity analysis by excluding the Ruhstaller et al. 2018,[13] Chen et al. 2014a,[23] Chen et al. 2014b,[19] Sun et al. 2014,[24] and Wu et al. 2014[25]

| Outcome | No. of studies | Ruhstaller et al. 2018[13] | | Chen et al. 2014a[23] | | Chen et al. 2014b[19] | | Sun et al. 2014[24] | | Wu et al. 2014[25] | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* |
| OS | 15 | 0.82 [0.75, 0.90] | <0.0001 | NR | NR | NR | NR | 0.82 [0.75, 0.89] | <0.00001 | NR | NR |
| PFS | 6 | 0.93 [0.76, 1.14] | 0.48 | NR | NR | NR | NR | 0.94 [0.79, 1.12] | 0.48 | NR | NR |
| ORR | 16 | NR | NR | NR | NR | 1.33 [1.14, 1.54] | 0.0002 | 1.32 [1.14, 1.52] | 0.0002 | 1.34 [1.19, 1.51] | <0.00001 |
| DCR | 15 | NR | NR | NR | NR | 1.24 [1.11, 1.38] | <0.0001 | 1.23 [1.13, 1.34] | <0.00001 | NR | NR |

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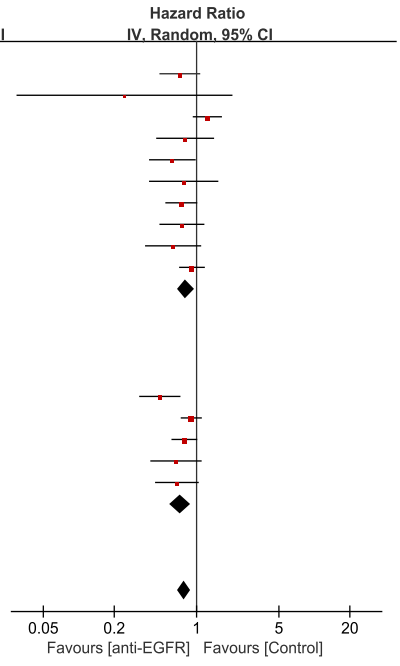
**A**

| Study or Subgroup | log[Hazard Ratio] | SE | anti-EGFR Total | Control Total | Weight | Hazard Ratio IV, Random, 95% CI |
|---|---|---|---|---|---|---|
| **2.1.1 EGFR+CT** | | | | | | |
| Lorenzen 2009 | -0.2236 | 0.2821 | 32 | 30 | 3.9% | 0.80 [0.46, 1.39] |
| Lu 2017 | -0.4734 | 0.226 | 40 | 40 | 5.5% | 0.62 [0.40, 0.97] |
| **Subtotal (95% CI)** | | | **72** | **70** | **9.4%** | **0.69 [0.49, 0.97]** |
| Heterogeneity: Tau² = 0.00; Chi² = 0.48, df = 1 (P = 0.49); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.13 (P = 0.03) | | | | | | |
| **2.1.2 EGFR+CRT** | | | | | | |
| Castro 2018 | -0.3275 | 0.1968 | 53 | 54 | 6.6% | 0.72 [0.49, 1.06] |
| Chen 2016a | -0.7186 | 0.199 | 68 | 68 | 6.5% | 0.49 [0.33, 0.72] |
| Chen 2016b | -1.4092 | 1.0701 | 32 | 32 | 0.3% | 0.24 [0.03, 1.99] |
| Crosby 2017 | 0.2103 | 0.1389 | 129 | 129 | 9.8% | 1.23 [0.94, 1.62] |
| Dutton 2014 | -0.1036 | 0.0987 | 224 | 225 | 12.8% | 0.90 [0.74, 1.09] |
| Luo 2015 | -0.2521 | 0.3389 | 28 | 27 | 2.9% | 0.78 [0.40, 1.51] |
| Sun 2014 | -0.4586 | 0.2733 | 21 | 21 | 4.1% | 0.63 [0.37, 1.08] |
| Suntharalingam 2017 | -0.09 | 0.1217 | 159 | 169 | 11.0% | 0.91 [0.72, 1.16] |
| Wu 2018 | -0.239 | 0.122 | 176 | 176 | 11.0% | 0.79 [0.62, 1.00] |
| Ye 2015 | -0.4027 | 0.2494 | 37 | 36 | 4.8% | 0.67 [0.41, 1.09] |
| **Subtotal (95% CI)** | | | **927** | **937** | **69.8%** | **0.80 [0.67, 0.95]** |
| Heterogeneity: Tau² = 0.04; Chi² = 20.33, df = 9 (P = 0.02); I² = 56% | | | | | | |
| Test for overall effect: Z = 2.60 (P = 0.009) | | | | | | |
| **2.1.3 EGFR+IC+CRT+S** | | | | | | |
| Ruhstaller 2018 | -0.2989 | 0.1525 | 149 | 151 | 8.9% | 0.74 [0.55, 1.00] |
| **Subtotal (95% CI)** | | | **149** | **151** | **8.9%** | **0.74 [0.55, 1.00]** |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.96 (P = 0.05) | | | | | | |
| **2.1.5 EGFR+RT** | | | | | | |
| Shi 2014 | -0.2868 | 0.2176 | 32 | 32 | 5.8% | 0.75 [0.49, 1.15] |
| Zhou 2015 | -0.3845 | 0.2112 | 40 | 40 | 6.0% | 0.68 [0.45, 1.03] |
| **Subtotal (95% CI)** | | | **72** | **72** | **11.8%** | **0.71 [0.53, 0.96]** |
| Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 1 (P = 0.75); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.22 (P = 0.03) | | | | | | |
| **Total (95% CI)** | | | **1220** | **1230** | **100.0%** | **0.78 [0.69, 0.88]** |
| Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40% | | | | | | |
| Test for overall effect: Z = 3.95 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.84, df = 3 (P = 0.84), I² = 0% | | | | | | |




**B**

| Study or Subgroup | log[Hazard Ratio] | SE | anti-EGFR Total | Control Total | Weight | Hazard Ratio IV, Random, 95% CI |
|---|---|---|---|---|---|---|
| **2.2.1 mAb** | | | | | | |
| Castro 2018 | -0.3275 | 0.1968 | 53 | 54 | 6.6% | 0.72 [0.49, 1.06] |
| Chen 2016b | -1.4092 | 1.0701 | 32 | 32 | 0.3% | 0.24 [0.03, 1.99] |
| Crosby 2017 | 0.2103 | 0.1389 | 129 | 129 | 9.8% | 1.23 [0.94, 1.62] |
| Lorenzen 2009 | -0.2236 | 0.2821 | 32 | 30 | 3.9% | 0.80 [0.46, 1.39] |
| Lu 2017 | -0.4734 | 0.226 | 40 | 40 | 5.5% | 0.62 [0.40, 0.97] |
| Luo 2015 | -0.2521 | 0.3389 | 28 | 27 | 2.9% | 0.78 [0.40, 1.51] |
| Ruhstaller 2018 | -0.2989 | 0.1525 | 149 | 151 | 8.9% | 0.74 [0.55, 1.00] |
| Shi 2014 | -0.2868 | 0.2176 | 32 | 32 | 5.8% | 0.75 [0.49, 1.15] |
| Sun 2014 | -0.4586 | 0.2733 | 21 | 21 | 4.1% | 0.63 [0.37, 1.08] |
| Suntharalingam 2017 | -0.09 | 0.1217 | 159 | 169 | 11.0% | 0.91 [0.72, 1.16] |
| **Subtotal (95% CI)** | | | **675** | **685** | **58.9%** | **0.81 [0.69, 0.96]** |
| Heterogeneity: Tau² = 0.02; Chi² = 13.92, df = 9 (P = 0.13); I² = 35% | | | | | | |
| Test for overall effect: Z = 2.47 (P = 0.01) | | | | | | |
| **2.2.2 TKI** | | | | | | |
| Chen 2016a | -0.7186 | 0.199 | 68 | 68 | 6.5% | 0.49 [0.33, 0.72] |
| Dutton 2014 | -0.1036 | 0.0987 | 224 | 225 | 12.8% | 0.90 [0.74, 1.09] |
| Wu 2018 | -0.239 | 0.122 | 176 | 176 | 11.0% | 0.79 [0.62, 1.00] |
| Ye 2015 | -0.4027 | 0.2494 | 37 | 36 | 4.8% | 0.67 [0.41, 1.09] |
| Zhou 2015 | -0.3845 | 0.2112 | 40 | 40 | 6.0% | 0.68 [0.45, 1.03] |
| **Subtotal (95% CI)** | | | **545** | **545** | **41.1%** | **0.73 [0.59, 0.89]** |
| Heterogeneity: Tau² = 0.03; Chi² = 8.52, df = 4 (P = 0.07); I² = 53% | | | | | | |
| Test for overall effect: Z = 3.04 (P = 0.002) | | | | | | |
| **Total (95% CI)** | | | **1220** | **1230** | **100.0%** | **0.78 [0.69, 0.88]** |
| Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40% | | | | | | |
| Test for overall effect: Z = 3.95 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.75, df = 1 (P = 0.39), I² = 0% | | | | | | |



**C**

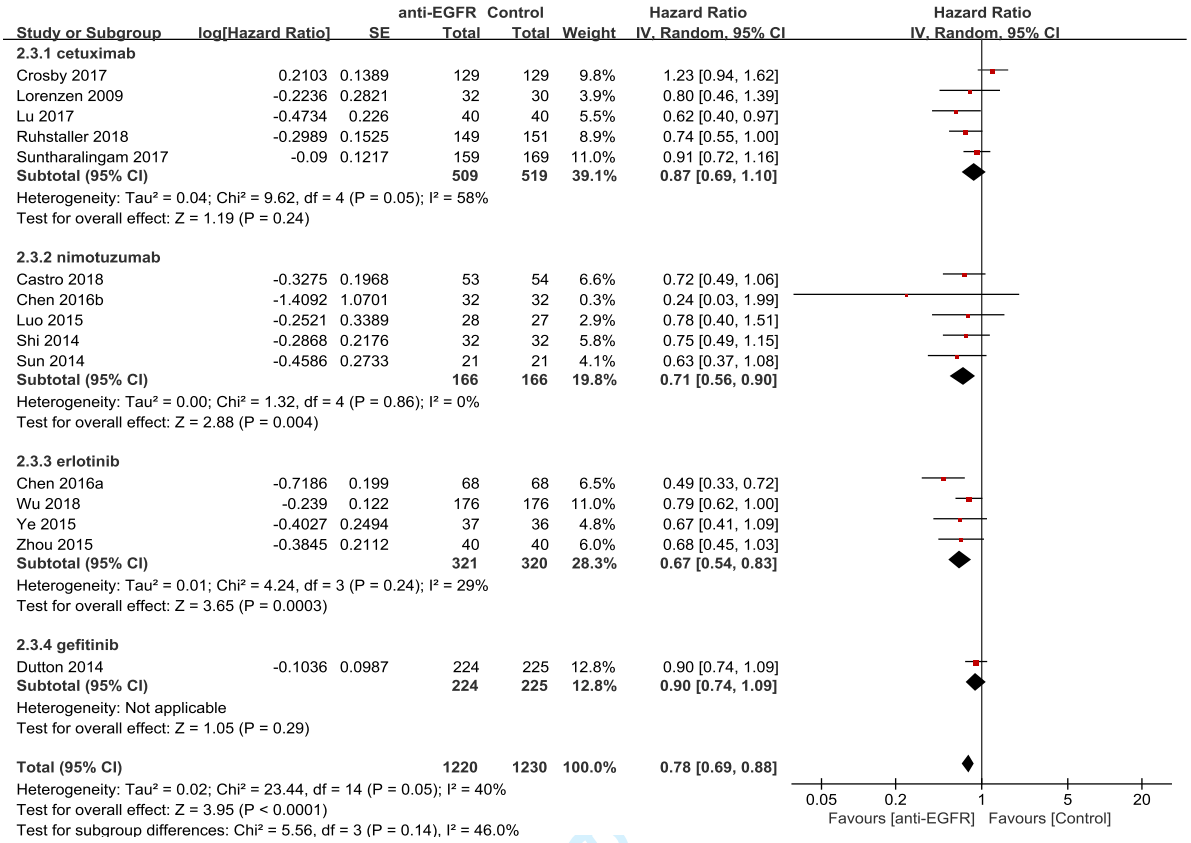
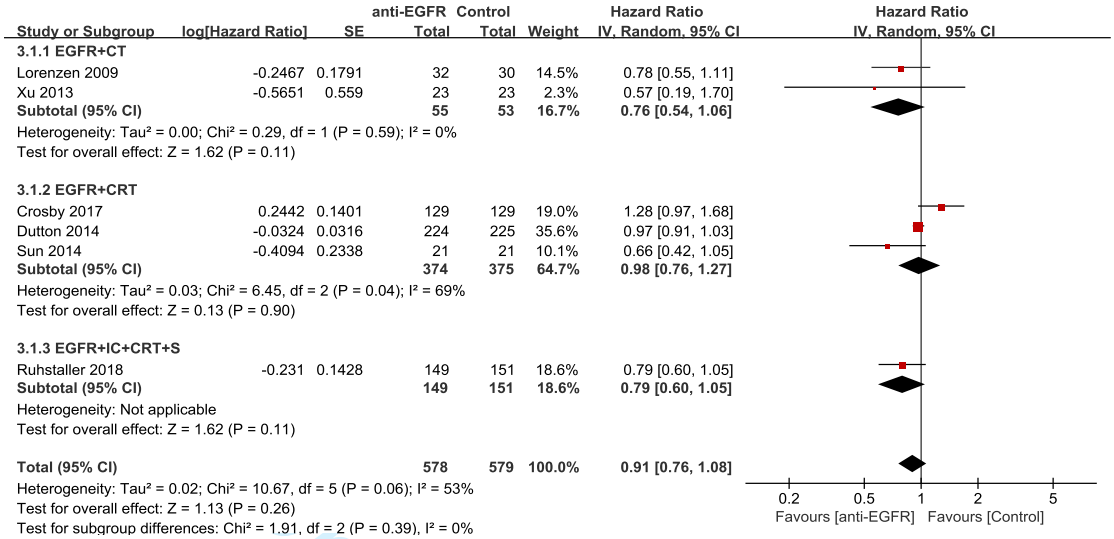


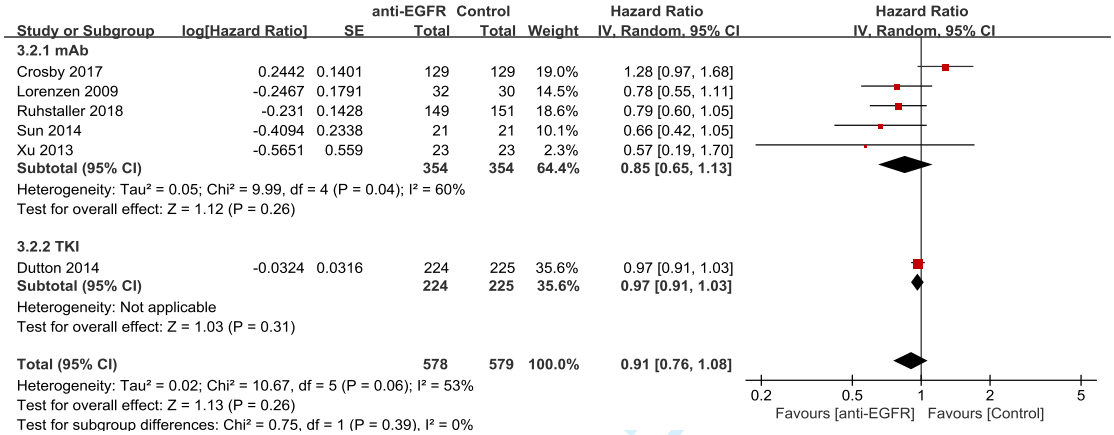

**Figure S1** Subgroup analysis of the overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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**A**




**B**




**C**

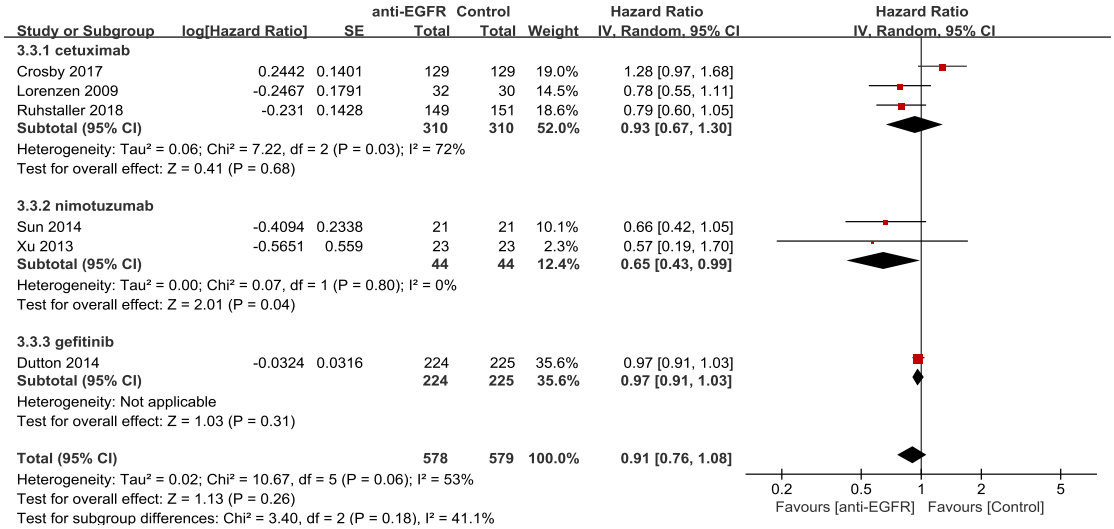



**Figure S2** Subgroup analysis of the progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

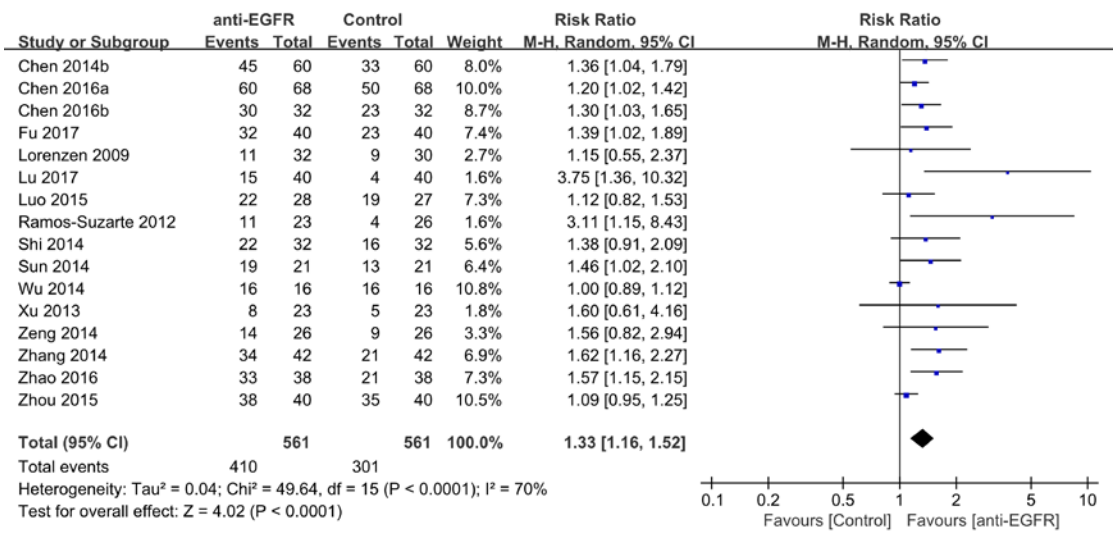
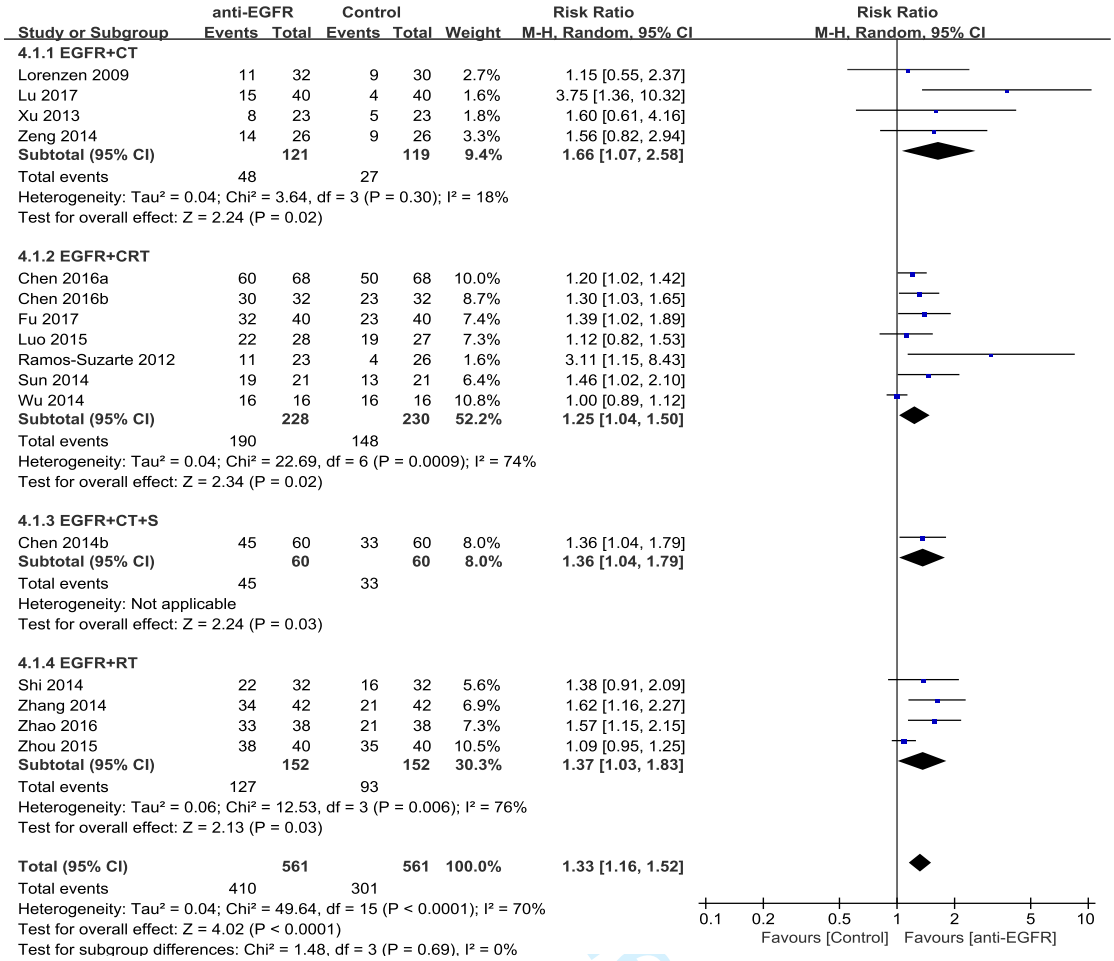


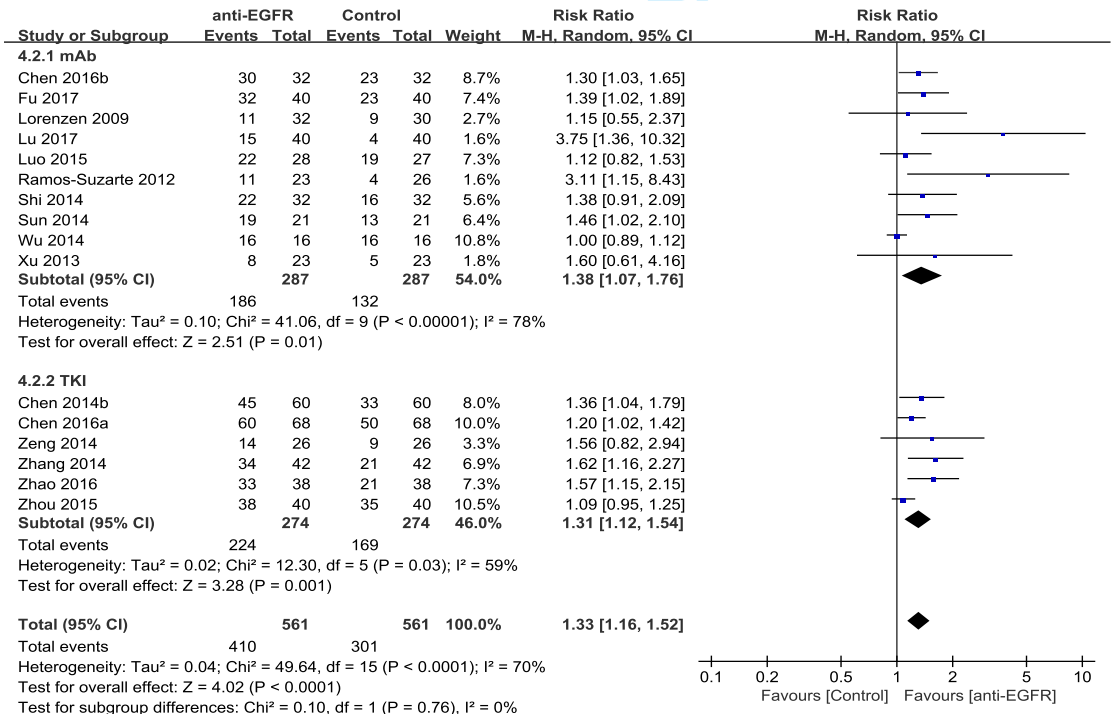

**Figure S3** Forest plot of the objective response rate (ORR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.1.1 EGFR+CT
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Subtotal (95% CI) 121 119 9.4% 1.66 [1.07, 2.58]
Total events 48 27
Heterogeneity: Tau² = 0.04; Chi² = 3.64, df = 3 (P = 0.30); I² = 18%
Test for overall effect: Z = 2.24 (P = 0.02)
4.1.2 EGFR+CRT
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Subtotal (95% CI) 228 230 52.2% 1.25 [1.04, 1.50]
Total events 190 148
Heterogeneity: Tau² = 0.04; Chi² = 22.69, df = 6 (P = 0.0009); I² = 74%
Test for overall effect: Z = 2.34 (P = 0.02)
4.1.3 EGFR+CT+S
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Subtotal (95% CI) 60 60 8.0% 1.36 [1.04, 1.79]
Total events 45 33
Heterogeneity: Not applicable
Test for overall effect: Z = 2.24 (P = 0.03)
4.1.4 EGFR+RT
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 152 152 30.3% 1.37 [1.03, 1.83]
Total events 127 93
Heterogeneity: Tau² = 0.06; Chi² = 12.53, df = 3 (P = 0.006); I² = 76%
Test for overall effect: Z = 2.13 (P = 0.03)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 1.48, df = 3 (P = 0.69), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.2.1 mAb
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Subtotal (95% CI) 287 287 54.0% 1.38 [1.07, 1.76]
Total events 186 132
Heterogeneity: Tau² = 0.10; Chi² = 41.06, df = 9 (P < 0.00001); I² = 78%
Test for overall effect: Z = 2.51 (P = 0.01)
4.2.2 TKI
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 274 274 46.0% 1.31 [1.12, 1.54]
Total events 224 169
Heterogeneity: Tau² = 0.02; Chi² = 12.30, df = 5 (P = 0.03); I² = 59%
Test for overall effect: Z = 3.28 (P = 0.001)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 0.10, df = 1 (P = 0.76), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**C**

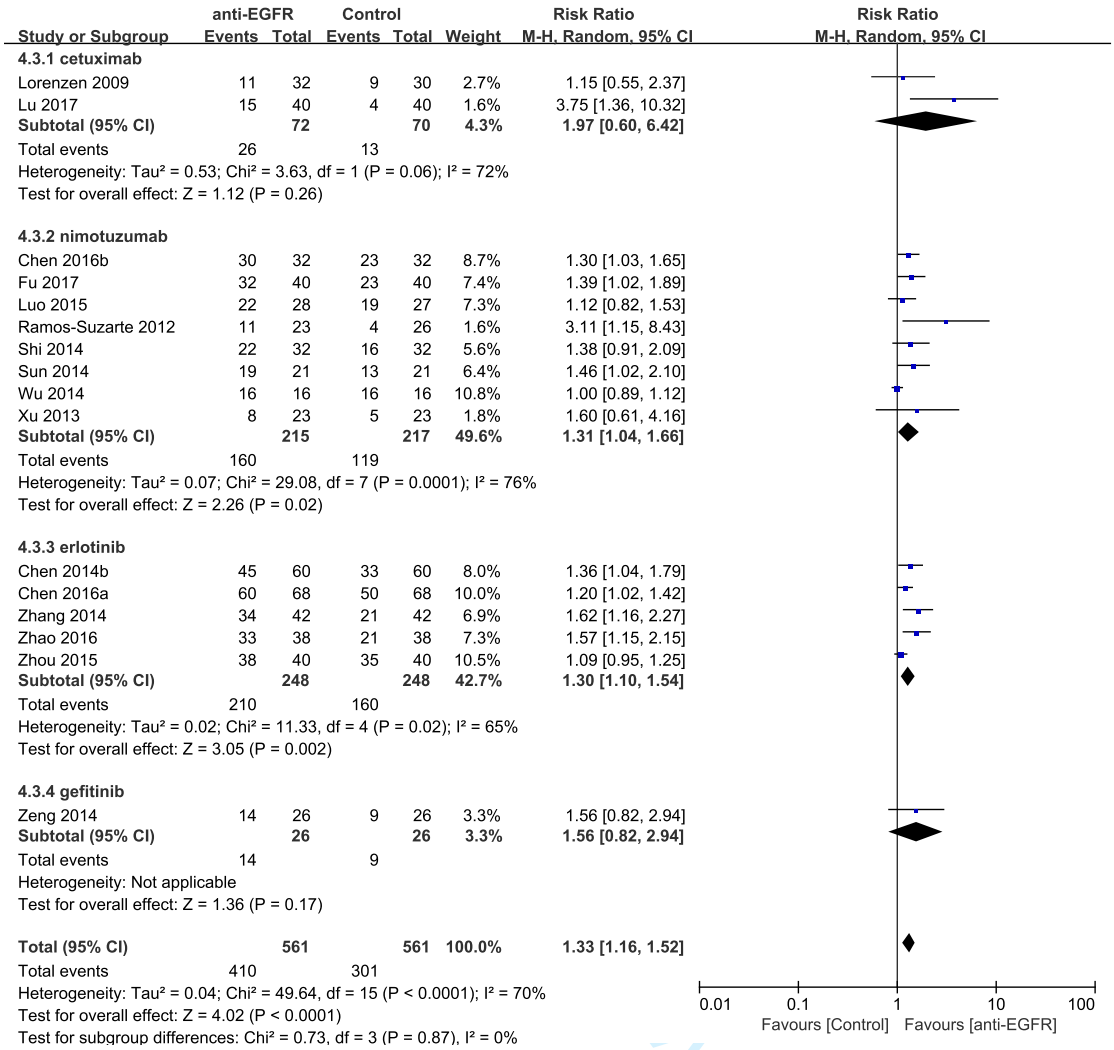



**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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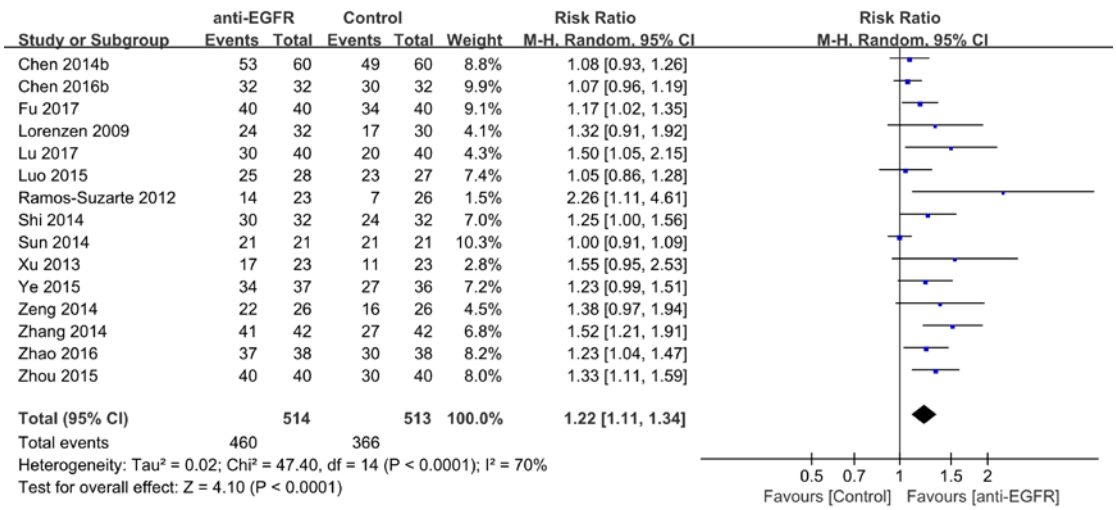
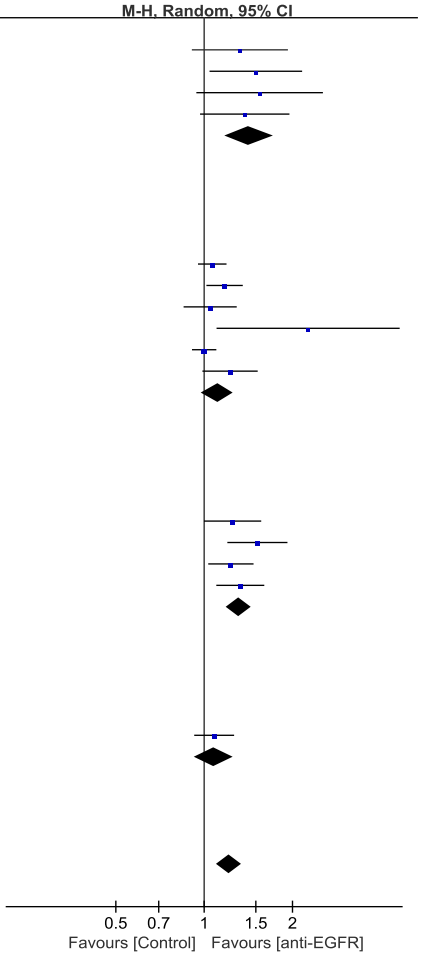



**Figure S5** Forest plot of the disease control rate (DCR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

peer review only

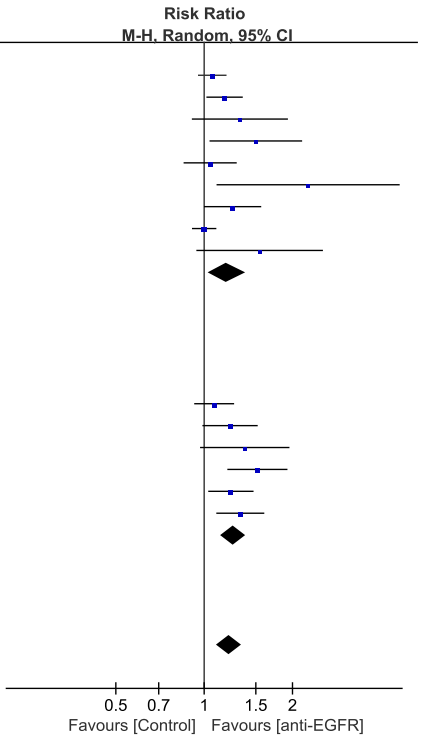
**A**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.1.1 EGFR+CT** | | | | | | |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| **Subtotal (95% CI)** | | **121** | | **119** | **15.7%** | **1.42 [1.17, 1.72]** |
| Total events | 93 | | 64 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 0.38, df = 3 (P = 0.94); I² = 0% | | | | | | |
| Test for overall effect: Z = 3.61 (P = 0.0003) | | | | | | |
| **5.1.2 EGFR+CRT** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| **Subtotal (95% CI)** | | **181** | | **182** | **45.5%** | **1.11 [0.98, 1.26]** |
| Total events | 166 | | 142 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 18.13, df = 5 (P = 0.003); I² = 72% | | | | | | |
| Test for overall effect: Z = 1.69 (P = 0.09) | | | | | | |
| **5.1.3 EGFR+RT** | | | | | | |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **152** | | **152** | **30.0%** | **1.31 [1.19, 1.45]** |
| Total events | 148 | | 111 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 2.33, df = 3 (P = 0.51); I² = 0% | | | | | | |
| Test for overall effect: Z = 5.41 (P < 0.00001) | | | | | | |
| **5.1.4 EGFR+CT+S** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| **Subtotal (95% CI)** | | **60** | | **60** | **8.8%** | **1.08 [0.93, 1.26]** |
| Total events | 53 | | 49 | | | |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.02 (P = 0.31) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 9.05, df = 3 (P = 0.03), I² = 66.9% | | | | | | |




**B**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.2.1 mAb** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| **Subtotal (95% CI)** | | **271** | | **271** | **56.5%** | **1.20 [1.04, 1.38]** |
| Total events | 233 | | 187 | | | |
| Heterogeneity: Tau² = 0.03; Chi² = 37.25, df = 8 (P < 0.0001); I² = 79% | | | | | | |
| Test for overall effect: Z = 2.43 (P = 0.01) | | | | | | |
| **5.2.2 TKI** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **243** | | **242** | **43.5%** | **1.26 [1.14, 1.38]** |
| Total events | 227 | | 179 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 7.16, df = 5 (P = 0.21); I² = 30% | | | | | | |
| Test for overall effect: Z = 4.54 (P < 0.00001) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.29, df = 1 (P = 0.59), I² = 0% | | | | | | |



**C**

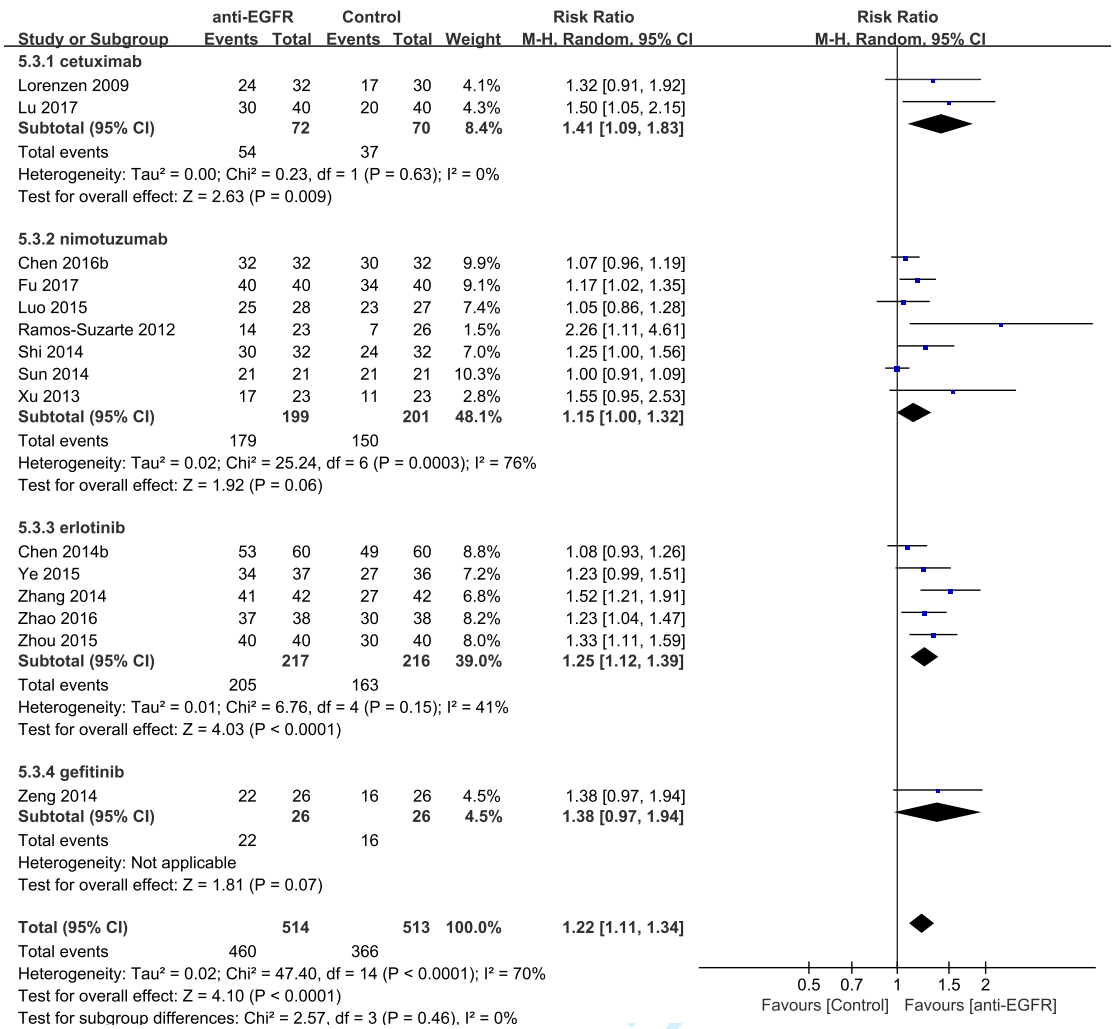



**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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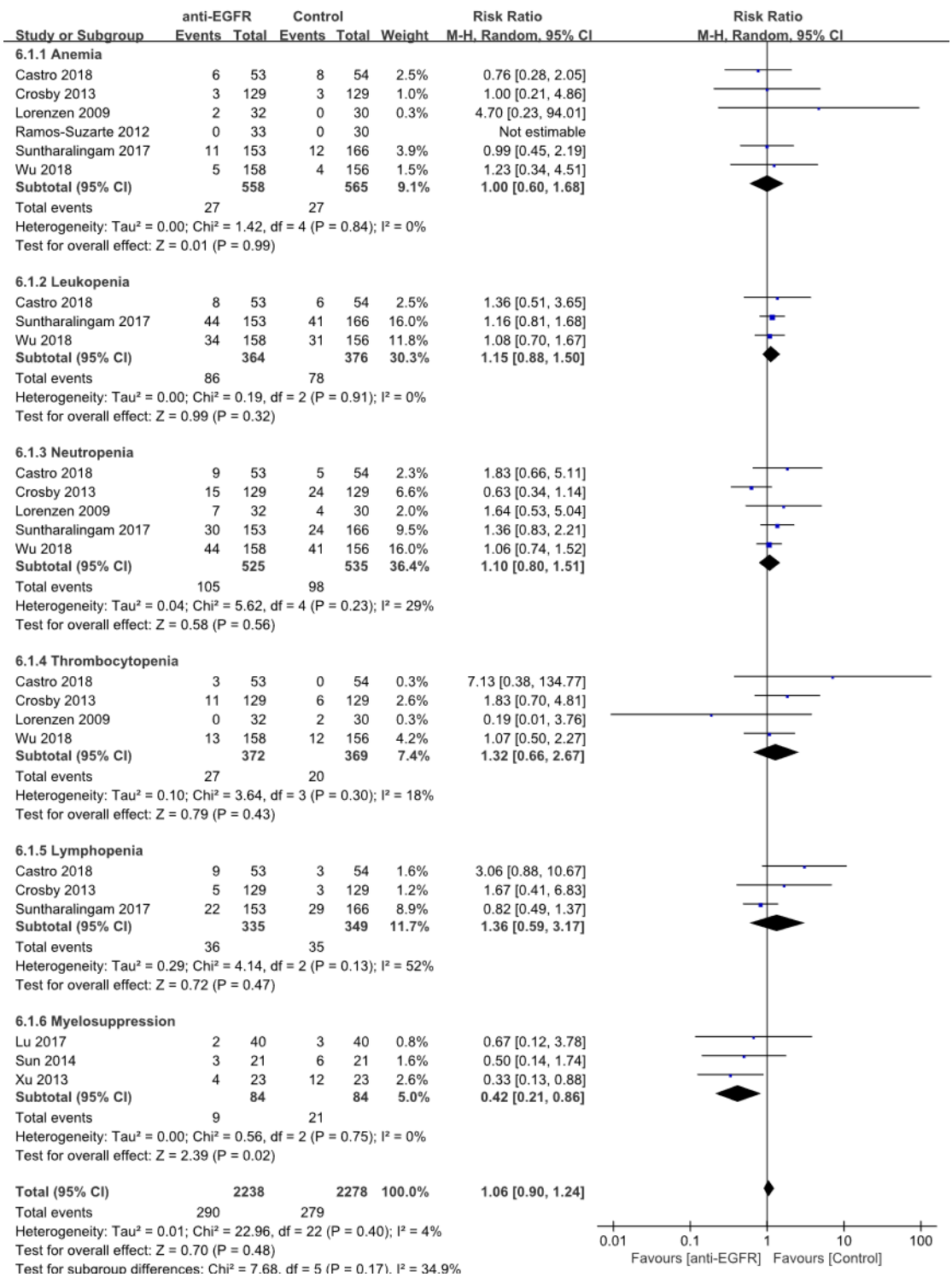
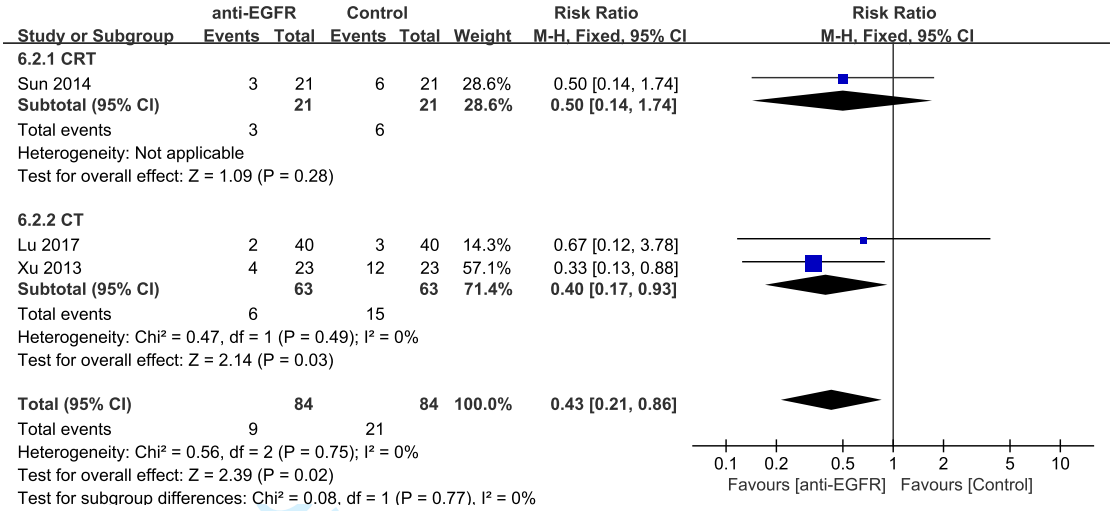



**Figure S7** Forest plot of grade 3–5 haematological toxicities in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer



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**A**




**B**

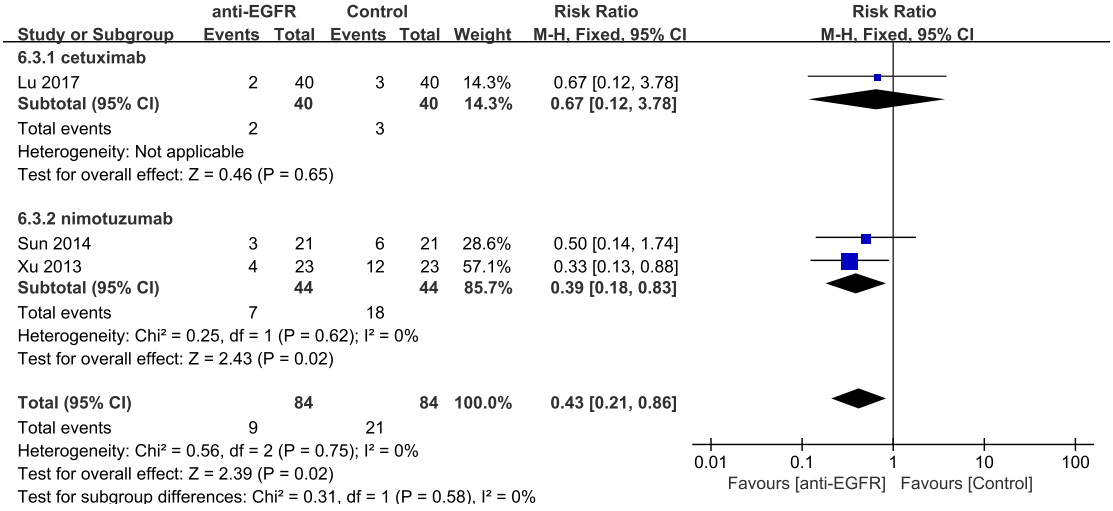
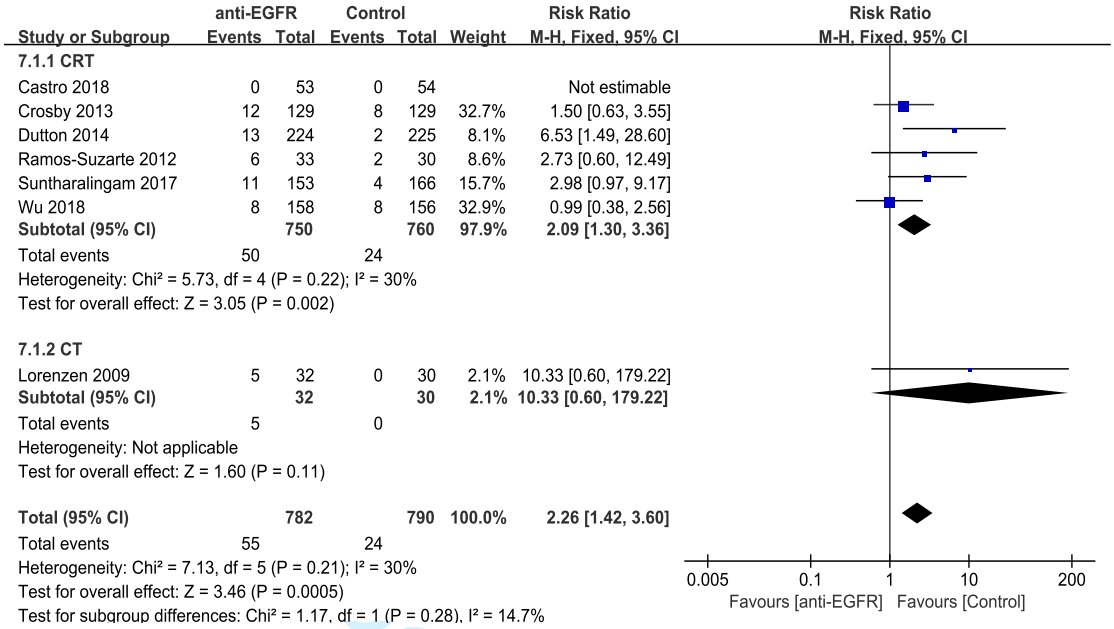


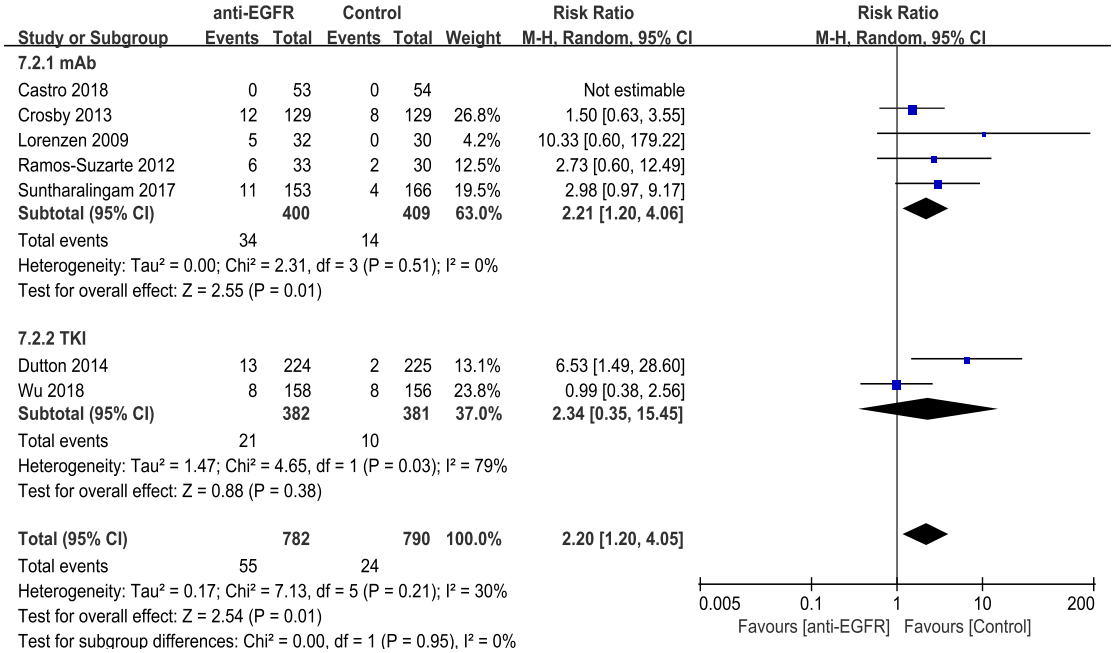

**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

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**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
7.1.1 CRT
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 32.7% 1.50 [0.63, 3.55]
Dutton 2014 13 224 2 225 8.1% 6.53 [1.49, 28.60]
Ramos-Suzarte 2012 6 33 2 30 8.6% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 15.7% 2.98 [0.97, 9.17]
Wu 2018 8 158 8 156 32.9% 0.99 [0.38, 2.56]
Subtotal (95% CI) 750 760 97.9% 2.09 [1.30, 3.36]
Total events 50 24
Heterogeneity: Chi² = 5.73, df = 4 (P = 0.22); I² = 30%
Test for overall effect: Z = 3.05 (P = 0.002)
7.1.2 CT
Lorenzen 2009 5 32 0 30 2.1% 10.33 [0.60, 179.22]
Subtotal (95% CI) 32 30 2.1% 10.33 [0.60, 179.22]
Total events 5 0
Heterogeneity: Not applicable
Test for overall effect: Z = 1.60 (P = 0.11)
Total (95% CI) 782 790 100.0% 2.26 [1.42, 3.60]
Total events 55 24
Heterogeneity: Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 3.46 (P = 0.0005)
Test for subgroup differences: Chi² = 1.17, df = 1 (P = 0.28), I² = 14.7%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
7.2.1 mAb
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 26.8% 1.50 [0.63, 3.55]
Lorenzen 2009 5 32 0 30 4.2% 10.33 [0.60, 179.22]
Ramos-Suzarte 2012 6 33 2 30 12.5% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 19.5% 2.98 [0.97, 9.17]
Subtotal (95% CI) 400 409 63.0% 2.21 [1.20, 4.06]
Total events 34 14
Heterogeneity: Tau² = 0.00; Chi² = 2.31, df = 3 (P = 0.51); I² = 0%
Test for overall effect: Z = 2.55 (P = 0.01)
7.2.2 TKI
Dutton 2014 13 224 2 225 13.1% 6.53 [1.49, 28.60]
Wu 2018 8 158 8 156 23.8% 0.99 [0.38, 2.56]
Subtotal (95% CI) 382 381 37.0% 2.34 [0.35, 15.45]
Total events 21 10
Heterogeneity: Tau² = 1.47; Chi² = 4.65, df = 1 (P = 0.03); I² = 79%
Test for overall effect: Z = 0.88 (P = 0.38)
Total (95% CI) 782 790 100.0% 2.20 [1.20, 4.05]
Total events 55 24
Heterogeneity: Tau² = 0.17; Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 2.54 (P = 0.01)
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.95), I² = 0%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**C**

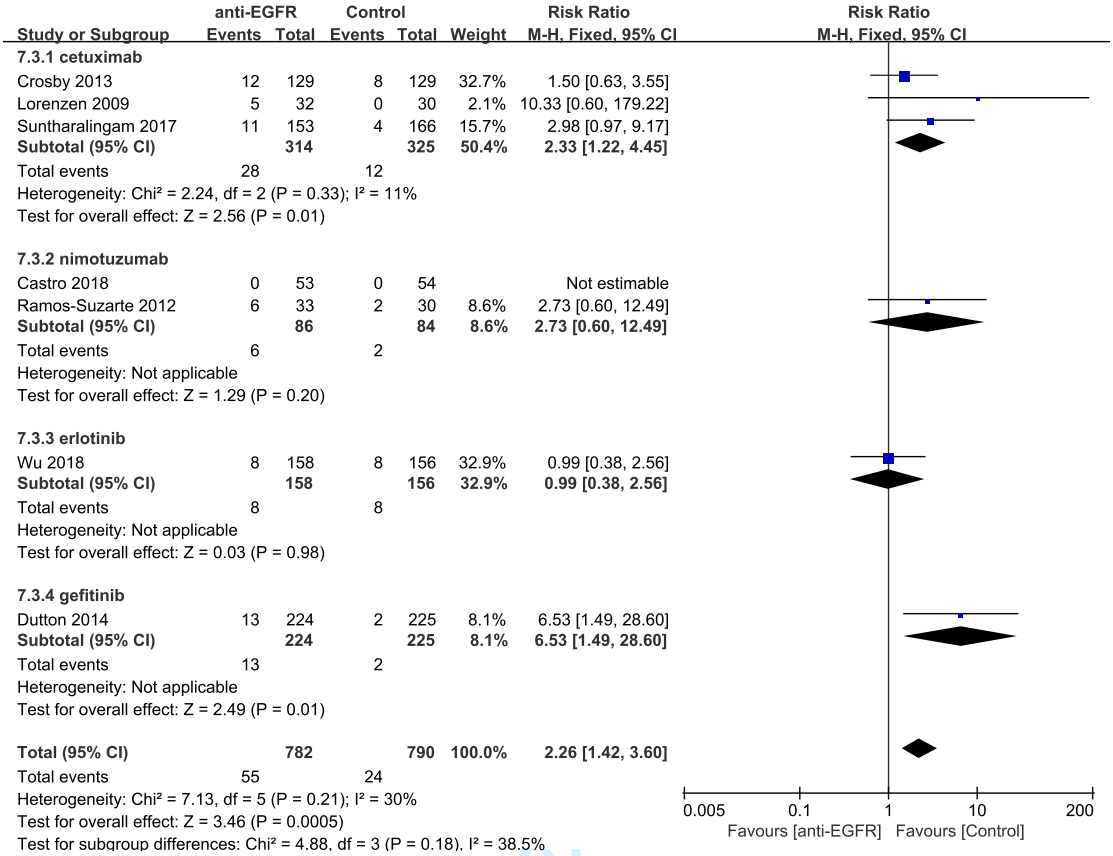
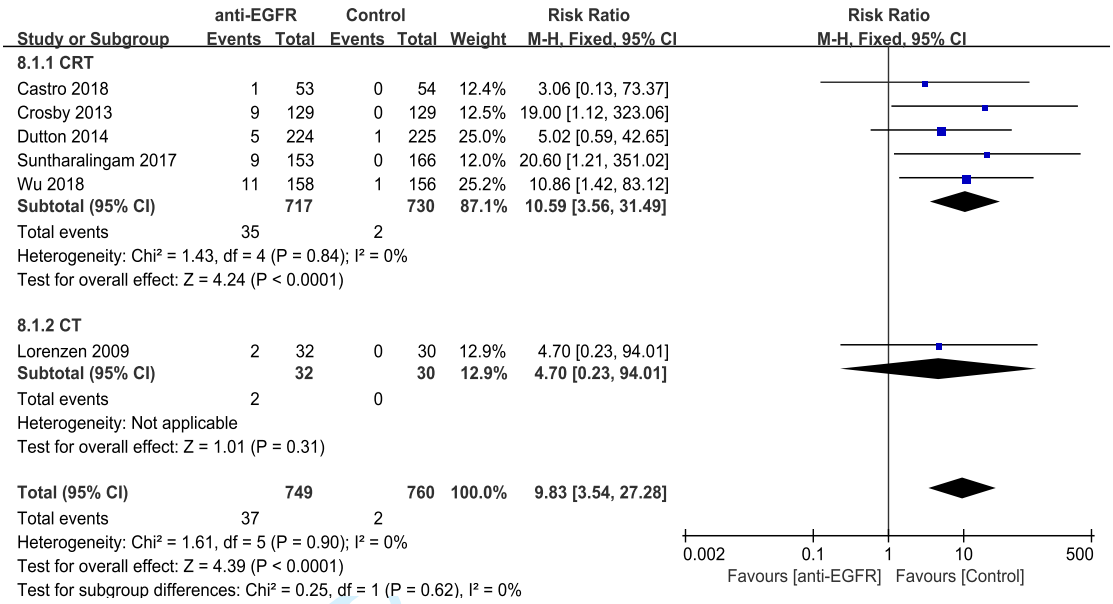


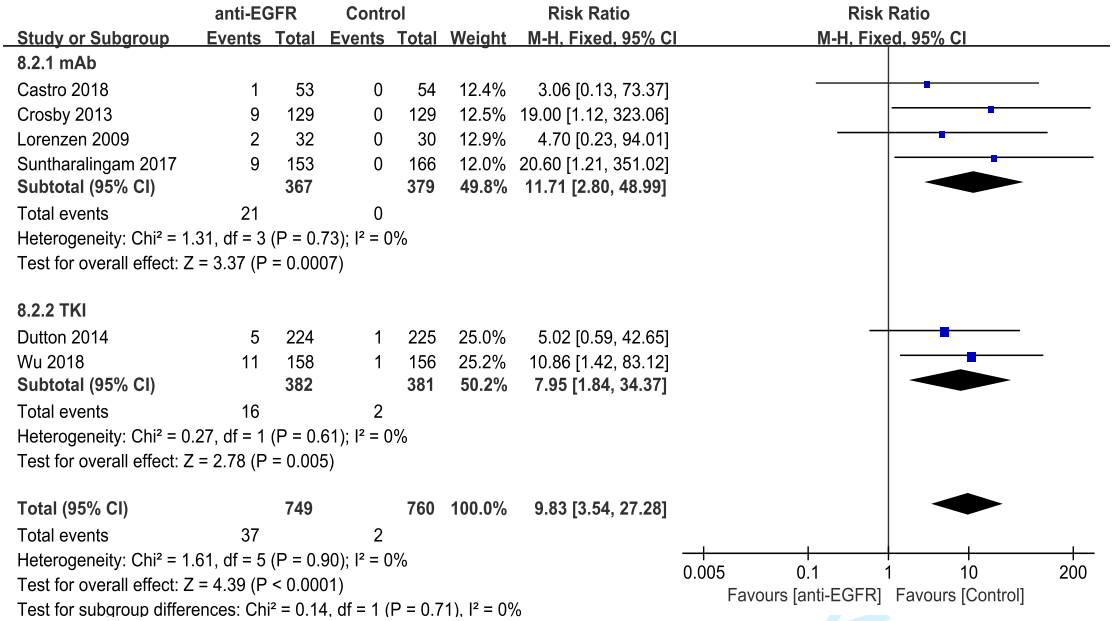

**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.1.1 CRT
Castro 2018 1 53 0 54 12.4% 3.06 [0.13, 73.37]
Crosby 2013 9 129 0 129 12.5% 19.00 [1.12, 323.06]
Dutton 2014 5 224 1 225 25.0% 5.02 [0.59, 42.65]
Suntharalingam 2017 9 153 0 166 12.0% 20.60 [1.21, 351.02]
Wu 2018 11 158 1 156 25.2% 10.86 [1.42, 83.12]
Subtotal (95% CI) 717 730 87.1% 10.59 [3.56, 31.49]
Total events 35 2
Heterogeneity: Chi² = 1.43, df = 4 (P = 0.84); I² = 0%
Test for overall effect: Z = 4.24 (P < 0.0001)
8.1.2 CT
Lorenzen 2009 2 32 0 30 12.9% 4.70 [0.23, 94.01]
Subtotal (95% CI) 32 30 12.9% 4.70 [0.23, 94.01]
Total events 2 0
Heterogeneity: Not applicable
Test for overall effect: Z = 1.01 (P = 0.31)
Total (95% CI) 749 760 100.0% 9.83 [3.54, 27.28]
Total events 37 2
Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0%
Test for overall effect: Z = 4.39 (P < 0.0001)
Test for subgroup differences: Chi² = 0.25, df = 1 (P = 0.62), I² = 0%
0.002 0.1 1 10 500
Favours [anti-EGFR] Favours [Control]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.2.1 mAb
Castro 2018 1 53 0 54 12.4% 3.06 [0.13, 73.37]
Crosby 2013 9 129 0 129 12.5% 19.00 [1.12, 323.06]
Lorenzen 2009 2 32 0 30 12.9% 4.70 [0.23, 94.01]
Suntharalingam 2017 9 153 0 166 12.0% 20.60 [1.21, 351.02]
Subtotal (95% CI) 367 379 49.8% 11.71 [2.80, 48.99]
Total events 21 0
Heterogeneity: Chi² = 1.31, df = 3 (P = 0.73); I² = 0%
Test for overall effect: Z = 3.37 (P = 0.0007)
8.2.2 TKI
Dutton 2014 5 224 1 225 25.0% 5.02 [0.59, 42.65]
Wu 2018 11 158 1 156 25.2% 10.86 [1.42, 83.12]
Subtotal (95% CI) 382 381 50.2% 7.95 [1.84, 34.37]
Total events 16 2
Heterogeneity: Chi² = 0.27, df = 1 (P = 0.61); I² = 0%
Test for overall effect: Z = 2.78 (P = 0.005)
Total (95% CI) 749 760 100.0% 9.83 [3.54, 27.28]
Total events 37 2
Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0%
Test for overall effect: Z = 4.39 (P < 0.0001)
Test for subgroup differences: Chi² = 0.14, df = 1 (P = 0.71), I² = 0%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**C**

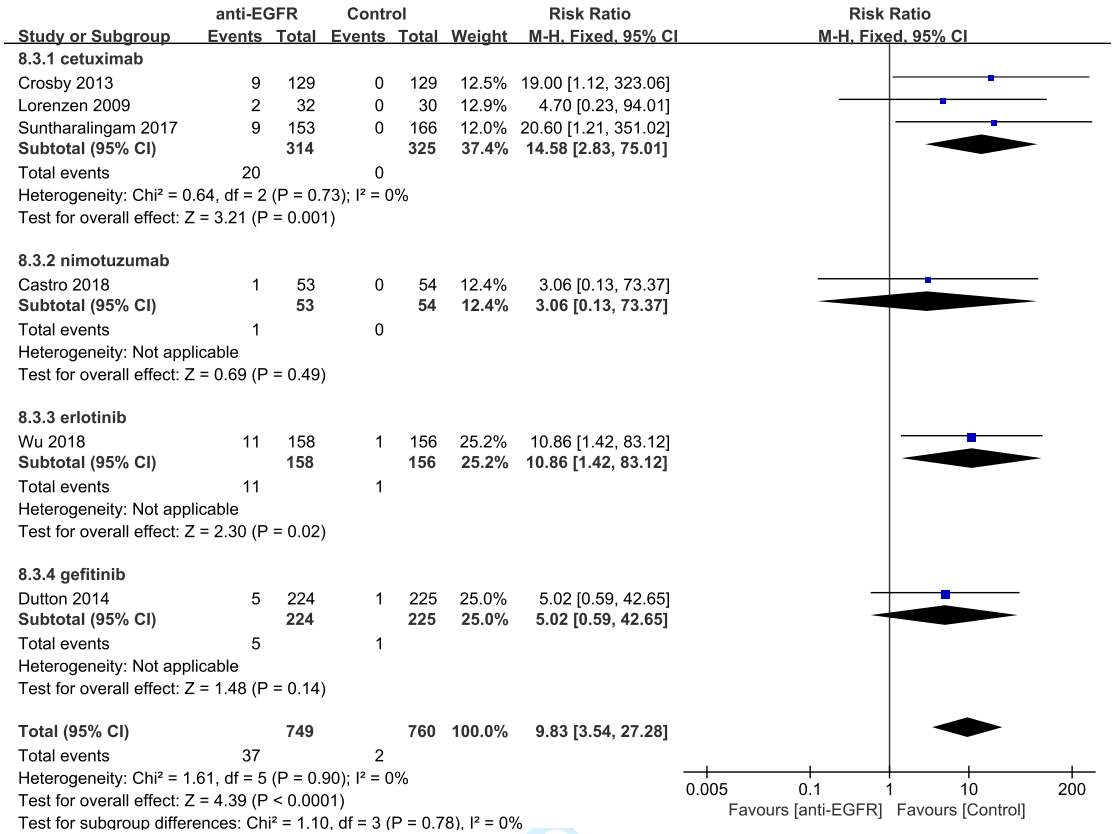



**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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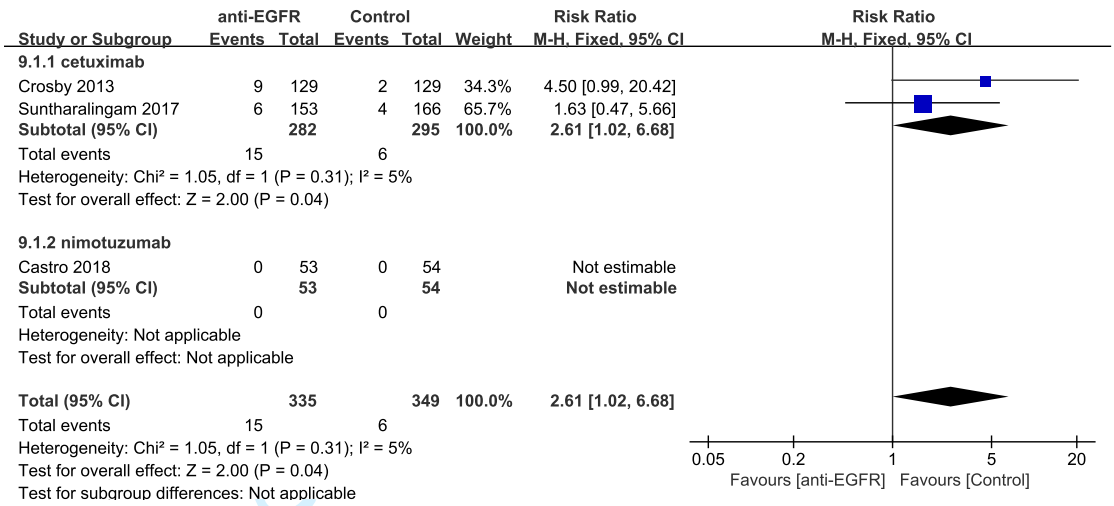



**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

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**Supplementary Data S1**

**Appendix 1. PubMed search strategy (N = 86)**

#1 “esophageal neoplasms”.[Mesh]

#2 “esophageal neoplasm*”.ti,ab.

#3 “esophagus neoplasm*”.ti,ab.

#4 “oesophageal neoplasm*”.ti,ab.

#5 “esophageal cancer*”.ti,ab.

#6 “esophagus cancer*”.ti,ab.

#7 “oesophageal cancer*”.ti,ab.

#8 “cancer of esophagus”.ti,ab.

#9 “cancer of the esophagus”.ti,ab.

#10 “esophageal tumor*”.ti,ab.

#11 “esophagus tumor*”.ti,ab.

#12 “oesophageal tumor*”.ti,ab.

#13 “esophageal carcinoma*”.ti,ab.

#14 “esophagus carcinoma*”.ti,ab.

#15 “oesophageal carcinoma*”.ti,ab.

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 “EGFR inhibitor*”.ti,ab.

#18 “epidermal growth factor receptor inhibitor*”.ti,ab.

#19 “anti-human epidermal growth factor receptor”.ti,ab.

#20 #17 OR #18 OR #19

#21 “cetuximab”.[Mesh]

#22 “erbitux”.ti,ab.

#23 “IMC C225”.ti,ab.

#24 “IMC-C225”.ti,ab.

#25 “MAb C225”.ti,ab.

#26 “C225”.ti,ab.

#27 “C-225”.ti,ab.



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#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 “panitumumab”.[Mesh]

#30 “ABX-EGF monoclonal antibody”.ti,ab.

#31 “ABX-EGF MAb”.ti,ab.

#32 “vectibix”.ti,ab.

#33 #29 OR #30 OR #31 OR #32

#34 “nimotuzumab”.ti,ab.

#35 “theraloc”.ti,ab.

#36 #34 OR #35

#37 “erlotinib hydrochloride”.[Mesh]

#38 “erlotinib HCl”.ti,ab.

#39 “erlotinib”.ti,ab.

#40 “tarceva”.ti,ab.

#41 “11C-erlotinib”.ti,ab.

#42 “11C erlotinib”.ti,ab.

#43 “OSI-774”.ti,ab.

#44 “OSI 774”.ti,ab.

#45 “OSI774”.ti,ab.

#46 “CP 358774”.ti,ab.

#47 “CP-358774”.ti,ab.

#48 “CP358774”.ti,ab.

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 “gefitinib”.[Mesh]

#51 “iressa”.ti,ab.

#52 “ZD1839”.ti,ab.

#53 “ZD-1839”.ti,ab.

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 “randomized controlled trials as topic”.[Mesh]

#57 “randomized controlled trial[publication type]”



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#58 “random*”.ti,ab.

#59 “clinical trials as topic”.[Mesh]

#60 “clinical trial[publication type]”

#61 “controlled clinical trials as topic”.[Mesh]

#62 “controlled clinical trial[publication type]”

#63 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62

#64 #16 AND #55 AND #63

**Appendix 2. EMBASE search strategy (N = 831)**

#1 'esophagus tumor'/exp

#2 'esophageal neoplasm*':ab,ti

#3 'esophagus neoplasm*':ab,ti

#4 'oesophageal neoplasm*':ab,ti

#5 'esophageal cancer*':ab,ti

#6 'esophagus cancer*':ab,ti

#7 'oesophageal cancer*':ab,ti

#8 'cancer of esophagus':ab,ti

#9 'cancer of the esophagus':ab,ti

#10 'esophageal tumor*':ab,ti

#11 'esophagus tumor*':ab,ti

#12 'oesophageal tumor*':ab,ti

#13 'esophageal carcinoma*':ab,ti

#14 'esophagus carcinoma*':ab,ti

#15 'oesophageal carcinoma*':ab,ti

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 'egfr inhibitor*':ab,ti

#18 'epidermal growth factor receptor inhibitor*':ab,ti

#19 'anti-human epidermal growth factor receptor':ab,ti

#20 #17 OR #18 OR #19



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#21 'cetuximab'/exp

#22 'erbitux':ab,ti

#23 'imc c225':ab,ti

#24 'imc-c225':ab,ti

#25 'mab c225':ab,ti

#26 'c225':ab,ti

#27 'c-225':ab,ti

#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 'panitumumab'/exp

#30 'abx-egf monoclonal antibody':ab,ti

#31 'abx-egf mab':ab,ti

#32 'vectibix':ab,ti

#33 #29 OR #30 OR #31 OR #32

#34 'nimotuzumab'/exp

#35 'theraloc':ab,ti

#36 #34 OR #35

#37 'erlotinib'/exp

#38 'erlotinib hcl':ab,ti

#39 'erlotinib':ab,ti

#40 'tarceva':ab,ti

#41 '11c-erlotinib':ab,ti

#42 '11c erlotinib':ab,ti

#43 'osi-774':ab,ti

#44 'osi 774':ab,ti

#45 'osi774':ab,ti

#46 'cp 358774':ab,ti

#47 'cp-358774':ab,ti

#48 'cp358774':ab,ti

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 'gefitinib'/exp



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#51 'iressa':ab,ti

#52 'zd1839':ab,ti

#53 'zd-1839':ab,ti

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 'randomized controlled trial'/exp

#57 'randomized controlled trial (topic)'/exp

#58 'random*':ab,ti

#59 'clinical trial (topic)'/exp

#60 'clinical trial'/exp

#61 'clinical trial*':ab,ti

#62 'controlled clinical trial'/exp

#63 'controlled clinical trial (topic)'/exp

#64 'controlled clinical trial*':ab,ti

#65 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62 OR #63 OR #64

#66 #16 AND #55 AND #65

**Appendix 3. CENTRAL search strategy (N = 94)**

#1 MeSH descriptor: [esophageal neoplasms] explode all trees

#2 (esophageal neoplasm*):ti,ab,kw OR (esophagus neoplasm*):ti,ab,kw OR (oesophageal neoplasm*):ti,ab,kw OR (esophageal cancer*):ti,ab,kw OR (esophagus cancer*):ti,ab,kw. (Word variations have been searched)

#3 (oesophageal cancer*):ti,ab,kw OR (cancer of esophagus):ti,ab,kw OR (cancer of the esophagus):ti,ab,kw OR (esophageal tumor*):ti,ab,kw OR (esophagus tumor*):ti,ab,kw. (Word variations have been searched)

#4 (oesophageal tumor*):ti,ab,kw OR (esophageal carcinoma*):ti,ab,kw OR (esophagus carcinoma*):ti,ab,kw OR (oesophageal carcinoma*):ti,ab,kw. (Word variations have been searched)

#5 #1 OR #2 OR #3 OR #4

#6 (EGFR inhibitor*):ti,ab,kw OR (epidermal growth factor receptor inhibitor*):ti,ab,kw OR (anti-human epidermal growth factor receptor):ti,ab,kw. (Word variations have been searched)



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#7 MeSH descriptor: [cetuximab] explode all trees

#8 (erbitux):ti,ab,kw OR (IMC C225):ti,ab,kw OR (IMC-C225):ti,ab,kw. (Word variations have been searched)

#9 (MAb C225):ti,ab,kw OR (C225):ti,ab,kw OR (C-225):ti,ab,kw. (Word variations have been searched)

#10 #7 OR #8 OR #9

#11 MeSH descriptor: [panitumumab] explode all trees

#12 (ABX-EGF monoclonal antibody):ti,ab,kw OR (ABX-EGF MAb):ti,ab,kw OR (vectibix):ti,ab,kw. (Word variations have been searched)

#13 #11 OR #12

#14 (nimotuzumab):ti,ab,kw OR (theraloc):ti,ab,kw. (Word variations have been searched)

#15 MeSH descriptor: [erlotinib hydrochloride] explode all trees

#16 (erlotinib HCl):ti,ab,kw OR (erlotinib):ti,ab,kw OR (tarceva):ti,ab,kw OR (11C-erlotinib):ti,ab,kw OR (11C erlotinib):ti,ab,kw. (Word variations have been searched)

#17 (OSI-774):ti,ab,kw OR (OSI 774):ti,ab,kw OR (OSI774):ti,ab,kw. (Word variations have been searched)

#18 (CP 358774):ti,ab,kw OR (CP-358774):ti,ab,kw OR (CP358774):ti,ab,kw. (Word variations have been searched)

#19 #15 OR #16 #17 OR #18

#20 MeSH descriptor: [gefitinib] explode all trees

#21 (iressa):ti,ab,kw OR (ZD1839):ti,ab,kw OR (ZD-1839):ti,ab,kw. (Word variations have been searched)

#22 #20 OR #21

#23 #6 OR #10 OR #13 OR #14 OR #19 OR #22

#24 MeSH descriptor: [randomized controlled trials as topic] explode all trees

#25 MeSH descriptor: [randomized controlled trial] explode all trees

#26 (random*):ti,ab,kw. (Word variations have been searched)

#27 MeSH descriptor: [clinical trials as topic] explode all trees

#28 MeSH descriptor: [clinical trial] explode all trees

#29 (clinical trial*):ti,ab,kw. (Word variations have been searched)



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#30 MeSH descriptor: [controlled clinical trial] explode all trees

#31 MeSH descriptor: [controlled clinical trials as topic] explode all trees

#32 (controlled clinical trial*):ti,ab,kw. (Word variations have been searched)

#33 #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32

#34 #5 AND #23 AND #33

**Appendix 4. Web of Science search strategy (N = 191)**

#1 TS=(esophageal neoplasms OR esophageal neoplasm* OR esophagus neoplasm* OR oesophageal neoplasm* OR esophageal cancer* OR esophagus cancer* OR oesophageal cancer* OR cancer of esophagus OR cancer of the esophagus OR esophageal tumor* OR esophagus tumor* OR oesophageal tumor* OR esophageal carcinoma* OR esophagus carcinoma* OR oesophageal carcinoma*)

#2 TS=(EGFR inhibitor* OR epidermal growth factor receptor inhibitor* OR anti-human epidermal growth factor receptor)

#3 TS=(cetuximab OR erbitux OR IMC C225 OR IMC-C225 OR MAb C225 OR C225 OR C-225) #4 TS=(panitumumab OR ABX-EGF monoclonal antibody OR ABX-EGF MAb OR vectibix)

#5 TS=(nimotuzumab OR theraloc)

#6 TS=(Erlotinib Hydrochloride OR Erlotinib HCl OR Erlotinib OR Tarceva OR 11C-erlotinib OR 11C erlotinib OR OSI-774 OR OSI 774 OR OSI774 OR CP 358774 OR CP-358774 OR CP358774)

#7 TS=(Gefitinib OR Iressa OR ZD1839 OR ZD-1839)

#8 #7 OR #6 OR #5 OR #4 OR #3 OR #2

#9 TS=(randomized controlled trials as topic OR randomized controlled trial OR clinical trials as topic OR clinical trial OR controlled clinical trial OR controlled clinical trials as topic OR random*)

#10 #9 AND #8 AND #1

**Appendix 5. ClinicalTrials.gov search strategy (N = 24)**

#1 Condition or disease: esophageal neoplasms

#2 Intervention/treatment: cetuximab OR panitumumab OR nimotuzumab OR erlotinib hydrochloride OR gefitinib

#3 Study Results: studies with results

#4 #1 AND #2 AND #3



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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 4-5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 1,6 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 6 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5-6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6-7 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 7-8 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 7-8 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7-8 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 7-8 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 8-9 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 9 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 10-11 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 11-14 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 11-14 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 11-14 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 11-14 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 14-18 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 18-19 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 19 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 20 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment for oesophageal cancer: a systematic review and meta-analysis



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| | |



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# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment for oesophageal cancer: a systematic review and meta-analysis

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LJZ, YLS, and NJ contributed equally to this work.

**Trial registration:** The study protocol has been registered in the International Prospective Register of Systematic Reviews (PROSPERO, CRD42020169230, 28/04/2020).

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## Abstract

**Objectives:** Despite remarkable advances in the treatment of oesophageal cancer (OC), the role of anti-epidermal growth factor receptor (anti-EGFR) agents in treating OC remains controversial. A systematic review and meta-analysis were performed to elucidate the efficacy and safety of anti-EGFR agents.

**Design:** A meta-analysis of randomised controlled trials (RCTs).

**Setting:** RCTs from any country and healthcare setting.

**Participants:** Patients with OC.

**Interventions:** We searched PubMed, EMBASE, Web of Science, ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine, China National Knowledge Infrastructure, and Wanfang Data Knowledge Service Platform for RCTs of anti-EGFR treatment versus conventional treatment in patients with OC, from inception to December 2019.

**Primary and secondary outcome measures:** Overall survival (OS) and progression-free survival (PFS) were primary outcome measures, while objective response rate (ORR), disease control rate (DCR), and toxicities were secondary outcome measures.

**Results:** We followed the Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) guidelines. In total, 25 RCTs with 3406 OC patients were included. Overall, anti-EGFR treatment significantly improved the OS (HR 0.81, 95% CI 0.74–0.89, $P < 0.00001$), ORR (RR 1.33, 95% CI 1.16–1.52, $P < 0.0001$), and DCR (RR 1.22, 95% CI 1.11–1.34, $P < 0.0001$), but not PFS (HR 0.91, 95% CI 0.76–1.08, $P = 0.26$). Anti-EGFR treatment was significantly associated with higher incidences of myelosuppression (RR 0.42, 95% CI 0.21–




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0.86, $P = 0.02$), diarrhoea (RR 2.26, 95% CI 1.42–3.60, $P = 0.0005$), acne-like rash (RR 9.83, 95% CI 3.54–27.28, $P < 0.0001$), and hypomagnesaemia (RR 2.61, 95% CI 1.02–6.68, $P = 0.04$).

**Conclusions:** Overall, anti-EGFR treatment has positive effects on the OS, ORR, and DCR in OC. However, considering the high incidence of adverse effects such as myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia following anti-EGFR treatment, careful monitoring is recommended.

**Strengths and limitations of this study**

- This systematic review and meta-analysis utilised a comprehensive study design to evaluate the efficacy and safety of anti-EGFR agents combined with conventional treatments versus conventional treatments alone for oesophageal cancer (OC).
- The combinations of Medical Subject Headings (MeSH)/Emtree terms and free-text terms were used to do literature search.
- Eight databases, even a manual search and unpublished studies were retrieved to identify the eligible studies.
- The quality of each eligible study was assessed by the Cochrane Handbook for Systematic Reviews ana Meta-analysis.
- This study has been registered a priori and reported in line with Preferred Reporting Items for Systematic Reviews and Meta-Analysis.



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## Introduction

Oesophageal cancer (OC) is the seventh most common cancer type and the sixth leading cause of cancer mortality, accounting for approximately one in every twenty cancer-related deaths worldwide in 2018.[1] However, its incidence and mortality rates vary considerably, as demonstrated by a two- to three-fold difference in these statistics between regions all over the world.[2] OC is common in Eastern and Southern Africa, and the highest incidence rates have been reported in East Asia, with rates in Mongolia and China being among the top five worldwide.[1 2] The highest-risk area, called "oesophageal cancer belt", where squamous cell cancers dominated all cases by 90%, and adenocarcinoma is largely a disease of white. [1]

OC is diagnosed in the later stages in 80% of the patients. Typically, few patients benefit from single-modality therapies, such as surgery (15%-20% rates of 5-year overall survival), chemotherapy (CT, 15%-45% rates of 5-year overall survival), and radiotherapy (RT, 0%-15% rates of 5-year overall survival).[3] Therefore, chemoradiotherapy (CRT, 26% rates of 5-year overall survival) has become a standard of care for OC. However, controlling locoregional recurrence or persistent disease is still a major challenge.[4] Attempts such as the use of higher radiation doses, combination of various CT regimens and RT or other combination therapies (e.g. surgery, neoadjuvant CT) have failed to achieve any improvement.[5-8] Therefore, the addition of novel therapies to conventional treatments (referred to any single-modality or multimodality therapies were mentioned above) is urgently needed.

The epidermal growth factor receptor (EGFR) signalling pathway has been implicated in the proliferation, invasion, angiogenesis, and survival of cancer cells.[9] Moreover, overexpression



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of EGFR in OC has been associated with poor prognosis.[10] Previous studies have shown the synergistic effects of EGFR inhibitors and conventional treatments.[11 12] Therefore, there exists a strong rationale for testing drugs targeting EGFR as an effective treatment strategy against OC. Tyrosine kinase inhibitors (TKIs) and monoclonal antibodies (mAbs) are two promising anti-EGFR drugs. mAbs bind to the extracellular domain of EGFR and compete with endogenous ligands to block the ligand-induced EGFR tyrosine kinase activation by blocking the ligand-binding region.[13] TKIs compete reversibly with adenosine 5' triphosphate to bind to the intracellular catalytic domain of EGFR tyrosine kinase and inhibit the EGFR autophosphorylation and downstream signaling.[13] Various anti-EGFR agents, such as cetuximab,[14 15] nimotuzumab,[16 17] panitumumab,[18] erlotinib,[19 20] and gefitinib,[21 22] have been administered for treating OC. However, studies examining the role of anti-EGFR agents in combination with conventional treatments have produced controversial results, and the toxicities associated with these agents are variable and uncertain.[14-17 19-22] Moreover, there is no clear evidence whether the combination of anti-EGFR agents and conventional treatments is superior to conventional treatments alone for OC. To address this research gap, we conducted a systematic review and meta-analysis of randomised controlled trials (RCTs) to evaluate the efficacy and safety of anti-EGFR agents combined with conventional treatments versus conventional treatments alone in patients with OC.

## Methods

### Literature Search Strategy

A literature search of the following databases: PubMed, EMBASE, Web of Science,



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ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine (CBM), China National Knowledge Infrastructure (CNKI), and Wanfang Data Knowledge Service Platform was performed by two authors to identify studies from inception to December 2019. Furthermore, a manual search using the reference lists of all the retrieved articles and relevant journals was conducted to identify additional eligible studies. Unpublished studies were retrieved from the System for Information on Grey Literature database to reduce publication bias. The combinations of Medical Subject Headings (MeSH)/Emtree terms and free-text terms were used to search the above databases. Location of OC primary site, including typical endoscopic measurements of each region measured from the incisors (range from 15cm to 42cm), which is defined as beginning from the upper oesophageal sphincter to the proximal 2 cm of the cardia.[23] The main search terms used were as follows: oesophageal neoplasms, epidermal growth factor receptor inhibitor, cetuximab, panitumumab, nimotuzumab, erlotinib, gefitinib, randomised controlled trial, and clinical trial. The full search strategies utilised are presented in **Supplementary Data S1**.

Two authors independently screened the studies based on the selection criteria. An initial screening of each title and abstract was performed, and eligible studies were retrieved as full text. Any disagreements between the two authors during the selection process were settled by discussion and consensus with a third author.

**Inclusion and Exclusion Criteria**

Eligible studies were included if they met the following criteria: (1) patients with histologically confirmed OC; (2) anti-EGFR treatment versus conventional treatment (referred to any single-



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modality or multimodality treatments for OC, such as surgery, CT, RT, or any combinations of them); (3) RCT; (4) hazard ratios (HRs) with 95% confidence intervals (CIs) for overall survival (OS) and/or progression-free survival (PFS) were reported or sufficient data were provided to calculate these values.

Studies were excluded if they were repeated studies, reviews, comments, letters, case reports, meeting abstracts, study protocols, or animal studies. Duplicate and incomplete data were also excluded.

**Data Extraction and Quality Assessment**

Data extraction and quality assessment were carried out by two independent authors, and discrepancies were resolved by discussion and consensus with the third author.

The information extracted included the first author's name, publication year, country, study design, enrolment period, sample size, mean/median age (range, years), treatment protocol, and major clinical outcomes (OS, PFS, objective response rate (ORR), disease control rate (DCR), and grade 3-5 toxicities) of each eligible study.

The quality of each eligible study was assessed by the Cochrane Handbook for Systematic Reviews of Interventions Version 6.0 (www.training.cochrane.org/handbook). Random sequence generation, allocation concealment, blinding, incomplete outcome data, and selective outcome reporting were examined to categorise each eligible study as low risk of bias (A), moderate risk of bias (B), or high risk of bias (C).

**Statistical Analysis**

The primary outcomes were OS and PFS, expressed as HR with 95% CI that could be



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calculated from the number of observed deaths minus the number of estimated deaths and its variance.[24] OS was defined as the time from diagnosis to death or last follow-up and PFS was defined as the time from diagnosis to the first documented recurrence. Both the data of the OS and PFS were got from Kaplan–Meier curves, which were read using Engauge Digitizer version 10.8 (free software downloaded from http://sourceforge.net/projects/digitizer/). Secondary outcomes, including ORR, DCR, and haematological and non-haematological toxicities, were expressed as relative risks (RRs) ratio with 95% CIs using the Mantel-Haenszel method. ORR was defined as the rate of partial response (PR) and complete response (CR), and DCR was defined as the rate of PR, CR, and stabilisation.

Statistical heterogeneity across studies was evaluated by the chi-squared test, with a significant level at $P < 0.1$, and quantified with the $I^2$ statistic. The fixed-effects model was used if the heterogeneity test showed no statistical significance ($I^2 < 50\%$; $P > 0.1$); otherwise, the random-effects model was adopted. Publication bias was assessed by inspecting the funnel plots for asymmetries and outliers. Sensitivity analysis was conducted to explore the robustness of the results by excluding each study to recalculate the pooled HR and RR estimates. Five studies were excluded because of small sample sizes or treatments with surgery and/or induction chemotherapy.[14 20 25-27] Subgroup analysis was performed according to the different treatment regimens.

All analyses were conducted in Review Manager version 5.3 (Cochrane Collaboration, Oxford, UK). A two-sided $P$-value < 0.05 was considered significant for all analyses except heterogeneity tests. This study was reported according to the Preferred Reporting Items for



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Systematic Reviews and Meta-analyses (PRISMA) guidelines.[28]

**Patient and public involvement**

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

**Results**

**Study Selection**

A total of 1374 records were identified by searching the databases, and no additional records were found through other sources. After removing duplicates, 1298 records remained. Based on title screening, 952 records were excluded as they did not meet the eligibility criteria. Next, 346 abstract records were screened for further evaluation, of which 301 records were excluded as they were reviews (n = 114), non-RCTs (n = 108, including case reports, meeting abstracts, letters, editorials, and news), animal studies (n = 3), or studies on other cancers and drugs (n = 76) unrelated to our study. After screening 45 full-text articles, 20 records were excluded as they were non-RCTs (n = 5), ongoing studies (n = 2), studies on other cancers and drugs (n = 8), and studies without outcomes to be combined (n = 5). Finally, 25 RCTs were included in the quantitative synthesis for this meta-analysis. The flow diagram of the selection process is shown in **Figure 1**.

**Study Characteristics**

The eligible studies were published between 2009 and 2018, and a total of 3406 patients were included in this meta-analysis. The sample size of each study ranged from 16 to 225. In total,



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1699 patients were in the anti-EGFR arm and 1707 patients were in the non-anti-EGFR arm. The age of the included patients ranged from 26 to 87 years. Seventeen studies were conducted in China,[19-21 25-27 29-39] three in the United Kingdom of Great Britain and Northern Ireland (UK)[22 40 41], and one each in the United States of America (USA),[15] Germany,[11] Switzerland,[14] Cuba,[17] and Brazil.[16] Two studies from the UK were performed on the same group of study patients and had partly overlapping study periods.[40 41] However, different outcomes were analysed in each study. For the anti-EGFR arms, patients were given anti-EGFR agents, including cetuximab, nimotuzumab, erlotinib, or gefitinib in combination with surgery, CT, RT, or CRT. In case of the non-anti-EGFR arms, patients received only surgery, CT, RT, or CRT. The basic characteristics of the 25 selected studies are presented in **Table S1**.

**Quality Assessment**

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The quality assessment of each study is shown in **Table 1**. Briefly, among the 25 RCTs, one study was rated as A level,[22] 17 studies as B level,[15 20 21 25-27 29-39] and seven studies as C level.[1 14 16 17 19 40 41] Eleven of the included studies were assessed as being at low risk of random sequence generation,[14-16 22 29 31 37-41] and six studies used allocation concealment.[11 14 16 22 40 41] However, almost all the studies did not use blinding, except for one study which reported the blinding of participants and outcome assessment.[22] All the studies were assessed as having low risk of bias because they addressed incomplete outcome data and selective reporting. No other source of bias was identified from these studies.

**Table 1** Quality assessment of each study

| Study (year) | Random sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Other bias | Rank |
|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Unclear | Yes | No | No | Yes | Yes | Yes | C |
| Crosby et al. 2013[41] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014a[25] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Crosby et al. 2017[40] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Suntharalingam et al. 2017[15] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Lu et al. 2017[37] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ruhstaller et al. 2018[14] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Ramos-Suzarte et al. 2012[17] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Xu et al. 2013[36] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Shi et al. 2014[35] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Sun et al. 2014[26] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2014[27] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Luo et al. 2015[33] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016b[30] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |



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**Table 1** Continued

| Study (year) | Random sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Other bias | Rank |
|---|---|---|---|---|---|---|---|---|
| Fu et al. 2017[29] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Castro et al. 2018[16] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014b[20] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhang et al. 2014[34] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ye et al. 2015[32] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhou et al. 2015[38] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016a[39] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhao et al. 2016[31] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2018[19] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Zeng et al. 2014[21] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Dutton et al. 2014[22] | Yes | Yes | Yes | Yes | Yes | Yes | Yes | A |

## Efficacy Analysis

Fifteen studies[11 14-16 19 22 26 30 32 33 35 37-40] were included in the meta-analysis for analysing OS, which showed significant improvement in the anti-EGFR arm (HR 0.81, 95% CI 0.74–0.89, $P < 0.00001$; $I^2 = 40\%$, $P = 0.05$) (**Figure 2**). Subgroup analysis indicated that OS benefit was observed when anti-EGFR agents were administered in combination with CT, CRT, or RT for the treatment of OC. Both anti-EGFR mAbs and TKIs, especially erlotinib and nimotuzumab, significantly improved the OS (**Figure S1**).

Six studies[11 14 22 26 36 40] were combined for evaluating PFS, which demonstrated no significant difference between the anti-EGFR arm and the control arm (HR 0.91, 95% CI 0.76–1.08, $P = 0.26$; $I^2 = 53\%$, $P = 0.06$) (**Figure 3**). Although longer PFS was identified in patients treated with



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nimotuzumab, no significant difference was found in other treatments (**Figure S2**).

Analysis of the 16 studies,[11 17 20 21 26 27 29-31 33-39] which evaluated ORR, indicated that ORR significantly increased in the anti-EGFR arm (RR 1.33, 95% CI 1.16–1.52, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S3**). Higher ORR was detected in patients with OC who received anti-EGFR agents in combination with CT, CRT, surgery, or RT. Both the categories of anti-EGFR drugs, mAbs and TKIs, especially nimotuzumab and erlotinib, significantly increased ORR (**Figure S4**).

Fifteen studies,[11 17 20 21 26 29-38] which analysed DCR, showed that DCR was significantly improved in the anti-EGFR arm (RR 1.22, 95% CI 1.11–1.34, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S5**). Treatment with the combination of anti-EGFR agents and CT or RT resulted in higher DCR. Anti-EGFR mAbs or TKIs, particularly cetuximab or erlotinib, decreased the risk of DCR (**Figure S6**).

**Safety Analysis**

Six different types of haematological toxicities were analysed in this meta-analysis, including anaemia, leucopoenia, neutropenia, thrombocytopenia, lymphopenia, and myelosuppression. Results from the pooled analysis demonstrated that there was no significant association between anti-EGFR agents and the increased risk of haematological toxicities, except for myelosuppression (RR 0.42, 95% CI 0.21-0.86, $P = 0.02$) (**Figure S7**). Subgroup analysis revealed that the addition of anti-EGFR agents to CT was associated with an increased risk of myelosuppression compared with CT alone, while no significant difference was found with CRT. Additionally, all patients who received anti-EGFR mAbs (e.g., cetuximab and



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nimotuzumab), and only nimotuzumab displayed a significant difference in terms of the incidence of myelosuppression (RR 0.39, 95% CI 0.18–0.83, $P = 0.02$) (**Figure S8**).

Eight non-haematological toxicities, including cardiac toxicity, gastrointestinal toxicity, dermatologic toxicity, pulmonary toxicity, renal toxicity, constitutional symptoms, infection, and laboratory findings, were reported. The results indicated statistically significant differences for dermatologic toxicity (RR 9.83, 95% CI 3.54–27.28, $P < 0.0001$) and laboratory findings (RR 1.59, 95% CI 1.05–2.42, $P = 0.03$) in the anti-EGFR arm, while no significant differences were observed for other non-haematological toxicities (**Table S2**). Subgroup analysis confirmed that patients receiving anti-EGFR treatment had higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia than those in the control group (**Table S2**). Higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia were observed in patients receiving anti-EGFR agents in combination with CRT than in those receiving CRT alone (**Figure S9, S10, S11**). Increased risk of diarrhoea, acne-like rashes, and hypomagnesaemia were found among patients who received anti-EGFR mAbs (**Figure S9, S10, S11**), especially cetuximab (**Figure S9, S10, S11**). Additionally, anti-EGFR TKIs, particularly erlotinib, increased the incidence of acne-like rash (**Figure S10**). Gefitinib was also associated with an increased risk of diarrhoea (**Figure S9**).

**Sensitivity Analysis**

Sensitivity analysis was executed by excluding the studies by Ruhstaller et al. 2018[14], Chen et al. 2014a,[25] Chen et al. 2014b,[20] Sun et al. 2014,[26] and Wu et al. 2014.[27] Any significant changes in the pooled estimate in terms of OS, PFS, ORR, and DCR indicated that the results were



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statistically stable and reliable (**Table S3**). No other sensitivity analysis had any effect on the results.

**Publication Bias**

A funnel plot of the included studies, plotted to assess the possibility of publication bias, showed asymmetry, thereby indicating that publication bias existed (**Figure 4, Figure S12, S13**).

**Discussion**

OC, a malignant digestive system neoplasm with poor prognosis, is regarded as a threat to public health. Currently, anti-EGFR agents are commonly used as valuable therapeutic options for OC. However, various studies have published contradictory results regarding these agents. Thus, a meta-analysis of RCTs was conducted to evaluate the efficacy and safety of anti-EGFR agents in combination with conventional treatment when compared with conventional treatment alone. To the best of our knowledge, this is the first meta-analysis to evaluate the benefits and toxicities of anti-EGFR and non-anti-EGFR treatment strategies for OC. The findings of this meta-analysis indicated that the anti-EGFR therapies might be more effective in terms of OS, ORR, and DCR than non-anti-EGFR therapies. However, anti-EGFR agents might be associated with higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia.

Adding anti-EGFR agents to CT, CRT, or RT significantly improved the OS. However, considering the fact that CRT is considered as the standard treatment approach for OC, the



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combination of anti-EGFR agents with CT or RT was not considered owing to limited availability of studies for this meta-analysis. According to this meta-analysis, the addition of anti-EGFR agents to CRT may be an important strategy to improve clinical outcomes in patients with OC. Moreover, anti-EGFR mAbs and TKIs, especially nimotuzumab and erlotinib, significantly improved the OS. The clinical benefits of nimotuzumab may be explained by the direct inhibition of the EGFR signalling pathway, thereby exerting an antiproliferative effect on tumour cells.[42] Furthermore, nimotuzumab enhanced the sensitivity of tumour cells to RT and CT.[43] These results might provide the potential explanation for the positive results obtained in this study. The combination of erlotinib with conventional treatment for OC is still controversial because of the limited number of RCTs on this TKI. G1/S cell cycle arrest induced by TKIs has already been described in various cancers. Studies have reported that TKIs inhibit the growth of OC cells,[44-47] which might be the reason for the significant improvement of OS in our study. Cetuximab did not significantly contribute to increased OS, which was consistent with a previous report.[48] It has been proposed that the reduced efficacy of cetuximab in combination with CRT is due to proinflammatory and antitumour proliferative effects.[49] Moreover, cetuximab might limit the free-radical damage by platinum drugs, which could again explain the non-significant outcomes in this meta-analysis.[50] For gefitinib, no improvement in OS was observed, which was in accordance with the results of a previous meta-analysis.[51] Currently, there is no evidence that gefitinib could suppress EGFR, thus indicating its limited clinical efficacy.

Considering the outcome of the PFS reported in this meta-analysis, there was no significant



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improvement in the anti-EGFR arm, which was consistent with the results of a previous meta-analysis.[52] Longer PFS was identified with nimotuzumab combination therapy, which was also reported by Jing et al.[53] The biological characteristics of nimotuzumab differed from those of other anti-EGFR agents, which might be the reason for its different clinical effects.

With regards to ORR and DCR, significant differences were noted between the anti-EGFR arm and non-anti-EGFR arm. Both anti-EGFR mAbs and TKIs, especially nimotuzumab, erlotinib, and cetuximab, had significant differences in ORR and DCR for OC. However, previous studies have reported different ORRs and DCRs.[54]

In terms of ORR, our results were in line with those of some studies that indicated a significant advantage of combining anti-EGFR agents with CT, CRT, or RT.[55 56] Furthermore, an ORR of 47.8% in the nimotuzumab group compared to 15.4% in the control group was noted.[17] Additionally, anti-EGFR TKIs were superior to conventional therapy in ORR.[57] Our meta-analysis indicated that erlotinib has a significant benefit in terms of ORR. However, combined data of ORR have not been reported in OC. Whether erlotinib therapy is better than other treatment strategies is still unknown owing to insufficient and conflicting homogeneity studies.

DCR improved greatly when anti-EGFR agents were combined with CT or RT; however, no confirmed conclusions could support our result as few studies reported this outcome. Erlotinib and cetuximab demonstrated a notable beneficial effect on DCR for treating OC, which was in agreement with previous studies.[48 57] Most meta-analysis studies have reported only the results of mAb and TKI therapy. However, as no specific drugs were reported, it was



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impossible to verify the effectiveness of each drug.

This meta-analysis confirmed that patients with OC in the anti-EGFR arm had higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia than the non-anti-EGFR arm, which was consistent with previous studies.[58 59] An increased risk of myelosuppression was found in anti-EGFR with CT. Moreover, this meta-analysis showed that nimotuzumab increased the risk of myelosuppression. These results might be related to the finding that nimotuzumab could enhance the sensitivity of tumour cells to CT.[43] However, these findings remain controversial due to the lack of sufficient high-quality, multi-centre, and large-sample RCTs, and thus should be interpreted cautiously.

Non-haematological toxicities, such as diarrhoea, acne-like rashes, and hypomagnesaemia occurred more often in patients with OC receiving anti-EGFR agents with CRT than in those receiving CRT alone. Furthermore, the combination of cetuximab and gefitinib and conventional treatment resulted in a higher risk of diarrhoea than conventional treatment alone. The underlying mechanism of diarrhoea caused by either anti-EGFR mAb or TKI remains poorly understood and might be induced by multiple factors, including altered gut motility, colonic crypt damage, changes to intestinal microflora, and altered transport in the colon.[60] A plausible explanation could be the inhibition of the EGFR pathway in enterocytes.[61] Thus, a better understanding of its mechanism may help in the identification of new approaches to prophylaxis, reduce risk, and develop more effective management.

Acne-like rash is the most common side effect of anti-EGFR agents,[62] and our meta-analysis confirmed that anti-EGFR mAbs and TKIs, such as cetuximab and erlotinib, could induce acne-



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like rashes. The mechanism leading to skin toxicity is not well known, but it is undoubtedly the result of modifications of the signals associated with its activation, particularly the RAS/RAF/MEK/ERK pathway, which affects cell cycle regulation, including the proliferation and differentiation of epidermis cells.[63] Skin toxicities often lead to physical and psychological discomforts, which might result in treatment interruptions, dose modifications, or even non-adherence to anti-cancer therapies.[62] Therefore, it is imperative to understand skin toxicities and develop proper approaches to control or relieve their adverse effects on patients.

An increased incidence of hypomagnesaemia was found in cetuximab, which was in accordance with another report.[48] Hypomagnesaemia, a frequent side effect of cetuximab, is a serious electrolyte abnormality associated with an increased risk of seizures and cardiac events.[64] The exact mechanism of cetuximab-induced hypomagnesaemia has not been well defined. Some researchers discovered that EGFR was strongly expressed in the kidney, particularly in the ascending limb of the loop of Henle, where 70% of the filtered magnesium was reabsorbed, and cetuximab might interfere with the magnesium transport as it blocks EGFR.[64 65] However, hypomagnesaemia is often ignored because of its non-specific symptoms, which are easily attributed to underlying tumours or previous CT regimens.[66] Consequently, serum magnesium should be monitored when cetuximab-based therapy is performed.

Several limitations should be considered while interpreting the results of this meta-analysis. First, we lacked access to individual patient data, which might have caused publication, reporting, or selection bias. Second, because of the insufficient number of studies conducted, we included different anti-EGFR agents and conventional treatments in meta-analysis, which



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increased clinical heterogeneity among the included studies. Subsequently, subgroup and sensitivity analyses were performed to increase the reliability of our results. Third, given the geographical variation incidence rates of OC, most of the included studies were conducted in China with small sample size in single institutions. Therefore, physicians from other countries should consider carefully when apply anti-EGFR agents in their clinical practices. Fourth, the methods were unclear, especially in the domain of 'allocation concealment and blinding', which might affect the overall methodological quality of the included studies. High methodology quality of RCTs is needed for further research. Finally, considering all these limitations, the results of this meta-analysis should be interpreted cautiously, and further research is necessary for verifying the effect of anti-EGFR therapy in OC.

In conclusion, this meta-analysis indicated that adding anti-EGFR agents to multimodal therapy benefited patients with OC in terms of the OS, ORR, and DCR. Therefore, anti-EGFR agents might be recommended as a treatment option for OC to improve treatment outcomes. The results from this study may help in establishing the optimum therapeutic regimen for OC. However, considering the higher incidence of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia when using the combination of anti-EGFR agents and multimodal therapy, careful monitoring and effective management is recommended. Further high quality large-scale and multi-centre studies are needed to confirm the efficacy and safety of anti-EGFR agents in OC.

**Contributors** LJZ and RY designed the study; LJZ, YLS, and NJ performed the literature



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search, extracted the data, and evaluated the quality of included studies; LJZ, YQH, BD, WL, YZH, YC, and HB- L conducted the statistical analysis; LJZ, YLS, and NJ were involved in the interpretation of the results; LJZ, YLS, NJ, and RY were responsible for the writing and critical revision of the manuscript. All authors read and approved the final manuscript.

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## Figure legends

**Figure 1** Flow diagram of the selection process of relevant studies

**Figure 2** Forest plot of the overall survival (OS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure 3** Forest plot of the progression-free survival (PFS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure 4** Funnel plot of publication bias

**Figure S1** Subgroup analysis of the overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S2** Subgroup analysis of the progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S3** Forest plot of the objective response rate (ORR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S5** Forest plot of the disease control rate (DCR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment



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regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S7** Forest plot of grade 3–5 haematological toxicities in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

**Figure S12** Funnel plot of publication bias for ORR

**Figure S13** Funnel plot of publication bias for DCR

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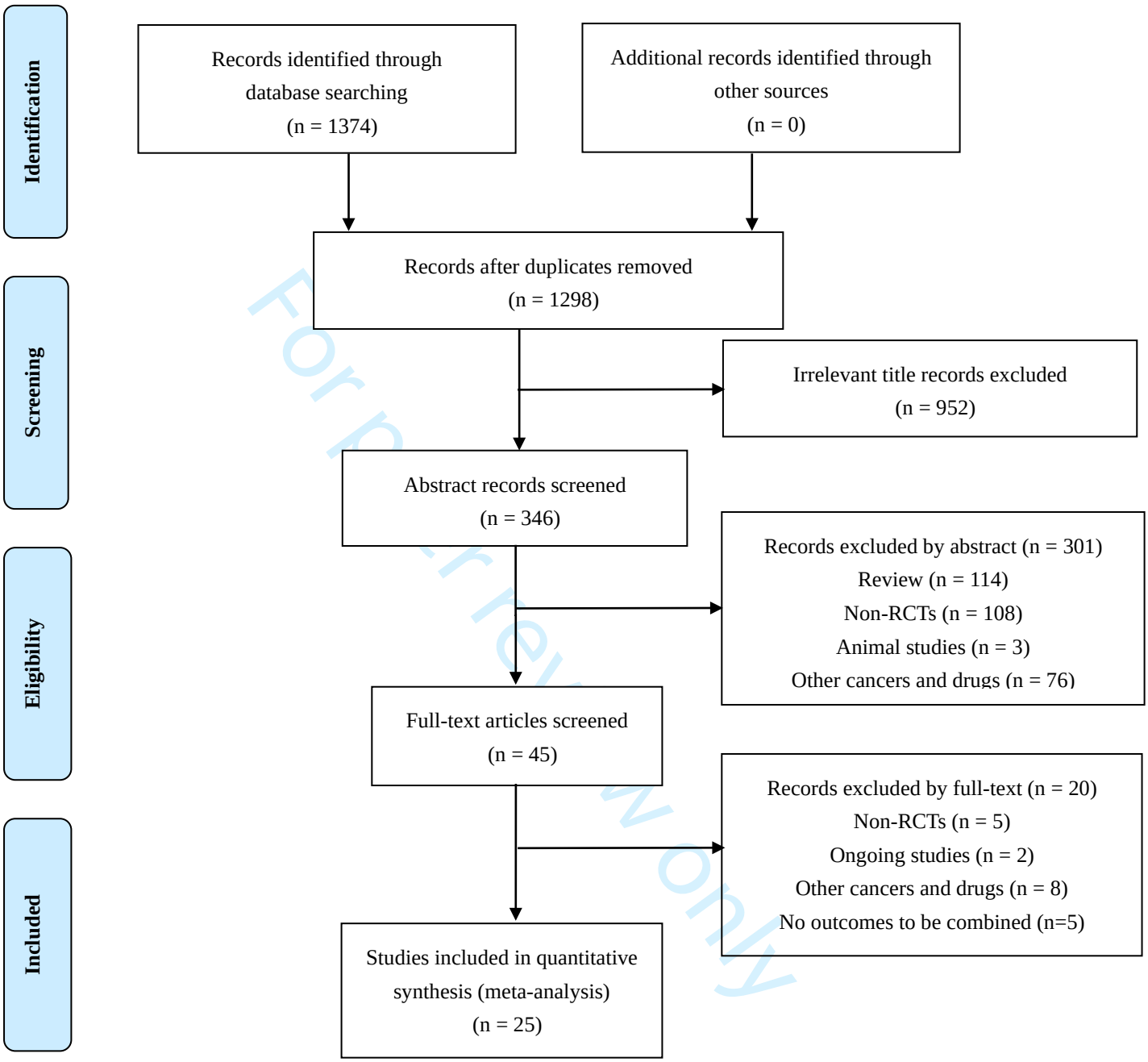



**Figure 1** Flow diagram of the selection process for relevant studies

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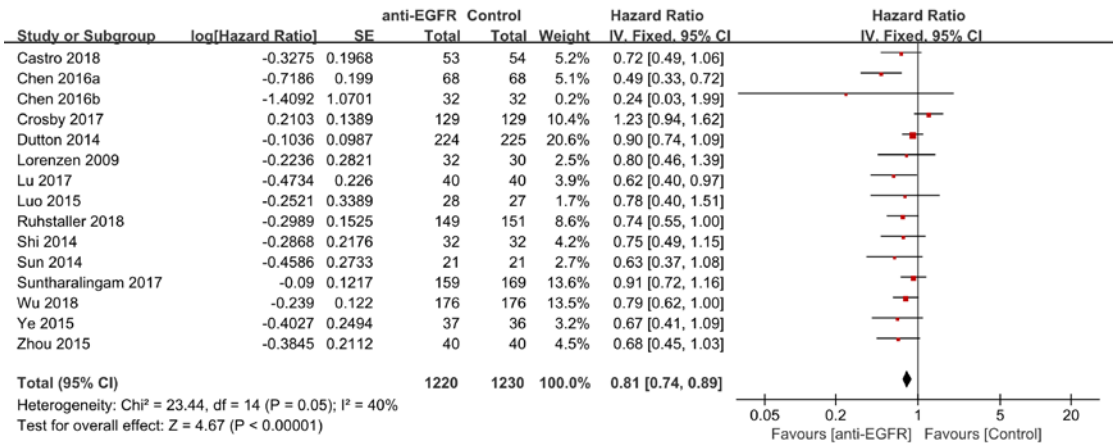



**Figure 2** Forest plot of the overall survival (OS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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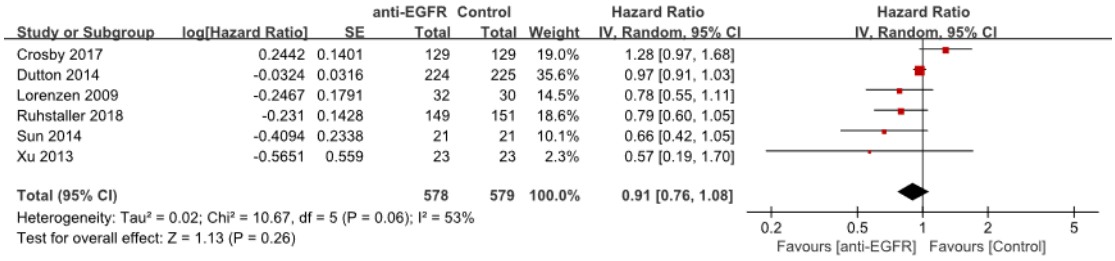



**Figure 3** Forest plot of the progression-free survival (PFS) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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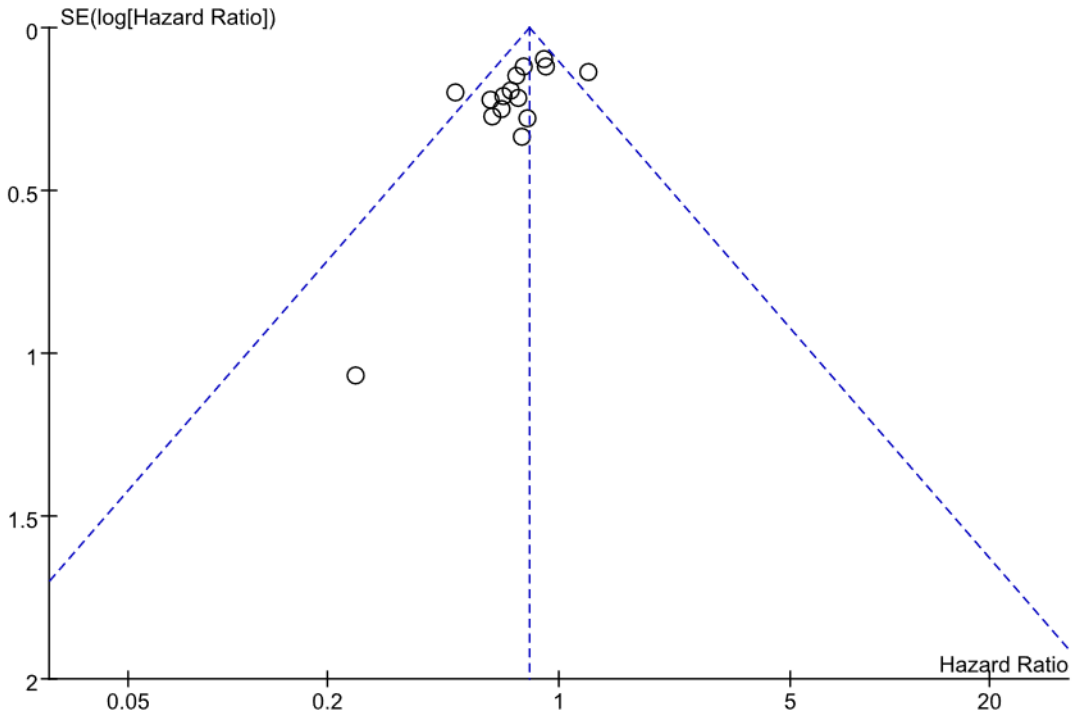



**Figure 4** Funnel plot of publication bias

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**Table S1** Basic characteristics of included studies in the meta-analysis

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Germany | RCT Multicenter | 2004.12-2006.12 | 32 | 30 | 61 (40-76) | 62 (40-74) | Cetuximab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | OS, PFS, ORR, DCR, Toxicities |
| Crosby et al. 2013[41] | UK | RCT Multicenter | 2008.02-2012.02 | 129 | 129 | 66.9 (44.9-84.1) | 66.6 (35.7-81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, Toxicities |
| Chen et al. 2014a[25] | China | RCT Single-institution | 2011.07-2012.01 | 20 | 20 | 57.3 (44-73) | | Surgery+Cetuximab+CT (Cisplatin+5-fluorouracil） | Surgery+CT (Cisplatin+5-fluorouracil) | OS, PFS, Toxicities |
| Crosby et al. 2017[40] | UK | RCT Multicenter | 2008.02-2012.02 | 129 | 129 | 66.9 (44.9-84.1) | 66.6 (35.7-81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, PFS, Toxicities |
| Suntharalingam et al. 2017[15] | USA | RCT Multicenter | 2008.06-2013.02 | 159 | 169 | 65 (39-87) | 63 (32-85) | Cetuximab+CRT (Cisplatin+Paclitaxel+RT) | CRT (Cisplatin+Paclitaxel+RT) | OS, Toxicities |
| Lu et al. 2017[37] | China | RCT Single-institution | 2013.01-2015.06 | 40 | 40 | 61.2 (45-80) | 61.3 (45-80) | Cetuximab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | OS, ORR, DCR, Toxicities |
| Ruhstaller et al. 2018[14] | Switzerland | RCT Multicenter | 2010.05-2016.10 | 149 | 151 | 61 (56-68) | 61 (53-68) | Cetuximab+IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+ RT) +Surgery | IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+RT) + Surgery | OS, PFS, Toxicities |
| Ramos-Suzarte et al. 2012[17] | Cuba | RCT Multicenter | 2005.12-2008.12 | 33 | 30 | 58.5 | 58.9 | Nimotuzumab+CRT (Cisplatin+5-fluorouracil+RT) | CRT (Cisplatin+5-fluorouracil+RT) | ORR, DCR, Toxicities |
| Xu et al. 2013[36] | China | RCT Single-institution | 2009.01-2012.12 | 23 | 23 | 63 (40-74) | 62 (41-75) | Nimotuzumab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | PFS, ORR, DCR, Toxicities |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Shi et al. 2014[35] | China | RCT Single-institution | 2009.10-2010.10 | 32 | 32 | 57 (42-68) | | Nimotuzumab+RT | RT | OS, ORR, DCR |
| Sun et al. 2014[26] | China | RCT Single-institution | 2008.01-2009.07 | 21 | 21 | 54 (36-78) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, PFS, ORR, DCR, Toxicities |
| Wu et al. 2014[27] | China | RCT Single-institution | 2009.10-2011.09 | 16 | 16 | 55 (38-68) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | ORR |
| Luo et al. 2015[33] | China | RCT Single-institution | NR | 28 | 27 | 62 (32-68) | 63 (31-69) | Nimotuzumab+CRT (Docetaxel+RT] | CRT (Docetaxel+RT] | OS, ORR, DCR, Toxicities |
| Chen et al. 2016b[30] | China | RCT Single-institution | 2012.01-2013.01 | 32 | 32 | 53.43 (39-68) | 54.52 (41-68) | Nimotuzumab+CRT (Nadaplatin+RT] | CRT (Nadaplatin+RT] | OS, ORR, DCR |
| Fu et al. 2017[29] | China | RCT Single-institution | 2013.12-2016.12 | 40 | 40 | 60.55 (31-76) | 61.38 (31-75) | Nimotuzumab+CRT (Docetaxel+Cisplatin++RT) | CRT (Docetaxel+Cisplatin++RT) | ORR, DCR |
| Castro et al. 2018[16] | Brazil | RCT Multicenter | 2009.02-2011.08 | 53 | 54 | 60.0 (40-75) | 58.5 (36-82) | Nimotuzumab+CRT (Cisplatin+Fluorouracil+RT) | CRT (Cisplatin+Fluorouracil+RT) | OS, Toxicities |
| Chen et al. 2014b[20] | China | RCT Single-institution | 2010.01-2012.01 | 60 | 60 | 58.7 (49-75) | | Surgery+Erlotinib+CT (Cisplatin+5-fluorouracil) | Surgery+CT (Cisplatin+5-fluorouracil) | CR, ORR, DCR |
| Zhang et al. 2014[34] | China | RCT Single-institution | 2009.10-2012.10 | 42 | 42 | 59.1 (49-74) | 58.4 (47-72) | Erlotinib+RT | RT | CR, ORR, DCR |
| Ye et al. 2015[32] | China | RCT Single-institution | 2011.02-2012.04 | 37 | 36 | 53.8 (NR) | 55.2 (NR) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, DCR |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Zhou et al. 2015[38] | China | RCT Single-institution | 2009.05-2011.11 | 40 | 40 | 55 (46-70) | 54 (45-68) | Erlotinib+RT | CT+RT (Docetaxel+Cisplatin)+RT | OS, ORR, DCR |
| Chen et al. 2016a[39] | China | RCT Single-institution | 2012.11-2014.11 | 68 | 68 | 47.3 (28-76) | 47.9 (26-78) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, ORR |
| Zhao et al. 2016[31] | China | RCT Single-institution | 2013.08-2014.04 | 38 | 38 | 64.5 (61-83) | 63.4 (59-81) | Erlotinib+RT | RT | CR, ORR, DCR |
| Wu et al. 2018[19] | China | RCT Multicenter | 2007.12-2015.06 | 176 | 176 | 61 (35-70) | 61.5 (40-70) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, Toxicities |
| Zeng et al. 2014[21] | China | RCT Single-institution | 2018.03-2010.08 | 26 | 26 | NR (40-80) | | Gefitinib+CT (Oxaliplatin+5-fluorouracil) | CT (Oxaliplatin+5-fluorouracil) | ORR, DCR |
| Dutton et al. 2014[22] | UK | RCT Multicenter | 2009.03-2011.11 | 224 | 225 | 64.7 (58.0-70.1) | 64.9 (58.2-70.7) | Gefitinib+CRT (NR) | Placebo+CRT (NR) | OS, PFS, Toxicities |

*RCT randomized controlled trial, NR Not reported, OS Overall survival, PFS Progression-free survival, ORR Objective response rate, DCR Disease control rate, RT Radiotherapy, CRT Chemoradiotherapy, IC Induction chemotherapy, CT Chemotherapy*

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**Table S2** Non-haematological toxicities of grade 3–5

| Toxicities | Subgroups | No. of studies | Sample Size | | RR [95% CI] | *P* | Heterogeneity | |
|---|---|---|---|---|---|---|---|---|
| | | | anti-EGFR (events/total) | Control (events/total) | | | $I^2$ | *P* |
| Cardiac toxicity** | ALL | 4 | 19/664 | 9/676 | 2.13 [0.97, 4.67] | 0.06 | 0% | 0.72 |
| Gastrointestinal toxicity | ALL | 28 | 359/2927 | 330/2990 | 1.11 [0.97, 1.28] | 0.13 | 16% | 0.23 |
| | Nausea and Vomiting | 6 | 74/558 | 81/565 | 0.94 [0.70, 1.26] | 0.67 | 43% | 0.12 |
| | Diarrhoea | 7 | 55/782 | 24/790 | 2.26 [1.42, 3.60] | 0.0005* | 30% | 0.21 |
| | Dehydration | 2 | 37/206 | 28/220 | 1.42 [0.91, 2.23] | 0.12 | 40% | 0.20 |
| | Dysphagia | 4 | 100/493 | 104/505 | 0.97 [0.76, 1.24] | 0.83 | 0% | 0.79 |
| | Oesophagitis | 5 | 64/520 | 63/531 | 1.03 [0.75, 1.42] | 0.84 | 0% | 0.70 |
| | Anorexia | 4 | 29/368 | 30/379 | 1.00 [0.61, 1.63] | 1.00 | 0% | 0.99 |
| Dermatologic toxicity | Acne-like rash | 6 | 37/749 | 2/760 | 9.83 [3.54, 27.28] | <0.0001* | 0% | 0.90 |
| Pulmonary toxicity | Dyspnoea | 3 | 10/335 | 6/349 | 1.72 [0.63, 4.68] | 0.29 | 26% | 0.25 |
| Renal toxicity | Renal insufficiency | 4 | 7/588 | 6/601 | 1.17 [0.41, 3.34] | 0.77 | 0% | 0.54 |
| Constitutional symptoms | ALL | 9 | 120/1084 | 104/1109 | 1.17 [0.92, 1.49] | 0.20 | 0% | 0.67 |
| | Fatigue | 6 | 109/749 | 93/760 | 1.19 [0.92, 1.53] | 0.19 | 0% | 0.51 |
| | Weight loss | 3 | 11/335 | 11/349 | 1.05 [0.47, 2.35] | 0.90 | 0% | 0.47 |
| Infection | ALL | 9 | 15/817 | 7/839 | 2.21 [0.91, 5.36] | 0.08 | 0% | 0.97 |
| | Fever | 4 | 2/271 | 1/280 | 2.17 [0.20, 23.69] | 0.53 | Not applicable | Not applicable |
| | Pneumonitis | 3 | 6/364 | 2/376 | 3.14 [0.64, 15.34] | 0.16 | 0% | 0.70 |
| | Stomatitis | 2 | 7/182 | 4/183 | 1.77 [0.53, 5.92] | 0.36 | 0% | 0.83 |
| Laboratory findings | ALL | 9 | 52/1005 | 34/1047 | 1.59 [1.05, 2.42] | 0.03* | 0% | 0.70 |
| | Hypomagnesaemia | 3 | 15/335 | 6/349 | 2.61 [1.02, 6.68] | 0.04* | 5% | 0.31 |
| | Hypokalaemia | 3 | 20/335 | 19/349 | 1.10 [0.60, 2.01] | 0.75 | 0% | 0.70 |
| | Hyponatraemia | 3 | 17/335 | 9/349 | 1.97 [0.91, 4.25] | 0.08 | 0% | 0.80 |

*$P < 0.05$, **including cardiac disorder/ischaemia/infarction



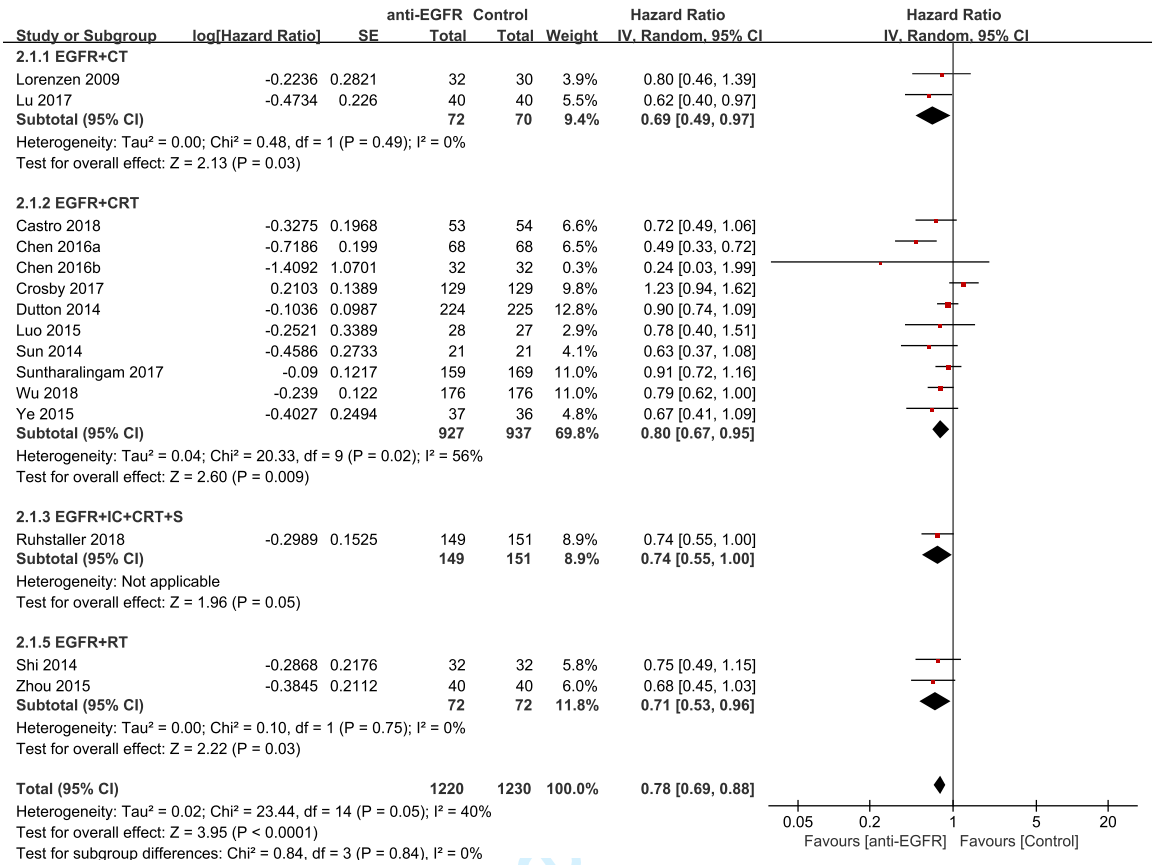
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**Table S3** Sensitivity analysis by excluding the Ruhstaller et al. 2018[14], Chen et al. 2014a[25], Chen et al. 2014b[20], Sun et al. 2014[26] and Wu et al. 2014[27]

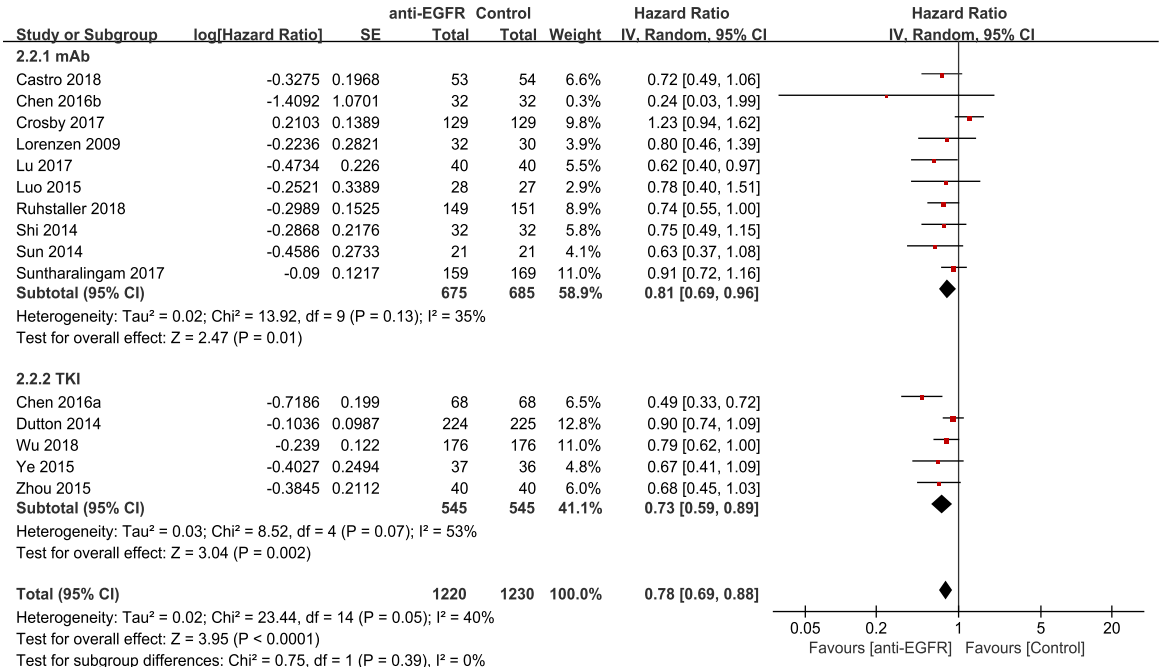
| Outcome | No. of studies | Ruhstaller et al. 2018[14] | | Chen et al. 2014a[25] | | Chen et al. 2014b[20] | | Sun et al. 2014[26] | | Wu et al. 2014[27] | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* |
| OS | 15 | 0.82 [0.75, 0.90] | <0.0001 | NR | NR | NR | NR | 0.82 [0.75, 0.89] | <0.00001 | NR | NR |
| PFS | 6 | 0.93 [0.76, 1.14] | 0.48 | NR | NR | NR | NR | 0.94 [0.79, 1.12] | 0.48 | NR | NR |
| ORR | 16 | NR | NR | NR | NR | 1.33 [1.14, 1.54] | 0.0002 | 1.32 [1.14, 1.52] | 0.0002 | 1.34 [1.19, 1.51] | <0.00001 |
| DCR | 15 | NR | NR | NR | NR | 1.24 [1.11, 1.38] | <0.0001 | 1.23 [1.13, 1.34] | <0.00001 | NR | NR |

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**A**


anti-EGFR Control Hazard Ratio Hazard Ratio
Study or Subgroup log[Hazard Ratio] SE Total Total Weight IV, Random, 95% CI IV, Random, 95% CI
2.1.1 EGFR+CT
Lorenzen 2009 -0.2236 0.2821 32 30 3.9% 0.80 [0.46, 1.39]
Lu 2017 -0.4734 0.226 40 40 5.5% 0.62 [0.40, 0.97]
Subtotal (95% CI) 72 70 9.4% 0.69 [0.49, 0.97]
Heterogeneity: Tau² = 0.00; Chi² = 0.48, df = 1 (P = 0.49); I² = 0%
Test for overall effect: Z = 2.13 (P = 0.03)
2.1.2 EGFR+CRT
Castro 2018 -0.3275 0.1968 53 54 6.6% 0.72 [0.49, 1.06]
Chen 2016a -0.7186 0.199 68 68 6.5% 0.49 [0.33, 0.72]
Chen 2016b -1.4092 1.0701 32 32 0.3% 0.24 [0.03, 1.99]
Crosby 2017 0.2103 0.1389 129 129 9.8% 1.23 [0.94, 1.62]
Dutton 2014 -0.1036 0.0987 224 225 12.8% 0.90 [0.74, 1.09]
Luo 2015 -0.2521 0.3389 28 27 2.9% 0.78 [0.40, 1.51]
Sun 2014 -0.4586 0.2733 21 21 4.1% 0.63 [0.37, 1.08]
Suntharalingam 2017 -0.09 0.1217 159 169 11.0% 0.91 [0.72, 1.16]
Wu 2018 -0.239 0.122 176 176 11.0% 0.79 [0.62, 1.00]
Ye 2015 -0.4027 0.2494 37 36 4.8% 0.67 [0.41, 1.09]
Subtotal (95% CI) 927 937 69.8% 0.80 [0.67, 0.95]
Heterogeneity: Tau² = 0.04; Chi² = 20.33, df = 9 (P = 0.02); I² = 56%
Test for overall effect: Z = 2.60 (P = 0.009)
2.1.3 EGFR+IC+CRT+S
Ruhstaller 2018 -0.2989 0.1525 149 151 8.9% 0.74 [0.55, 1.00]
Subtotal (95% CI) 149 151 8.9% 0.74 [0.55, 1.00]
Heterogeneity: Not applicable
Test for overall effect: Z = 1.96 (P = 0.05)
2.1.5 EGFR+RT
Shi 2014 -0.2868 0.2176 32 32 5.8% 0.75 [0.49, 1.15]
Zhou 2015 -0.3845 0.2112 40 40 6.0% 0.68 [0.45, 1.03]
Subtotal (95% CI) 72 72 11.8% 0.71 [0.53, 0.96]
Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 1 (P = 0.75); I² = 0%
Test for overall effect: Z = 2.22 (P = 0.03)
Total (95% CI) 1220 1230 100.0% 0.78 [0.69, 0.88]
Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40%
Test for overall effect: Z = 3.95 (P < 0.0001)
Test for subgroup differences: Chi² = 0.84, df = 3 (P = 0.84), I² = 0%
0.05 0.2 1 5 20
Favours [anti-EGFR] Favours [Control]

**B**


anti-EGFR Control Hazard Ratio Hazard Ratio
Study or Subgroup log[Hazard Ratio] SE Total Total Weight IV, Random, 95% CI IV, Random, 95% CI
2.2.1 mAb
Castro 2018 -0.3275 0.1968 53 54 6.6% 0.72 [0.49, 1.06]
Chen 2016b -1.4092 1.0701 32 32 0.3% 0.24 [0.03, 1.99]
Crosby 2017 0.2103 0.1389 129 129 9.8% 1.23 [0.94, 1.62]
Lorenzen 2009 -0.2236 0.2821 32 30 3.9% 0.80 [0.46, 1.39]
Lu 2017 -0.4734 0.226 40 40 5.5% 0.62 [0.40, 0.97]
Luo 2015 -0.2521 0.3389 28 27 2.9% 0.78 [0.40, 1.51]
Ruhstaller 2018 -0.2989 0.1525 149 151 8.9% 0.74 [0.55, 1.00]
Shi 2014 -0.2868 0.2176 32 32 5.8% 0.75 [0.49, 1.15]
Sun 2014 -0.4586 0.2733 21 21 4.1% 0.63 [0.37, 1.08]
Suntharalingam 2017 -0.09 0.1217 159 169 11.0% 0.91 [0.72, 1.16]
Subtotal (95% CI) 675 685 58.9% 0.81 [0.69, 0.96]
Heterogeneity: Tau² = 0.02; Chi² = 13.92, df = 9 (P = 0.13); I² = 35%
Test for overall effect: Z = 2.47 (P = 0.01)
2.2.2 TKI
Chen 2016a -0.7186 0.199 68 68 6.5% 0.49 [0.33, 0.72]
Dutton 2014 -0.1036 0.0987 224 225 12.8% 0.90 [0.74, 1.09]
Wu 2018 -0.239 0.122 176 176 11.0% 0.79 [0.62, 1.00]
Ye 2015 -0.4027 0.2494 37 36 4.8% 0.67 [0.41, 1.09]
Zhou 2015 -0.3845 0.2112 40 40 6.0% 0.68 [0.45, 1.03]
Subtotal (95% CI) 545 545 41.1% 0.73 [0.59, 0.89]
Heterogeneity: Tau² = 0.03; Chi² = 8.52, df = 4 (P = 0.07); I² = 53%
Test for overall effect: Z = 3.04 (P = 0.002)
Total (95% CI) 1220 1230 100.0% 0.78 [0.69, 0.88]
Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40%
Test for overall effect: Z = 3.95 (P < 0.0001)
Test for subgroup differences: Chi² = 0.75, df = 1 (P = 0.39), I² = 0%
0.05 0.2 1 5 20
Favours [anti-EGFR] Favours [Control]

**C**

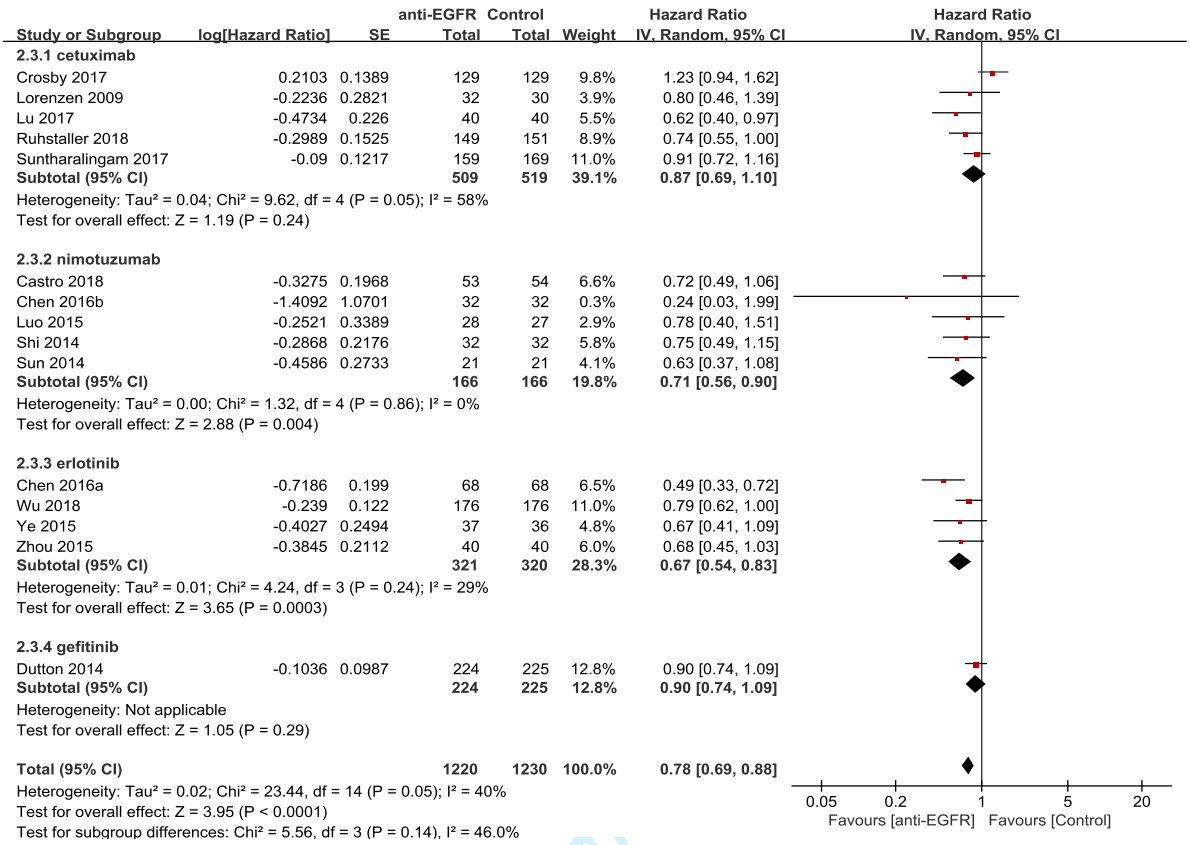
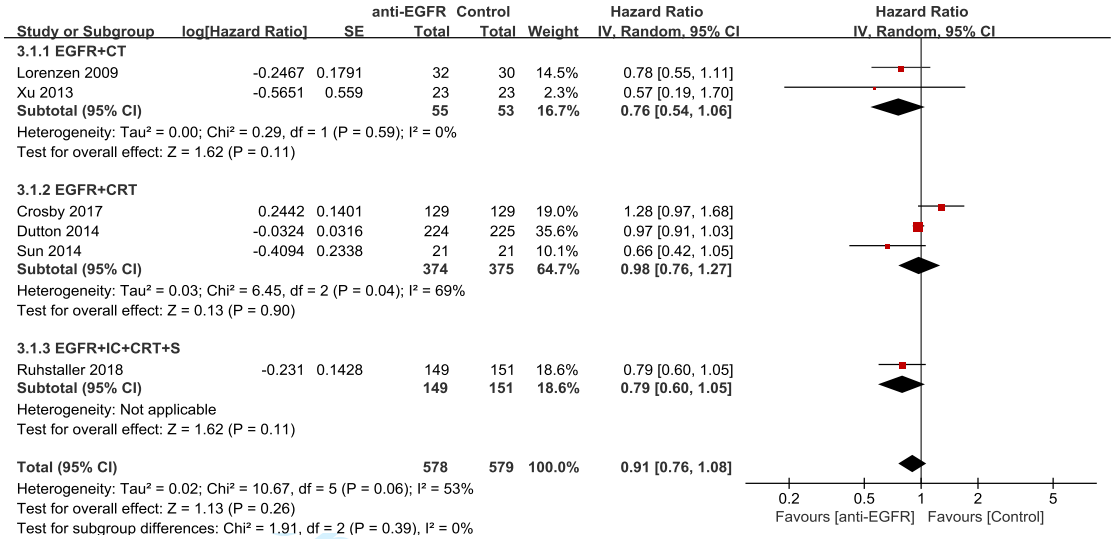


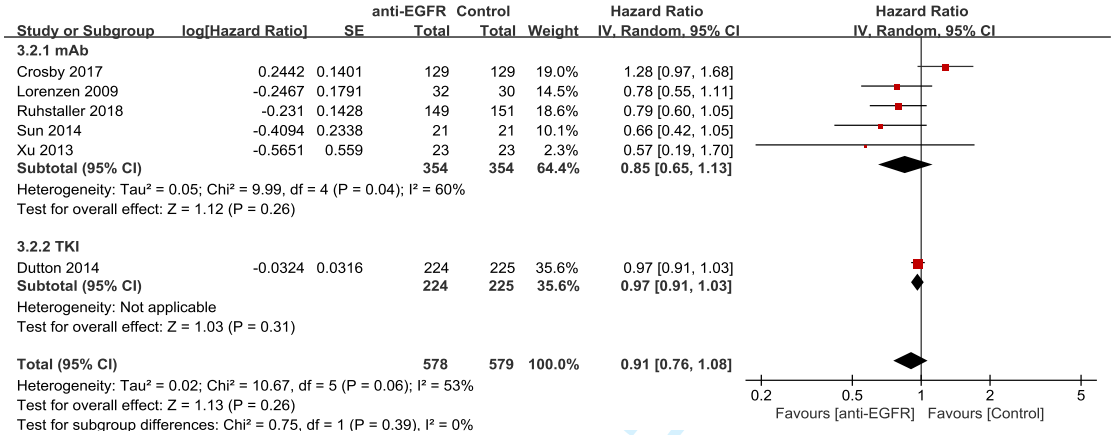

**Figure S1** Subgroup analysis of the overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

eview only

**A**




**B**




**C**

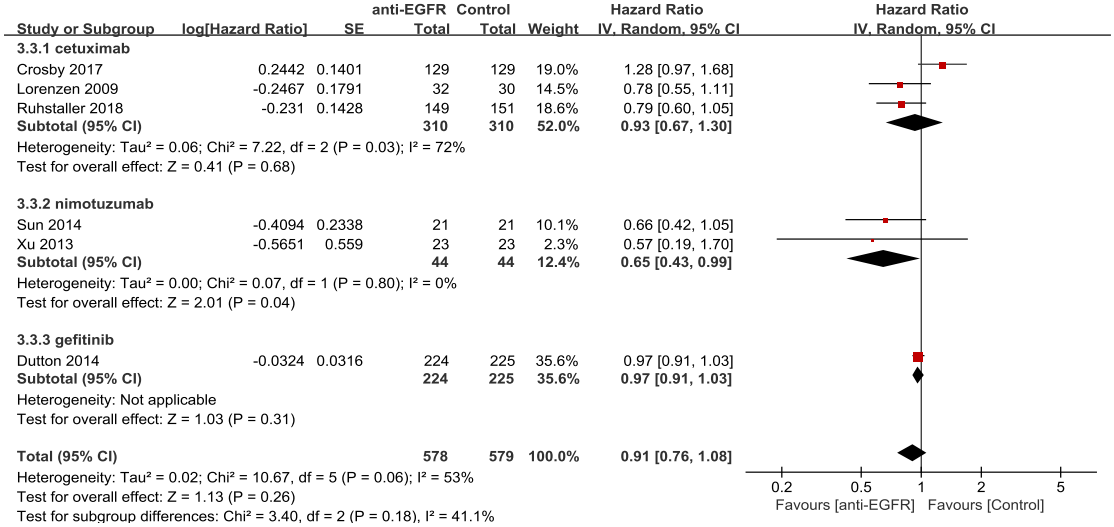



**Figure S2** Subgroup analysis of the progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

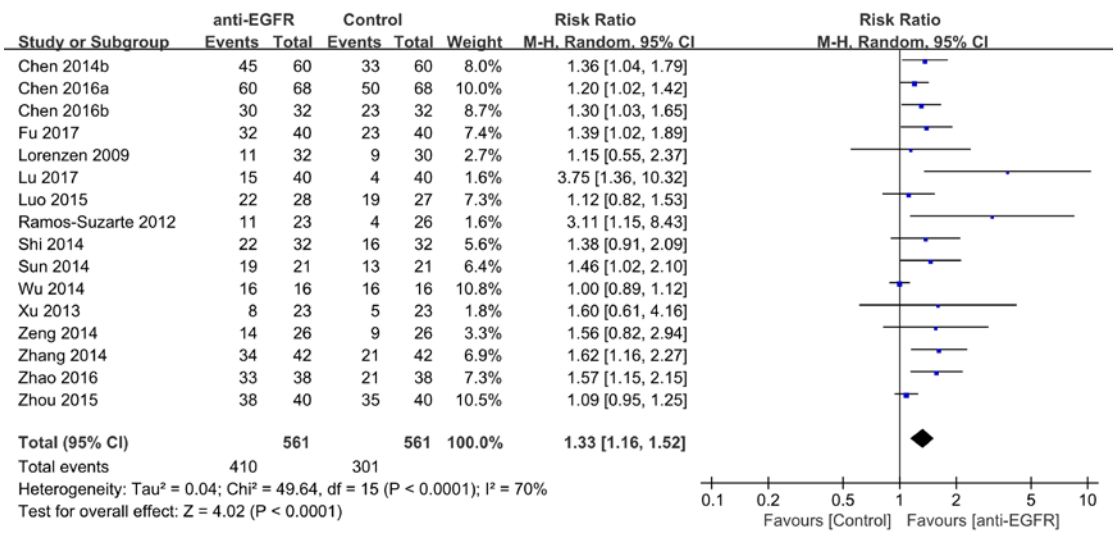
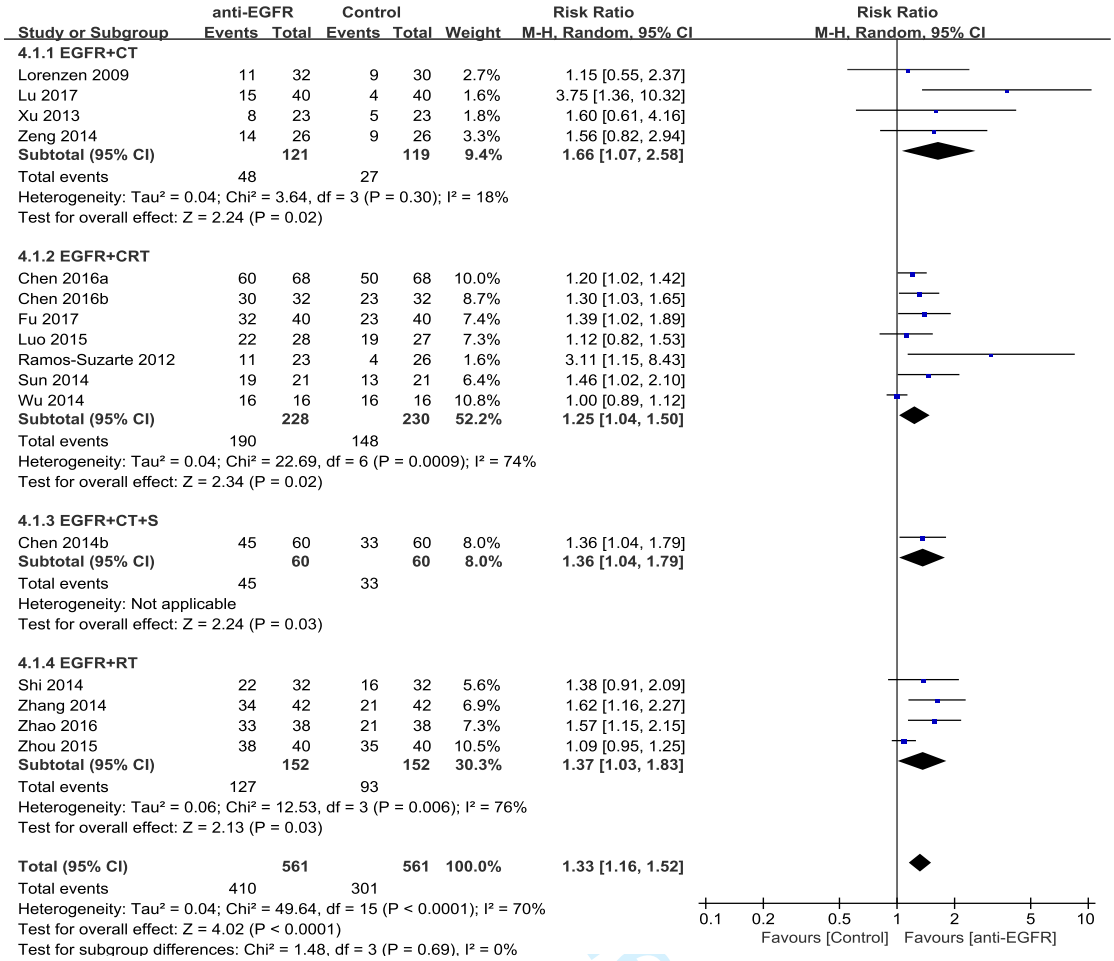


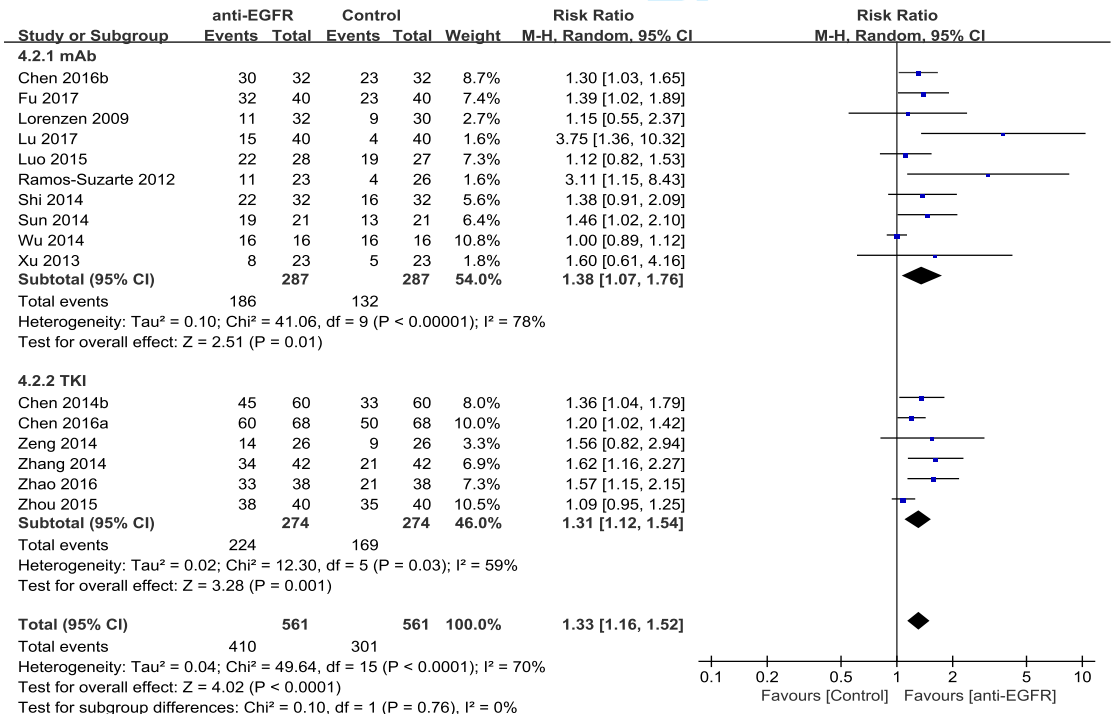

**Figure S3** Forest plot of the objective response rate (ORR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.1.1 EGFR+CT
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Subtotal (95% CI) 121 119 9.4% 1.66 [1.07, 2.58]
Total events 48 27
Heterogeneity: Tau² = 0.04; Chi² = 3.64, df = 3 (P = 0.30); I² = 18%
Test for overall effect: Z = 2.24 (P = 0.02)
4.1.2 EGFR+CRT
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Subtotal (95% CI) 228 230 52.2% 1.25 [1.04, 1.50]
Total events 190 148
Heterogeneity: Tau² = 0.04; Chi² = 22.69, df = 6 (P = 0.0009); I² = 74%
Test for overall effect: Z = 2.34 (P = 0.02)
4.1.3 EGFR+CT+S
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Subtotal (95% CI) 60 60 8.0% 1.36 [1.04, 1.79]
Total events 45 33
Heterogeneity: Not applicable
Test for overall effect: Z = 2.24 (P = 0.03)
4.1.4 EGFR+RT
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 152 152 30.3% 1.37 [1.03, 1.83]
Total events 127 93
Heterogeneity: Tau² = 0.06; Chi² = 12.53, df = 3 (P = 0.006); I² = 76%
Test for overall effect: Z = 2.13 (P = 0.03)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 1.48, df = 3 (P = 0.69), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.2.1 mAb
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Subtotal (95% CI) 287 287 54.0% 1.38 [1.07, 1.76]
Total events 186 132
Heterogeneity: Tau² = 0.10; Chi² = 41.06, df = 9 (P < 0.00001); I² = 78%
Test for overall effect: Z = 2.51 (P = 0.01)
4.2.2 TKI
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 274 274 46.0% 1.31 [1.12, 1.54]
Total events 224 169
Heterogeneity: Tau² = 0.02; Chi² = 12.30, df = 5 (P = 0.03); I² = 59%
Test for overall effect: Z = 3.28 (P = 0.001)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 0.10, df = 1 (P = 0.76), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**C**

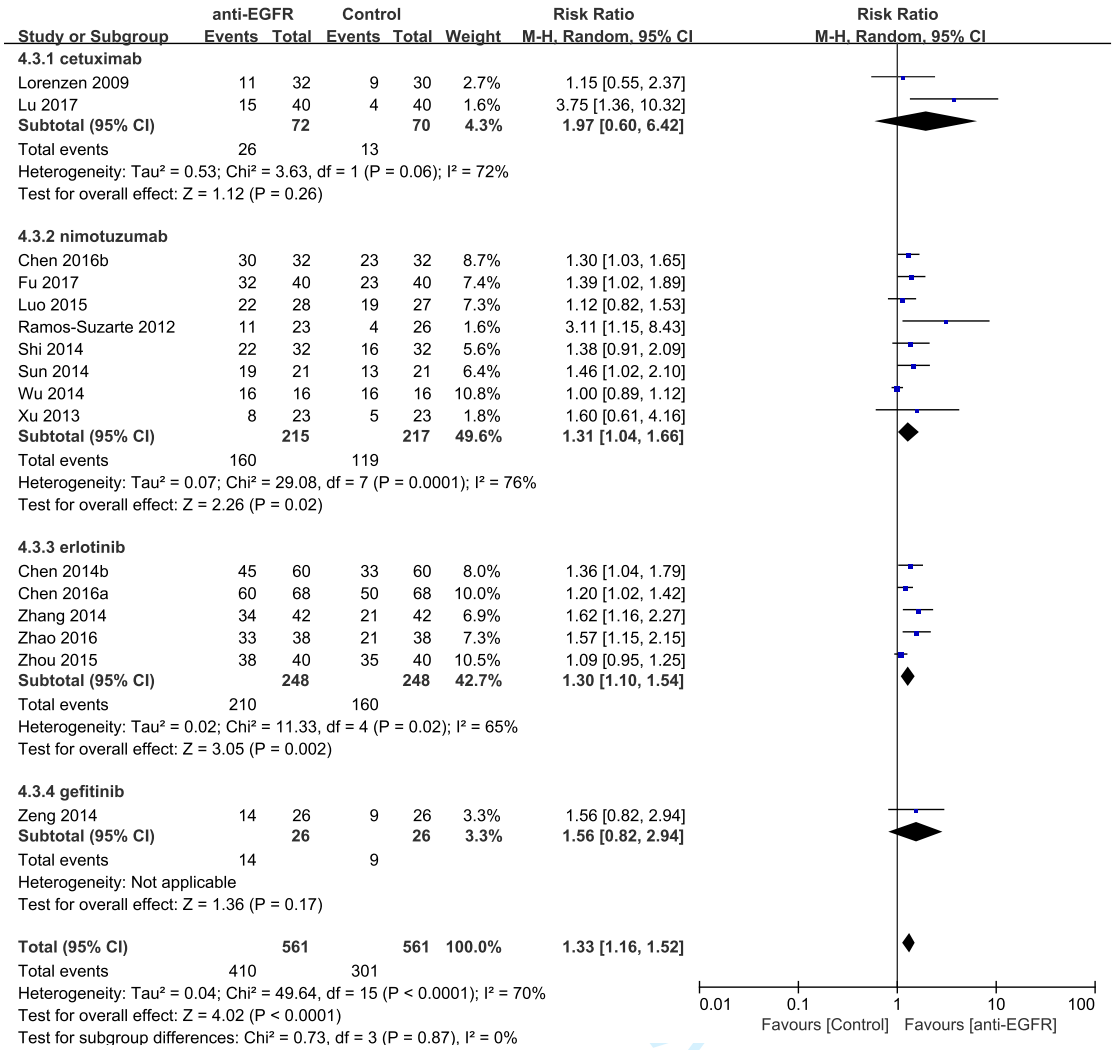



**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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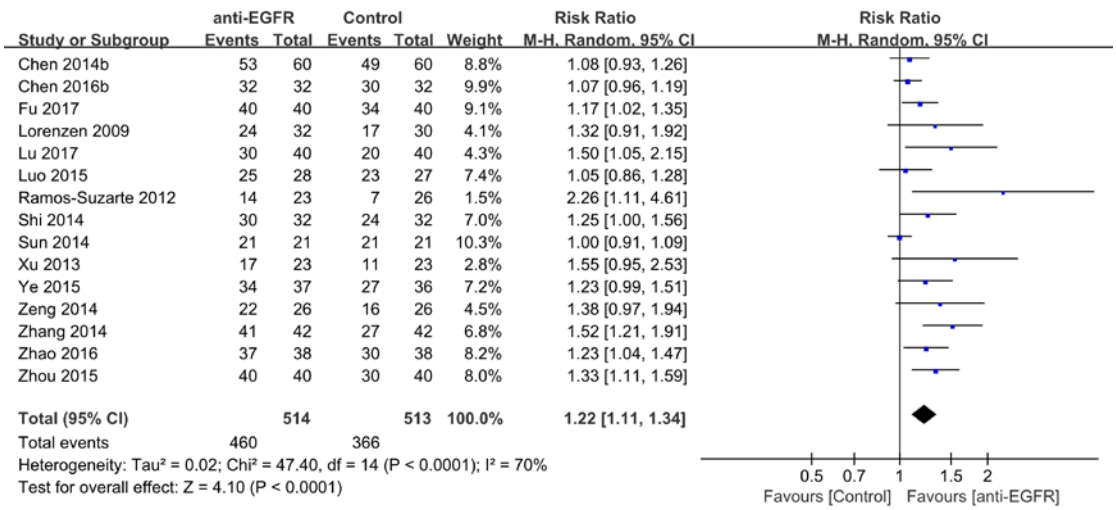
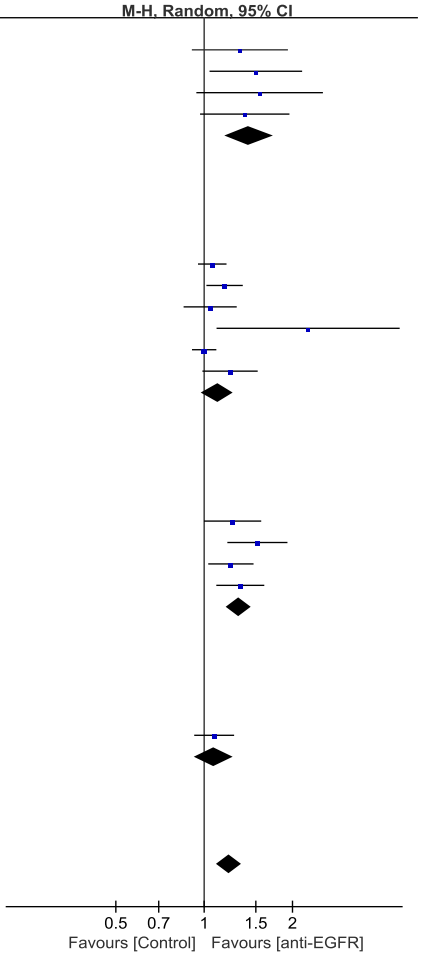



**Figure S5** Forest plot of the disease control rate (DCR) in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer

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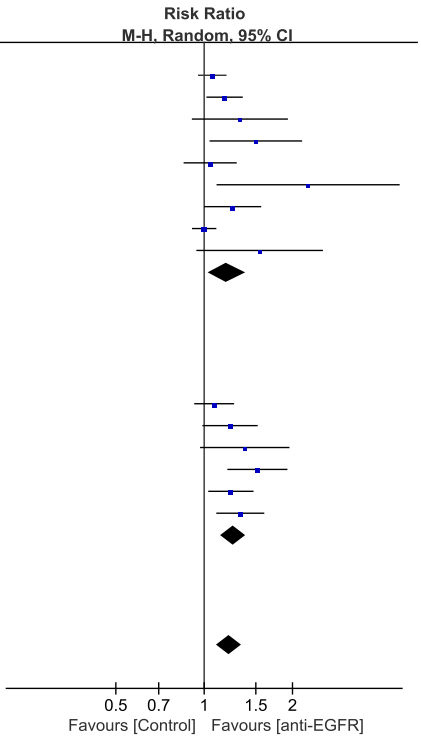
**A**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.1.1 EGFR+CT** | | | | | | |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| **Subtotal (95% CI)** | | **121** | | **119** | **15.7%** | **1.42 [1.17, 1.72]** |
| Total events | 93 | | 64 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 0.38, df = 3 (P = 0.94); I² = 0% | | | | | | |
| Test for overall effect: Z = 3.61 (P = 0.0003) | | | | | | |
| **5.1.2 EGFR+CRT** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| **Subtotal (95% CI)** | | **181** | | **182** | **45.5%** | **1.11 [0.98, 1.26]** |
| Total events | 166 | | 142 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 18.13, df = 5 (P = 0.003); I² = 72% | | | | | | |
| Test for overall effect: Z = 1.69 (P = 0.09) | | | | | | |
| **5.1.3 EGFR+RT** | | | | | | |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **152** | | **152** | **30.0%** | **1.31 [1.19, 1.45]** |
| Total events | 148 | | 111 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 2.33, df = 3 (P = 0.51); I² = 0% | | | | | | |
| Test for overall effect: Z = 5.41 (P < 0.00001) | | | | | | |
| **5.1.4 EGFR+CT+S** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| **Subtotal (95% CI)** | | **60** | | **60** | **8.8%** | **1.08 [0.93, 1.26]** |
| Total events | 53 | | 49 | | | |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.02 (P = 0.31) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 9.05, df = 3 (P = 0.03), I² = 66.9% | | | | | | |




**B**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.2.1 mAb** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| **Subtotal (95% CI)** | | **271** | | **271** | **56.5%** | **1.20 [1.04, 1.38]** |
| Total events | 233 | | 187 | | | |
| Heterogeneity: Tau² = 0.03; Chi² = 37.25, df = 8 (P < 0.0001); I² = 79% | | | | | | |
| Test for overall effect: Z = 2.43 (P = 0.01) | | | | | | |
| **5.2.2 TKI** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **243** | | **242** | **43.5%** | **1.26 [1.14, 1.38]** |
| Total events | 227 | | 179 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 7.16, df = 5 (P = 0.21); I² = 30% | | | | | | |
| Test for overall effect: Z = 4.54 (P < 0.00001) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.29, df = 1 (P = 0.59), I² = 0% | | | | | | |



**C**

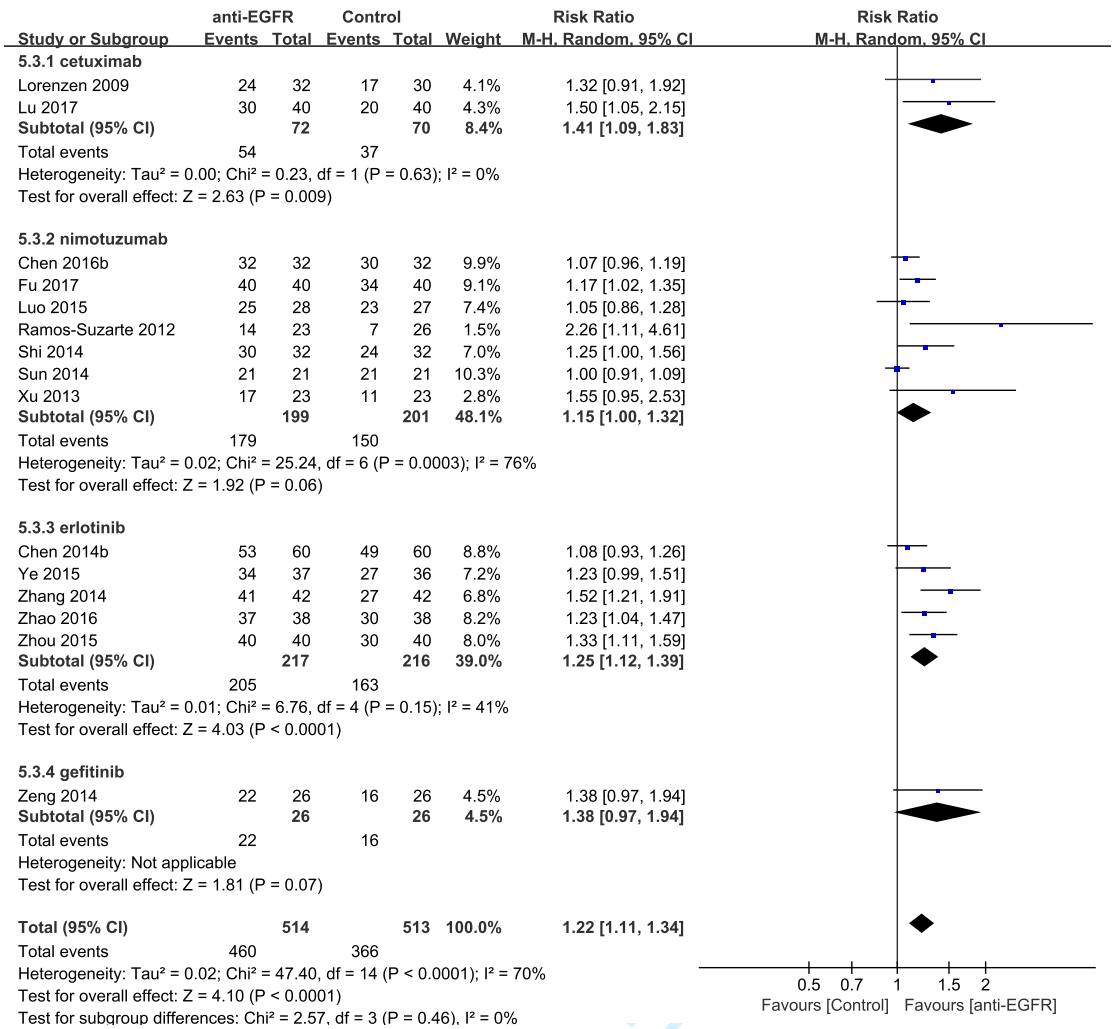



**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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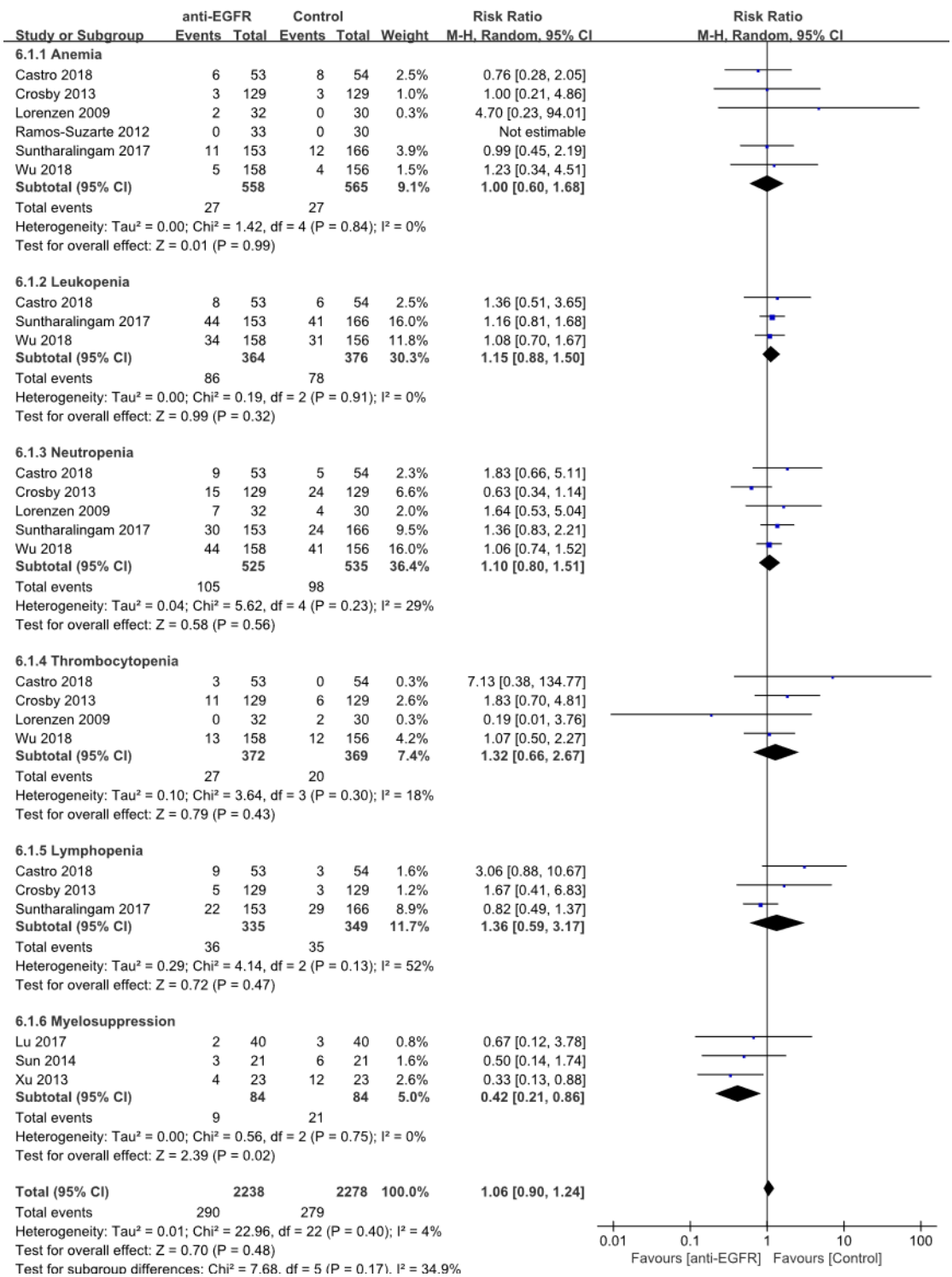
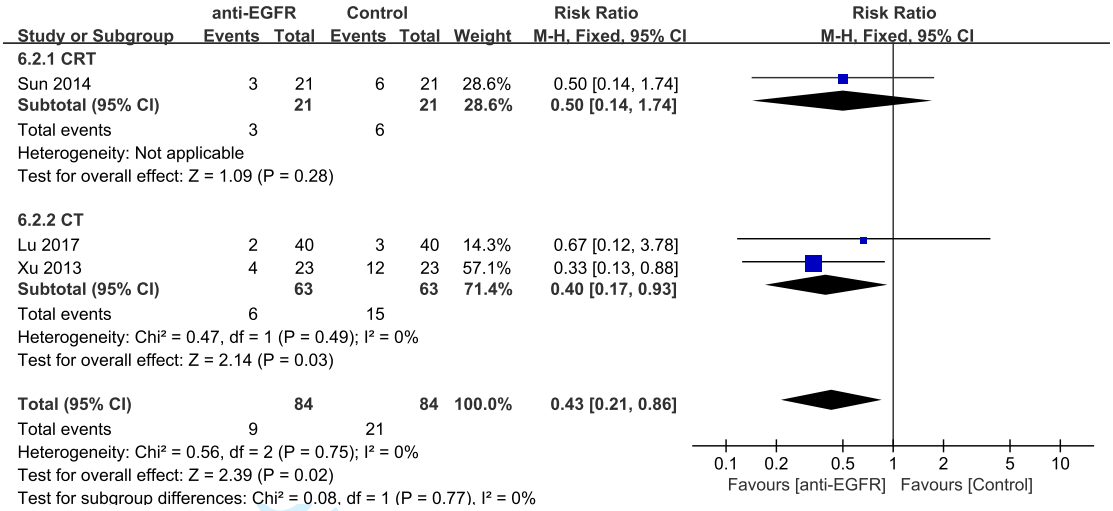



**Figure S7** Forest plot of grade 3–5 haematological toxicities in the anti-EGFR arm versus non-anti-EGFR arm in patients with oesophageal cancer



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**A**




**B**

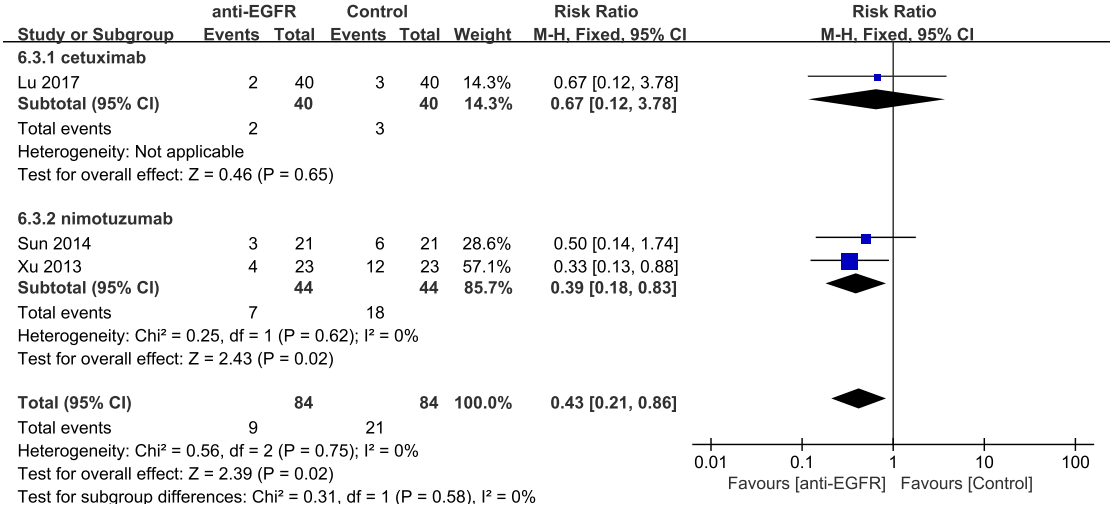
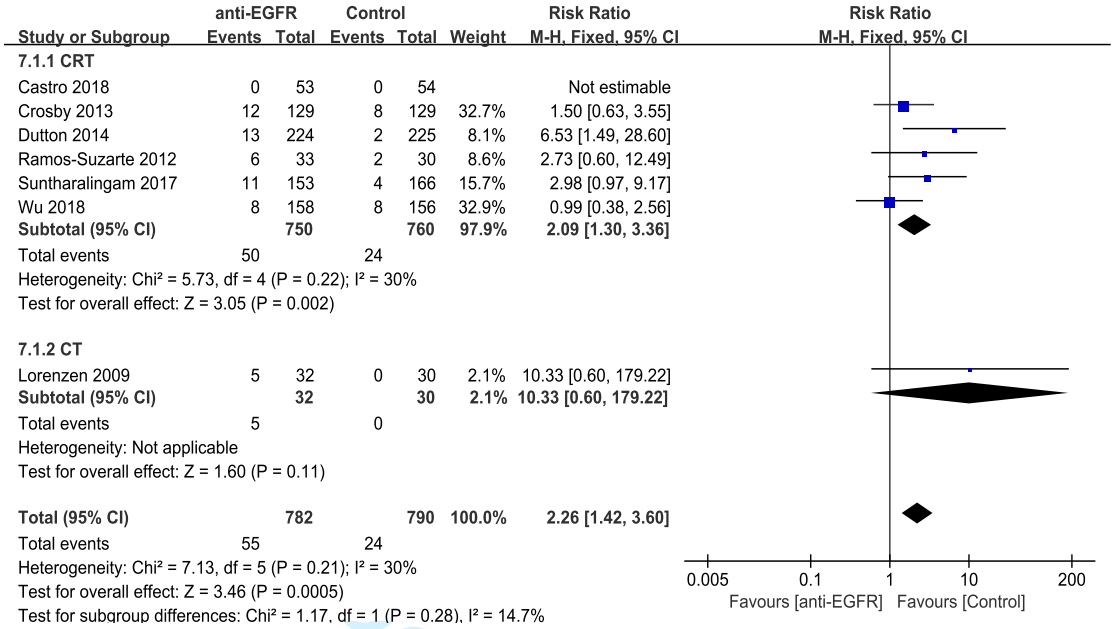
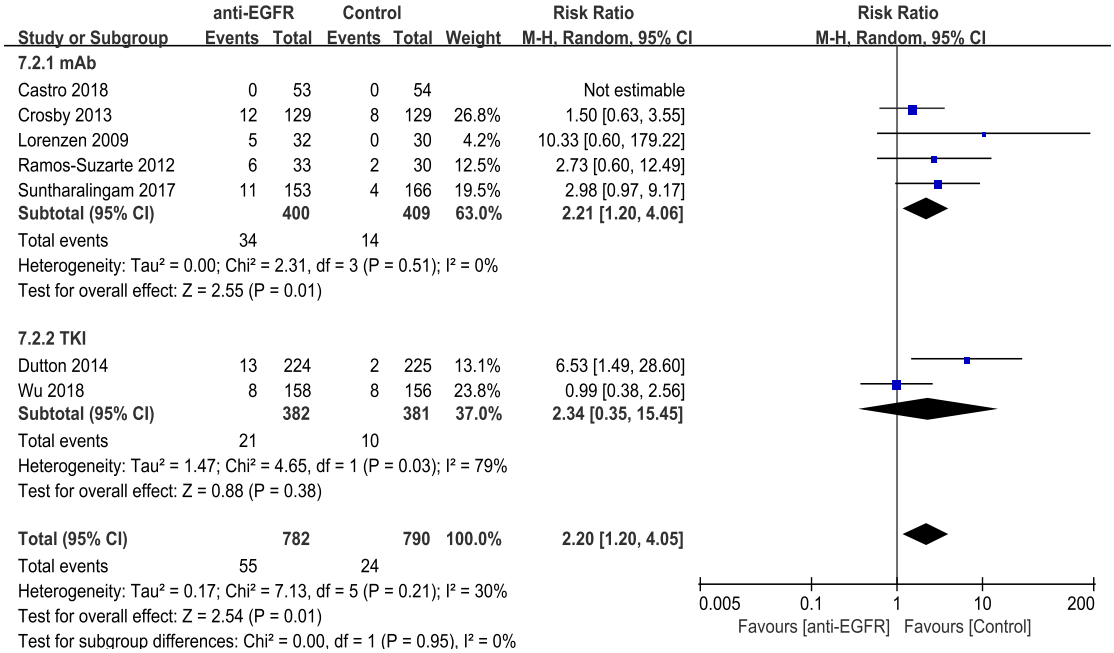



**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
7.1.1 CRT
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 32.7% 1.50 [0.63, 3.55]
Dutton 2014 13 224 2 225 8.1% 6.53 [1.49, 28.60]
Ramos-Suzarte 2012 6 33 2 30 8.6% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 15.7% 2.98 [0.97, 9.17]
Wu 2018 8 158 8 156 32.9% 0.99 [0.38, 2.56]
Subtotal (95% CI) 750 760 97.9% 2.09 [1.30, 3.36]
Total events 50 24
Heterogeneity: Chi² = 5.73, df = 4 (P = 0.22); I² = 30%
Test for overall effect: Z = 3.05 (P = 0.002)
7.1.2 CT
Lorenzen 2009 5 32 0 30 2.1% 10.33 [0.60, 179.22]
Subtotal (95% CI) 32 30 2.1% 10.33 [0.60, 179.22]
Total events 5 0
Heterogeneity: Not applicable
Test for overall effect: Z = 1.60 (P = 0.11)
Total (95% CI) 782 790 100.0% 2.26 [1.42, 3.60]
Total events 55 24
Heterogeneity: Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 3.46 (P = 0.0005)
Test for subgroup differences: Chi² = 1.17, df = 1 (P = 0.28), I² = 14.7%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
7.2.1 mAb
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 26.8% 1.50 [0.63, 3.55]
Lorenzen 2009 5 32 0 30 4.2% 10.33 [0.60, 179.22]
Ramos-Suzarte 2012 6 33 2 30 12.5% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 19.5% 2.98 [0.97, 9.17]
Subtotal (95% CI) 400 409 63.0% 2.21 [1.20, 4.06]
Total events 34 14
Heterogeneity: Tau² = 0.00; Chi² = 2.31, df = 3 (P = 0.51); I² = 0%
Test for overall effect: Z = 2.55 (P = 0.01)
7.2.2 TKI
Dutton 2014 13 224 2 225 13.1% 6.53 [1.49, 28.60]
Wu 2018 8 158 8 156 23.8% 0.99 [0.38, 2.56]
Subtotal (95% CI) 382 381 37.0% 2.34 [0.35, 15.45]
Total events 21 10
Heterogeneity: Tau² = 1.47; Chi² = 4.65, df = 1 (P = 0.03); I² = 79%
Test for overall effect: Z = 0.88 (P = 0.38)
Total (95% CI) 782 790 100.0% 2.20 [1.20, 4.05]
Total events 55 24
Heterogeneity: Tau² = 0.17; Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 2.54 (P = 0.01)
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.95), I² = 0%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**C**

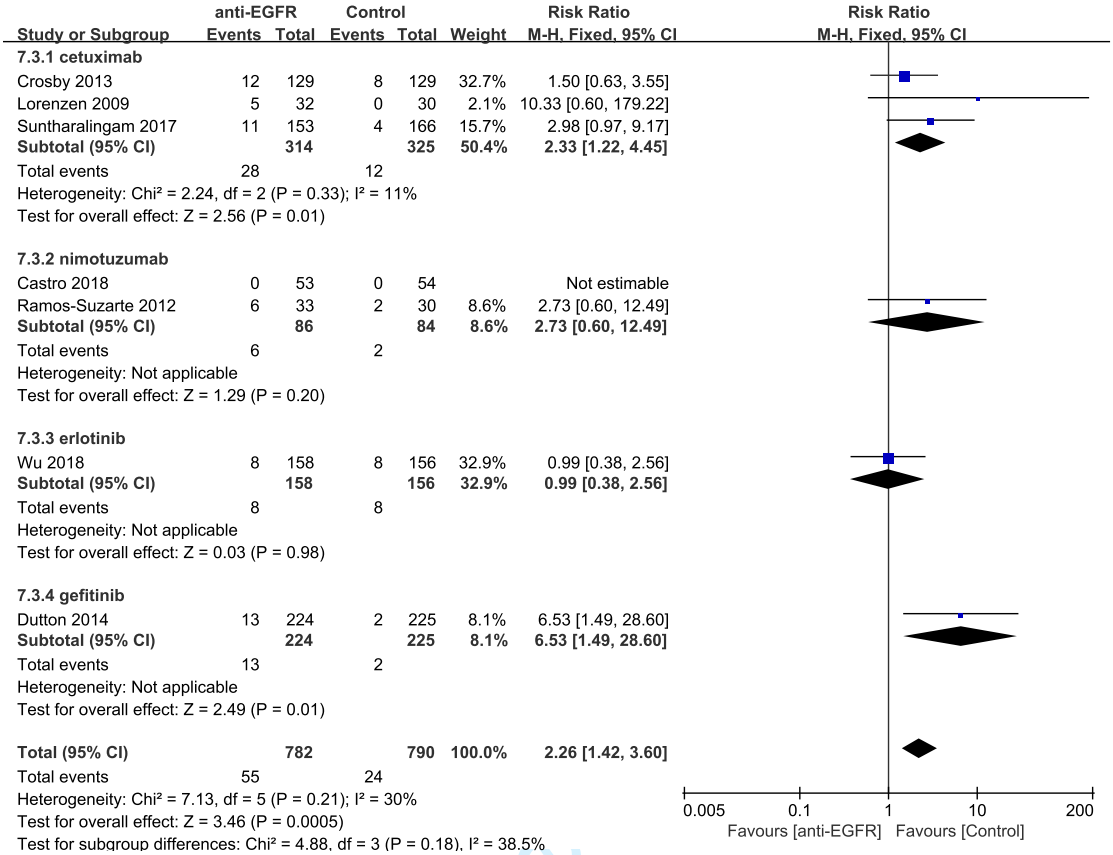
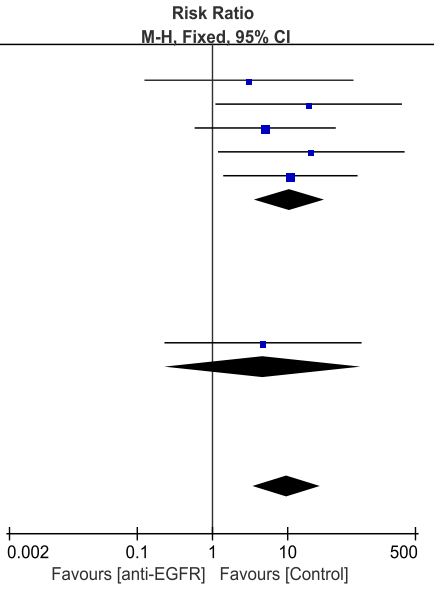



**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

eview only

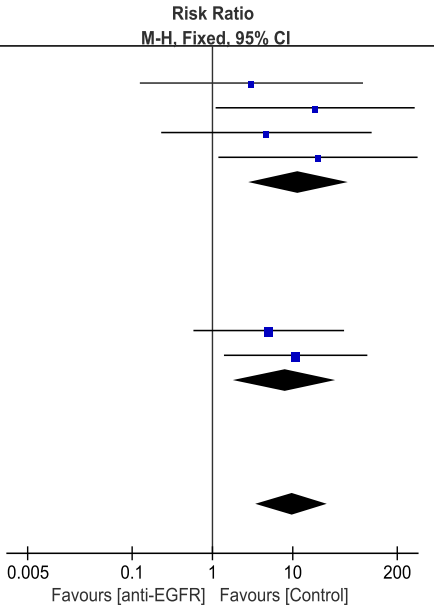
**A**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **8.1.1 CRT** | | | | | | |
| Castro 2018 | 1 | 53 | 0 | 54 | 12.4% | 3.06 [0.13, 73.37] |
| Crosby 2013 | 9 | 129 | 0 | 129 | 12.5% | 19.00 [1.12, 323.06] |
| Dutton 2014 | 5 | 224 | 1 | 225 | 25.0% | 5.02 [0.59, 42.65] |
| Suntharalingam 2017 | 9 | 153 | 0 | 166 | 12.0% | 20.60 [1.21, 351.02] |
| Wu 2018 | 11 | 158 | 1 | 156 | 25.2% | 10.86 [1.42, 83.12] |
| **Subtotal (95% CI)** | | **717** | | **730** | **87.1%** | **10.59 [3.56, 31.49]** |
| Total events | 35 | | 2 | | | |
| Heterogeneity: Chi² = 1.43, df = 4 (P = 0.84); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.24 (P < 0.0001) | | | | | | |
| **8.1.2 CT** | | | | | | |
| Lorenzen 2009 | 2 | 32 | 0 | 30 | 12.9% | 4.70 [0.23, 94.01] |
| **Subtotal (95% CI)** | | **32** | | **30** | **12.9%** | **4.70 [0.23, 94.01]** |
| Total events | 2 | | 0 | | | |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.01 (P = 0.31) | | | | | | |
| **Total (95% CI)** | | **749** | | **760** | **100.0%** | **9.83 [3.54, 27.28]** |
| Total events | 37 | | 2 | | | |
| Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.39 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.25, df = 1 (P = 0.62), I² = 0% | | | | | | |




**B**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **8.2.1 mAb** | | | | | | |
| Castro 2018 | 1 | 53 | 0 | 54 | 12.4% | 3.06 [0.13, 73.37] |
| Crosby 2013 | 9 | 129 | 0 | 129 | 12.5% | 19.00 [1.12, 323.06] |
| Lorenzen 2009 | 2 | 32 | 0 | 30 | 12.9% | 4.70 [0.23, 94.01] |
| Suntharalingam 2017 | 9 | 153 | 0 | 166 | 12.0% | 20.60 [1.21, 351.02] |
| **Subtotal (95% CI)** | | **367** | | **379** | **49.8%** | **11.71 [2.80, 48.99]** |
| Total events | 21 | | 0 | | | |
| Heterogeneity: Chi² = 1.31, df = 3 (P = 0.73); I² = 0% | | | | | | |
| Test for overall effect: Z = 3.37 (P = 0.0007) | | | | | | |
| **8.2.2 TKI** | | | | | | |
| Dutton 2014 | 5 | 224 | 1 | 225 | 25.0% | 5.02 [0.59, 42.65] |
| Wu 2018 | 11 | 158 | 1 | 156 | 25.2% | 10.86 [1.42, 83.12] |
| **Subtotal (95% CI)** | | **382** | | **381** | **50.2%** | **7.95 [1.84, 34.37]** |
| Total events | 16 | | 2 | | | |
| Heterogeneity: Chi² = 0.27, df = 1 (P = 0.61); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.78 (P = 0.005) | | | | | | |
| **Total (95% CI)** | | **749** | | **760** | **100.0%** | **9.83 [3.54, 27.28]** |
| Total events | 37 | | 2 | | | |
| Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.39 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.14, df = 1 (P = 0.71), I² = 0% | | | | | | |



**C**

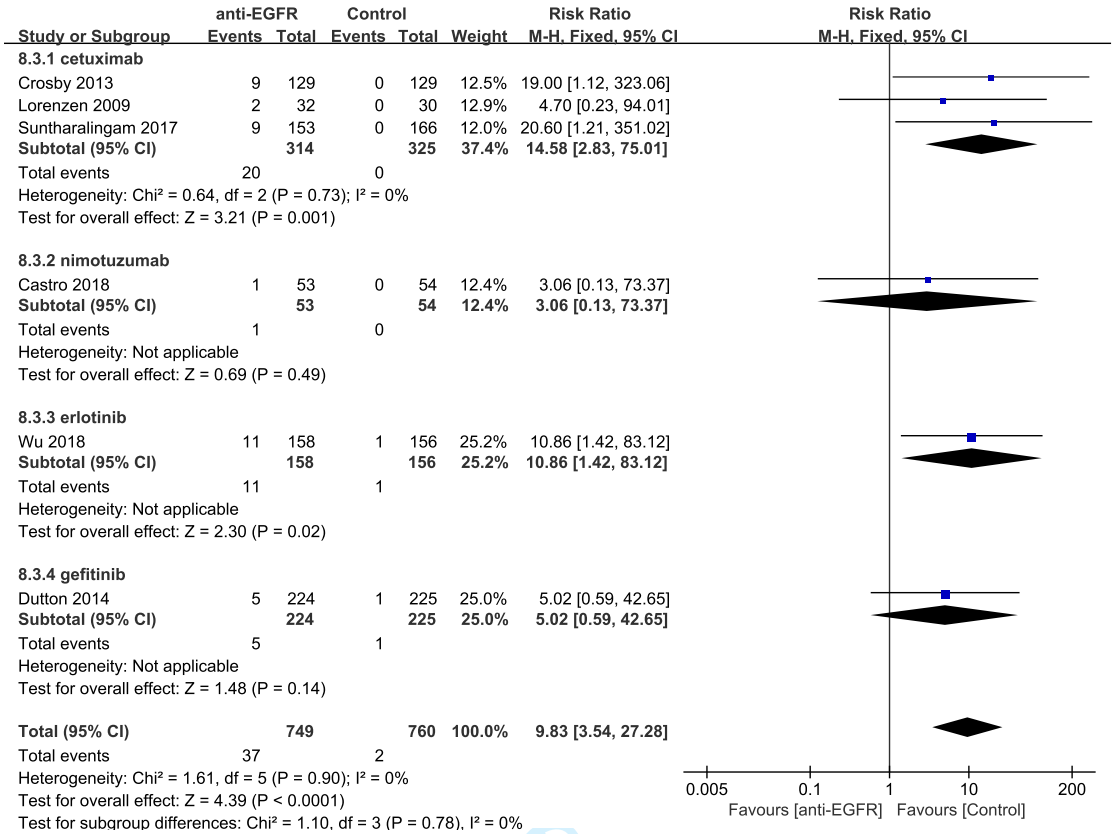



**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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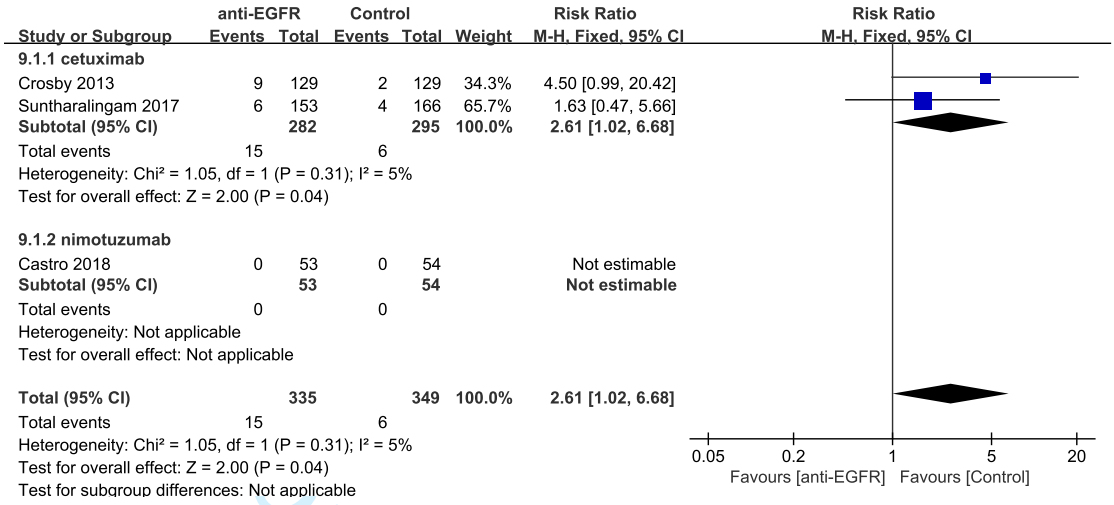



**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

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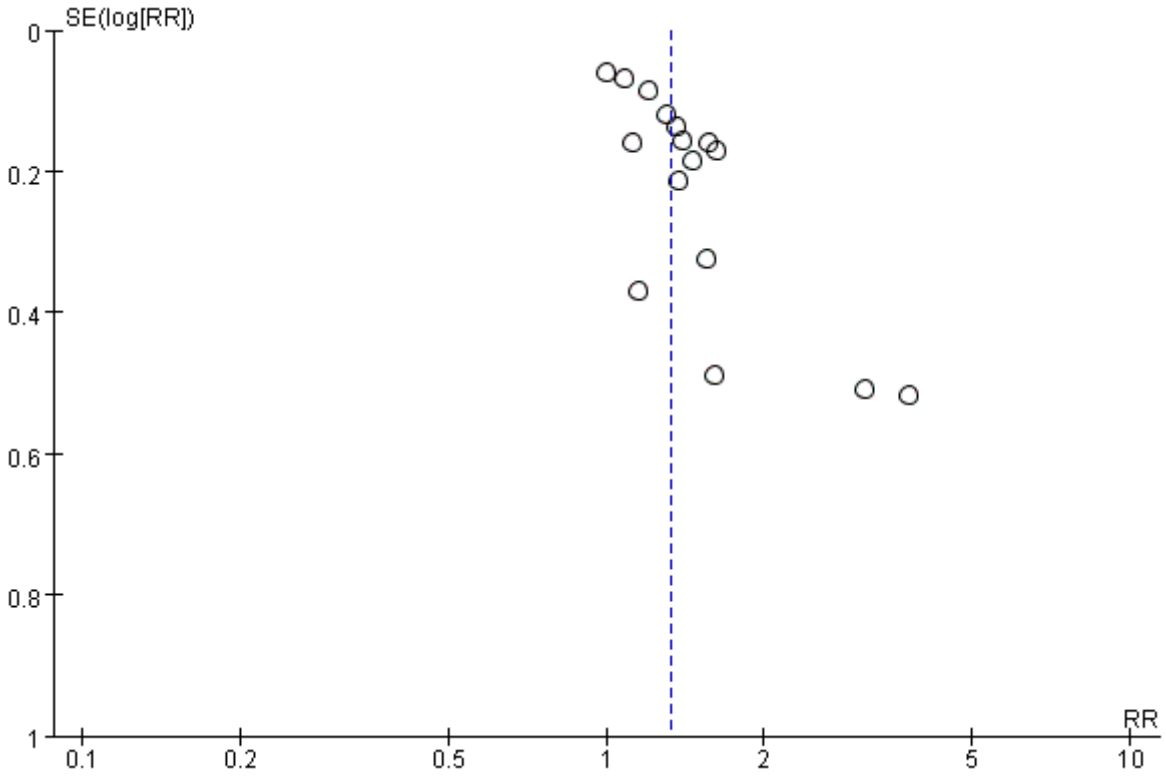



**Figure S12** Funnel plot of publication bias for ORR

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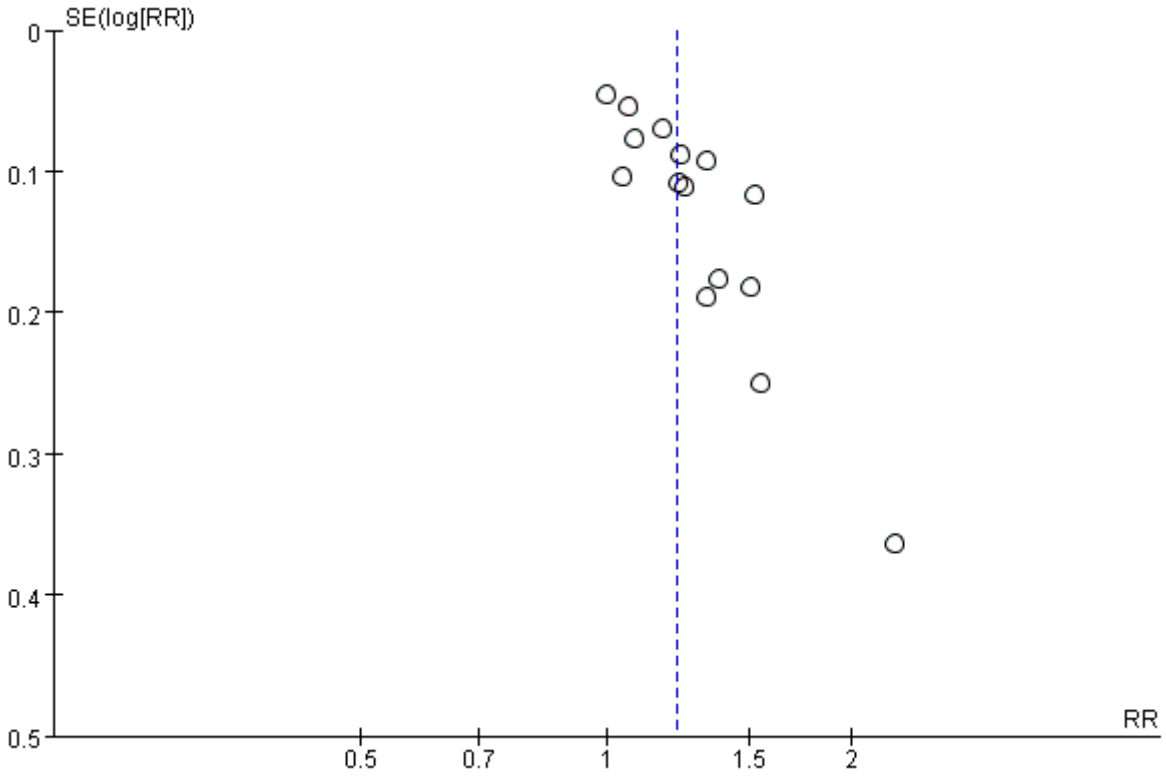



**Figure S13** Funnel plot of publication bias for DCR

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**Supplementary Data S1**

**Appendix 1. PubMed search strategy (N = 86)**

#1 “esophageal neoplasms”.[Mesh]

#2 “esophageal neoplasm*”.ti,ab.

#3 “esophagus neoplasm*”.ti,ab.

#4 “oesophageal neoplasm*”.ti,ab.

#5 “esophageal cancer*”.ti,ab.

#6 “esophagus cancer*”.ti,ab.

#7 “oesophageal cancer*”.ti,ab.

#8 “cancer of esophagus”.ti,ab.

#9 “cancer of the esophagus”.ti,ab.

#10 “esophageal tumor*”.ti,ab.

#11 “esophagus tumor*”.ti,ab.

#12 “oesophageal tumor*”.ti,ab.

#13 “esophageal carcinoma*”.ti,ab.

#14 “esophagus carcinoma*”.ti,ab.

#15 “oesophageal carcinoma*”.ti,ab.

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 “EGFR inhibitor*”.ti,ab.

#18 “epidermal growth factor receptor inhibitor*”.ti,ab.

#19 “anti-human epidermal growth factor receptor”.ti,ab.

#20 #17 OR #18 OR #19

#21 “cetuximab”.[Mesh]

#22 “erbitux”.ti,ab.

#23 “IMC C225”.ti,ab.

#24 “IMC-C225”.ti,ab.

#25 “MAb C225”.ti,ab.

#26 “C225”.ti,ab.

#27 “C-225”.ti,ab.



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#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 “panitumumab”.[Mesh]

#30 “ABX-EGF monoclonal antibody”.ti,ab.

#31 “ABX-EGF MAb”.ti,ab.

#32 “vectibix”.ti,ab.

#33 #29 OR #30 OR #31 OR #32

#34 “nimotuzumab”.ti,ab.

#35 “theraloc”.ti,ab.

#36 #34 OR #35

#37 “erlotinib hydrochloride”.[Mesh]

#38 “erlotinib HCl”.ti,ab.

#39 “erlotinib”.ti,ab.

#40 “tarceva”.ti,ab.

#41 “11C-erlotinib”.ti,ab.

#42 “11C erlotinib”.ti,ab.

#43 “OSI-774”.ti,ab.

#44 “OSI 774”.ti,ab.

#45 “OSI774”.ti,ab.

#46 “CP 358774”.ti,ab.

#47 “CP-358774”.ti,ab.

#48 “CP358774”.ti,ab.

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 “gefitinib”.[Mesh]

#51 “iressa”.ti,ab.

#52 “ZD1839”.ti,ab.

#53 “ZD-1839”.ti,ab.

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 “randomized controlled trials as topic”.[Mesh]

#57 “randomized controlled trial[publication type]”



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#58 “random*”.ti,ab.

#59 “clinical trials as topic”.[Mesh]

#60 “clinical trial[publication type]”

#61 “controlled clinical trials as topic”.[Mesh]

#62 “controlled clinical trial[publication type]”

#63 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62

#64 #16 AND #55 AND #63

**Appendix 2. EMBASE search strategy (N = 831)**

#1 'esophagus tumor'/exp

#2 'esophageal neoplasm*':ab,ti

#3 'esophagus neoplasm*':ab,ti

#4 'oesophageal neoplasm*':ab,ti

#5 'esophageal cancer*':ab,ti

#6 'esophagus cancer*':ab,ti

#7 'oesophageal cancer*':ab,ti

#8 'cancer of esophagus':ab,ti

#9 'cancer of the esophagus':ab,ti

#10 'esophageal tumor*':ab,ti

#11 'esophagus tumor*':ab,ti

#12 'oesophageal tumor*':ab,ti

#13 'esophageal carcinoma*':ab,ti

#14 'esophagus carcinoma*':ab,ti

#15 'oesophageal carcinoma*':ab,ti

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 'egfr inhibitor*':ab,ti

#18 'epidermal growth factor receptor inhibitor*':ab,ti

#19 'anti-human epidermal growth factor receptor':ab,ti

#20 #17 OR #18 OR #19



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#21 'cetuximab'/exp

#22 'erbitux':ab,ti

#23 'imc c225':ab,ti

#24 'imc-c225':ab,ti

#25 'mab c225':ab,ti

#26 'c225':ab,ti

#27 'c-225':ab,ti

#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 'panitumumab'/exp

#30 'abx-egf monoclonal antibody':ab,ti

#31 'abx-egf mab':ab,ti

#32 'vectibix':ab,ti

#33 #29 OR #30 OR #31 OR #32

#34 'nimotuzumab'/exp

#35 'theraloc':ab,ti

#36 #34 OR #35

#37 'erlotinib'/exp

#38 'erlotinib hcl':ab,ti

#39 'erlotinib':ab,ti

#40 'tarceva':ab,ti

#41 '11c-erlotinib':ab,ti

#42 '11c erlotinib':ab,ti

#43 'osi-774':ab,ti

#44 'osi 774':ab,ti

#45 'osi774':ab,ti

#46 'cp 358774':ab,ti

#47 'cp-358774':ab,ti

#48 'cp358774':ab,ti

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 'gefitinib'/exp



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#51 'iressa':ab,ti

#52 'zd1839':ab,ti

#53 'zd-1839':ab,ti

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 'randomized controlled trial'/exp

#57 'randomized controlled trial (topic)'/exp

#58 'random*':ab,ti

#59 'clinical trial (topic)'/exp

#60 'clinical trial'/exp

#61 'clinical trial*':ab,ti

#62 'controlled clinical trial'/exp

#63 'controlled clinical trial (topic)'/exp

#64 'controlled clinical trial*':ab,ti

#65 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62 OR #63 OR #64

#66 #16 AND #55 AND #65

**Appendix 3. CENTRAL search strategy (N = 94)**

#1 MeSH descriptor: [esophageal neoplasms] explode all trees

#2 (esophageal neoplasm*):ti,ab,kw OR (esophagus neoplasm*):ti,ab,kw OR (oesophageal neoplasm*):ti,ab,kw OR (esophageal cancer*):ti,ab,kw OR (esophagus cancer*):ti,ab,kw. (Word variations have been searched)

#3 (oesophageal cancer*):ti,ab,kw OR (cancer of esophagus):ti,ab,kw OR (cancer of the esophagus):ti,ab,kw OR (esophageal tumor*):ti,ab,kw OR (esophagus tumor*):ti,ab,kw. (Word variations have been searched)

#4 (oesophageal tumor*):ti,ab,kw OR (esophageal carcinoma*):ti,ab,kw OR (esophagus carcinoma*):ti,ab,kw OR (oesophageal carcinoma*):ti,ab,kw. (Word variations have been searched)

#5 #1 OR #2 OR #3 OR #4

#6 (EGFR inhibitor*):ti,ab,kw OR (epidermal growth factor receptor inhibitor*):ti,ab,kw OR (anti-human epidermal growth factor receptor):ti,ab,kw. (Word variations have been searched)



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#7 MeSH descriptor: [cetuximab] explode all trees

#8 (erbitux):ti,ab,kw OR (IMC C225):ti,ab,kw OR (IMC-C225):ti,ab,kw. (Word variations have been searched)

#9 (MAb C225):ti,ab,kw OR (C225):ti,ab,kw OR (C-225):ti,ab,kw. (Word variations have been searched)

#10 #7 OR #8 OR #9

#11 MeSH descriptor: [panitumumab] explode all trees

#12 (ABX-EGF monoclonal antibody):ti,ab,kw OR (ABX-EGF MAb):ti,ab,kw OR (vectibix):ti,ab,kw. (Word variations have been searched)

#13 #11 OR #12

#14 (nimotuzumab):ti,ab,kw OR (theraloc):ti,ab,kw. (Word variations have been searched)

#15 MeSH descriptor: [erlotinib hydrochloride] explode all trees

#16 (erlotinib HCl):ti,ab,kw OR (erlotinib):ti,ab,kw OR (tarceva):ti,ab,kw OR (11C-erlotinib):ti,ab,kw OR (11C erlotinib):ti,ab,kw. (Word variations have been searched)

#17 (OSI-774):ti,ab,kw OR (OSI 774):ti,ab,kw OR (OSI774):ti,ab,kw. (Word variations have been searched)

#18 (CP 358774):ti,ab,kw OR (CP-358774):ti,ab,kw OR (CP358774):ti,ab,kw. (Word variations have been searched)

#19 #15 OR #16 #17 OR #18

#20 MeSH descriptor: [gefitinib] explode all trees

#21 (iressa):ti,ab,kw OR (ZD1839):ti,ab,kw OR (ZD-1839):ti,ab,kw. (Word variations have been searched)

#22 #20 OR #21

#23 #6 OR #10 OR #13 OR #14 OR #19 OR #22

#24 MeSH descriptor: [randomized controlled trials as topic] explode all trees

#25 MeSH descriptor: [randomized controlled trial] explode all trees

#26 (random*):ti,ab,kw. (Word variations have been searched)

#27 MeSH descriptor: [clinical trials as topic] explode all trees

#28 MeSH descriptor: [clinical trial] explode all trees

#29 (clinical trial*):ti,ab,kw. (Word variations have been searched)



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#30 MeSH descriptor: [controlled clinical trial] explode all trees

#31 MeSH descriptor: [controlled clinical trials as topic] explode all trees

#32 (controlled clinical trial*):ti,ab,kw. (Word variations have been searched)

#33 #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32

#34 #5 AND #23 AND #33

**Appendix 4. Web of Science search strategy (N = 191)**

#1 TS=(esophageal neoplasms OR esophageal neoplasm* OR esophagus neoplasm* OR oesophageal neoplasm* OR esophageal cancer* OR esophagus cancer* OR oesophageal cancer* OR cancer of esophagus OR cancer of the esophagus OR esophageal tumor* OR esophagus tumor* OR oesophageal tumor* OR esophageal carcinoma* OR esophagus carcinoma* OR oesophageal carcinoma*)

#2 TS=(EGFR inhibitor* OR epidermal growth factor receptor inhibitor* OR anti-human epidermal growth factor receptor)

#3 TS=(cetuximab OR erbitux OR IMC C225 OR IMC-C225 OR MAb C225 OR C225 OR C-225) #4 TS=(panitumumab OR ABX-EGF monoclonal antibody OR ABX-EGF MAb OR vectibix)

#5 TS=(nimotuzumab OR theraloc)

#6 TS=(Erlotinib Hydrochloride OR Erlotinib HCl OR Erlotinib OR Tarceva OR 11C-erlotinib OR 11C erlotinib OR OSI-774 OR OSI 774 OR OSI774 OR CP 358774 OR CP-358774 OR CP358774)

#7 TS=(Gefitinib OR Iressa OR ZD1839 OR ZD-1839)

#8 #7 OR #6 OR #5 OR #4 OR #3 OR #2

#9 TS=(randomized controlled trials as topic OR randomized controlled trial OR clinical trials as topic OR clinical trial OR controlled clinical trial OR controlled clinical trials as topic OR random*)

#10 #9 AND #8 AND #1

**Appendix 5. ClinicalTrials.gov search strategy (N = 24)**

#1 Condition or disease: esophageal neoplasms

#2 Intervention/treatment: cetuximab OR panitumumab OR nimotuzumab OR erlotinib hydrochloride OR gefitinib

#3 Study Results: studies with results

#4 #1 AND #2 AND #3



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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 4-5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 1 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 6-7 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5-6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 7 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 7-8 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 7-8 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7-8 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 7-8 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 9 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 9-10 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 10-12 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 9-15 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 9-15 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 9-15 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 9-15 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 15-20 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 19-20 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 20 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 21 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment of oesophageal cancer: a systematic review and meta-analysis



| | |
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| Secondary Subject Heading: | Oncology, Pharmacology and therapeutics |
| Keywords: | ONCOLOGY, THERAPEUTICS, Oesophageal disease < GASTROENTEROLOGY |
| | |



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# Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment of oesophageal cancer: a systematic review and meta-analysis

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## Abstract

**Objectives:** Despite remarkable advances in the treatment of oesophageal cancer (OC), the role of anti-epidermal growth factor receptor (EGFR) agents in treating OC remains controversial. Herein, a systematic review and meta-analysis were conducted to elucidate the efficacy and safety of anti-EGFR agents in patients with OC.

**Design:** Meta-analysis of randomised controlled trials (RCTs) identified by searching the PubMed, EMBASE, Web of Science, ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine, China National Knowledge Infrastructure, and Wanfang Data Knowledge Service Platform databases from inception to December 2019. We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

**Setting:** RCTs from any country and healthcare setting.

**Participants:** Patients with OC.

**Interventions:** Combination therapy with anti-EGFR agents and conventional treatments versus conventional treatments alone in patients with OC.

**Primary and secondary outcome measures:** Overall survival (OS) and progression-free survival (PFS) were primary outcome measures, and objective response rate (ORR), disease control rate (DCR), and treatment toxicities were secondary outcome measures.

**Results:** In total, 25 RCTs comprising 3,406 patients with OC were included. Overall, anti-EGFR treatment significantly improved the OS [hazard ratio (HR): 0.81, 95% confidence interval (CI): 0.74–0.89, $P < 0.00001$], ORR [relative risk (RR): 1.33, 95% CI: 1.16–1.52, $P < 0.0001$), and DCR (RR: 1.22, 95% CI: 1.11–1.34, $P < 0.0001$) but not PFS (HR: 0.91, 95% CI:




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0.76–1.08, $P = 0.26$). Anti-EGFR treatment was significantly associated with higher incidences of myelosuppression, diarrhoea, acne-like rash, and hypomagnesaemia.

**Conclusions:** Overall, anti-EGFR agents have positive effects on OS, the ORR, and DCR in OC. However, considering the high incidence of adverse effects, such as myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia, careful monitoring of patients with OC is recommended during anti-EGFR treatment.

**Strengths and limitations of this study**

- This meta-analysis is the first attempt to summarise the evidence of the efficacy and safety of anti-EGFR agents in combination with conventional treatments versus conventional treatments alone for oesophageal cancer.
- Eight databases, a manual search and unpublished studies were retrieved to identify the eligible studies.
- Quality of the eligible studies was assessed using the Cochrane Handbook for Systematic Reviews and Meta-Analysis.
- This study has been registered a priori and followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.
- Publication, reporting, or selection bias cannot be ruled out because of the lack of access to individual patient data.



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## Introduction

Oesophageal cancer (OC) is the seventh most common cancer type and the sixth leading cause of cancer-related mortality, accounting for approximately 1 in 20 cancer-related deaths worldwide in 2018.[1] However, the incidences of OC and mortality rates vary considerably, as demonstrated by a two- to three-fold difference in statistical data between different geographical regions.[2] OC is common in Eastern and Southern Africa, but the highest incidence rates have been reported in East Asia, with those in Mongolia and China being among the top five worldwide.[1 2] These highest-risk areas are called the 'oesophageal cancer belt', with squamous cell cancers showing a dominance of 90%, while adenocarcinoma being predominant in the people of European descent.[1]

OC is diagnosed in later stages in 80% of the patients, and typically, few patients benefit from single-modality treatments, such as surgery [5-year overall survival (OS) of 15%–20%], chemotherapy (CT; 5-year OS of 15%–45%), and radiotherapy (RT; 5-year OS of 0%–15%).[3] Therefore, chemoradiotherapy (CRT; 5-year OS of 26%) has become a standard of care for OC. However, controlling locoregional recurrence or persistent disease remains a major challenge.[4] The use of higher radiation doses, combination of various CT regimens and RT, or other combination therapies (e.g. surgery and neoadjuvant CT) have failed to achieve any improvement.[5-8] Therefore, the addition of novel therapies to conventional single- or multimodality treatments is urgently needed.

The epidermal growth factor receptor (EGFR) signalling pathway is implicated in the proliferation, invasion, angiogenesis, and survival of cancer cells.[9] Moreover, overexpression



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of the EGFR in OC has been associated with a poor prognosis.[10] Previous studies have shown synergistic effects between EGFR inhibitors and conventional treatments.[11 12] Hence, there is a strong rationale for testing EGFR-targeting drugs to design an effective treatment strategy against OC. Tyrosine kinase inhibitors (TKIs) and monoclonal antibodies (mAbs) are two promising classes of anti-EGFR drugs. mAbs bind to the extracellular domain of the EGFR and compete with endogenous ligands to block the ligand-binding region and ligand-induced EGFR tyrosine kinase activation.[13] TKIs reversibly compete with ATP to bind to the intracellular catalytic domain of EGFR tyrosine kinase and inhibit EGFR autophosphorylation and downstream signalling.[13] Various anti-EGFR agents, such as cetuximab,[14 15] nimotuzumab,[16 17] panitumumab,[18] erlotinib,[19 20] and gefitinib,[21 22] are used for treating OC. However, the role of anti-EGFR agents in combination with conventional treatments is controversial, and toxicity associated with these agents is variable and uncertain.[14-17 19-22] Moreover, there is no clear evidence that combination of anti-EGFR agents and conventional treatments is superior to conventional treatments alone for OC. To address this research gap, we conducted a systematic review and meta-analysis of randomised controlled trials (RCTs) to evaluate the efficacy and safety of anti-EGFR agents in combination with conventional treatments versus conventional treatments alone in patients with OC.

## Methods

### Literature Search Strategy

A literature search of the following databases (from inception to December 2019) was performed by two authors to identify eligible studies: PubMed, EMBASE, Web of Science,



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ClinicalTrials.gov, the Cochrane Library, Chinese Biology Medicine, China National Knowledge Infrastructure, and the Wanfang Data Knowledge Service Platform. Furthermore, a manual search using the reference lists of all the retrieved articles and relevant journals was conducted to identify additional eligible studies. Unpublished studies were retrieved from the System for Information on Grey Literature database to reduce publication bias. Combinations of Medical Subject Headings/Emtree terms and free-text terms were used to search the above databases. The location of the primary OC site included typical endoscopic measurements of each region from the incisors (range: from 15 to 42 cm), starting from the upper oesophageal sphincter to the proximal 2 cm of the cardia.[23] The following main search terms were used: 'oesophageal neoplasms', 'epidermal growth factor receptor inhibitor', 'cetuximab', 'panitumumab', 'nimotuzumab', 'erlotinib', 'gefitinib', 'randomised controlled trial', and 'clinical trial'. The full search strategies utilised are presented in **Supplementary Data S1**.

Two authors independently screened the studies based on the selection criteria. Initially, each title and abstract were screened, and the full text of the eligible studies was retrieved. Any disagreements between the two authors during the selection process were settled by discussion and consensus with a third author.

**Inclusion and Exclusion Criteria**

Eligible studies were included if they met the following criteria: (1) patients with histologically confirmed OC; (2) anti-EGFR treatment versus conventional treatment (defined as any single- or multimodality treatment for OC, such as surgery, CT, RT, or any combination of these); (3) RCT; and (4) hazard ratios (HRs) with 95% confidence intervals (CIs) reported for OS and/or



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progression-free survival (PFS), or sufficient data provided to calculate these values.

Repeated studies, reviews, comments, letters, case reports, meeting abstracts, study protocols, or animal studies were excluded. Duplicate and incomplete data were also excluded.

**Data Extraction and Quality Assessment**

Data extraction and quality assessment were carried out by two independent authors, and discrepancies were resolved by discussion and consensus with the third author.

The extracted information for each eligible study included the name of the first author, publication year, country, study design, enrolment period, sample size, mean/median age of patients (range, years), treatment protocol, and major clinical outcomes [OS, PFS, objective response rate (ORR), disease control rate (DCR), and grade 3–5 toxicities].

The quality of each eligible study was assessed using the Cochrane Handbook for Systematic Reviews of Interventions version 6.0 (www.training.cochrane.org/handbook). Random sequence generation, allocation concealment, blinding, incomplete outcome data, and selective outcome reporting were examined to categorise each eligible study as having a low (A), moderate (B), or high (C) risk of bias.

**Statistical Analysis**

The primary outcomes were OS and PFS, expressed as the HR with a 95% CI, which were calculated from the number of observed deaths minus the number of estimated deaths and its variance.[24] OS was defined as the time from diagnosis to death or to the last follow-up, and PFS was defined as the time from diagnosis to the first documented recurrence. Both OS and PFS data were obtained from Kaplan–Meier curves using Engauge Digitizer version 10.8



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(http://sourceforge.net/projects/digitizer/). Secondary outcomes, including ORR, DCR, and haematological and nonhaematological toxicities, were expressed as relative risk (RR) ratios with 95% CIs using the Mantel–Haenszel method. ORR was defined as the rate of partial response (PR) and complete response (CR), and DCR was defined as the rate of PR, CR, and stabilisation.

Statistical heterogeneity across studies was evaluated using the chi-squared test, with a level of significance set at $P < 0.1$, and quantified with the $I^2$ statistic. The fixed-effects model was used if the heterogeneity test showed no statistical significance ($I^2 < 50\%$; $P > 0.1$); otherwise, the random-effects model was adopted. Publication bias was assessed by inspecting funnel plots for asymmetries and outliers. Sensitivity analysis was conducted to explore the robustness of the results by excluding certain studies to recalculate the pooled HR and RR estimates. Five studies were excluded because of small sample sizes or treatments with surgery and/or induction CT.[14 20 25-27] A subgroup analysis was performed based on the various treatment regimens.

All analyses were conducted using Review Manager version 5.3 (Cochrane Collaboration, Oxford, UK). A two-sided $P$-value < 0.05 was considered significant for all analyses, except heterogeneity tests. This study was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[28]

**Patient and Public Involvement**

Patients and/or the public were not involved in the design, conduct, reporting, or dissemination plans of this research.



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## Results

### Study Selection

A total of 1,374 records were identified by searching the databases, and no additional records were found using other sources. After removing duplicates, 1,298 records remained. Based on the screening of the titles, 952 records were excluded as not meeting the eligibility criteria. Next, the remaining 346 abstracts were further evaluated, of which 301 records were excluded as they were reviews (n = 114), non-RCTs (n = 108, including case reports, meeting abstracts, letters, editorials, and news), animal studies (n = 3), or studies on other cancers and drugs (n = 76) unrelated to this study. After screening the remaining 45 full-text articles, 20 records were excluded as they were non-RCTs (n = 5), ongoing studies (n = 2), studies on other cancers and drugs (n = 8), or studies without outcomes (n = 5). Finally, 25 RCTs were included in the quantitative synthesis for this meta-analysis. The flow diagram of the selection process is shown in **Figure 1**.

### Study Characteristics

The eligible studies were published between 2009 and 2018, and a total of 3,406 patients were included in this meta-analysis. The sample sizes of the studies ranged from 16 to 225 patients. In total, 1,699 patients were in the anti-EGFR arm and 1,707 patients were in the non-anti-EGFR arm. The age of the patients ranged from 26 to 87 years. Seventeen studies were conducted in China,[19-21 25-27 29-39] three in the UK,[22 40 41] and one each in the USA,[15] Germany,[11] Switzerland,[14] Cuba,[17] and Brazil.[16] Two studies from the UK were performed on the same group of patients, and the study periods partly overlapped.[40 41] However, each study analysed



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different outcomes. In the anti-EGFR arms, the patients were administered anti-EGFR agents, including cetuximab, nimotuzumab, erlotinib, or gefitinib, in combination with surgery, CT, RT, or CRT. In the non-anti-EGFR arms, the patients underwent only surgery, CT, RT, or CRT. The basic characteristics of the 25 selected studies are presented in **Table S1**.

**Quality Assessment**

The quality assessment data for each study are shown in **Table 1**. Briefly, among the 25 RCTs, one study was rated as level A,[22] seventeen studies as level B,[15 20 21 25-27 29-39] and seven studies as level C.[1 14 16 17 19 40 41] Eleven of the included studies were assessed to have a low risk of random sequence generation,[14-16 22 29 31 37-41] and six studies used allocation concealment.[11 14 16 22 40 41] Only one study reported the blinding of participants and outcome assessment.[22] All the studies were assessed to have a low risk of bias because they addressed incomplete outcome data and selective reporting. No other source of bias was identified in these studies.

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**Table 1** Quality assessment of each study

| Authors (year) | Random sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Other bias | Rank |
|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Unclear | Yes | No | No | Yes | Yes | Yes | C |
| Crosby et al. 2013[41] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014a[25] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Crosby et al. 2017[40] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Suntharalingam et al. 2017[15] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Lu et al. 2017[37] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ruhstaller et al. 2018[14] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Ramos-Suzarte et al. 2012[17] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Xu et al. 2013[36] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Shi et al. 2014[35] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Sun et al. 2014[26] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2014[27] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Luo et al. 2015[33] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016b[30] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Fu et al. 2017[29] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Castro et al. 2018[16] | Yes | Yes | No | No | Yes | Yes | Yes | C |
| Chen et al. 2014b[20] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhang et al. 2014[34] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Ye et al. 2015[32] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhou et al. 2015[38] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Chen et al. 2016a[39] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Zhao et al. 2016[31] | Yes | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Wu et al. 2018[19] | Unclear | Unclear | No | No | Yes | Yes | Yes | C |
| Zeng et al. 2014[21] | Unclear | Unclear | Unclear | Unclear | Yes | Yes | Yes | B |
| Dutton et al. 2014[22] | Yes | Yes | Yes | Yes | Yes | Yes | Yes | A |



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**Efficacy Analysis**

Fifteen studies[11 14-16 19 22 26 30 32 33 35 37-40] were included in the analysis of OS, which showed significant improvement in the anti-EGFR arm (HR: 0.81, 95% CI: 0.74–0.89, $P < 0.00001$; $I^2 = 40\%$, $P = 0.05$) (**Figure 2**). A subgroup analysis indicated that the administration of anti-EGFR agents in combination with CT, CRT, or RT improved the OS of patients with OC. Both anti-EGFR mAbs and TKIs, especially nimotuzumab and erlotinib, significantly improved the OS (**Figure S1**).

Six studies[11 14 22 26 36 40] were included in the evaluation of PFS, which demonstrated no significant difference between the anti-EGFR and control arms (HR: 0.91, 95% CI: 0.76–1.08, $P = 0.26$; $I^2 = 53\%$, $P = 0.06$) (**Figure 3**). A longer PFS was observed in patients treated with nimotuzumab, but no significant differences were found for other treatments (**Figure S2**).

Analysis of 16 studies[11 17 20 21 26 27 29-31 33-39] that evaluated the ORR indicated that the ORR significantly increased in the anti-EGFR arm (RR: 1.33, 95% CI: 1.16–1.52, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S3**). A higher ORR was reported in patients with OC who received anti-EGFR agents in combination with CT, CRT, surgery, or RT. Both anti-EGFR mAbs and TKIs, especially nimotuzumab and erlotinib, significantly increased the ORR (**Figure S4**).

Fifteen studies[11 17 20 21 26 29-38] that analysed the DCR showed that the DCR was significantly improved in the anti-EGFR arm (RR: 1.22, 95% CI: 1.11–1.34, $P < 0.0001$; $I^2 = 70\%$, $P < 0.0001$) (**Figure S5**). Combination treatment with anti-EGFR agents and CT or RT resulted in a higher DCR. Anti-EGFR mAbs and TKIs, particularly cetuximab and erlotinib, decreased the risk of DCR (**Figure S6**).



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**Safety Analysis**

Six types of haematological toxicities were analysed in this meta-analysis, including anaemia, leucopenia, neutropenia, thrombocytopenia, lymphopenia, and myelosuppression. The results of the pooled analysis demonstrated that there was no significant association between anti-EGFR agents and the increased risk of haematological toxicities, except for myelosuppression (RR: 0.42, 95% CI: 0.21–0.86, $P = 0.02$) (**Figure S7**). A subgroup analysis revealed that the addition of anti-EGFR agents to CT was associated with an increased risk of myelosuppression compared with that of CT alone, whereas no significant difference was found in the case of CRT. Additionally, among patients who received anti-EGFR mAbs, only those treated with nimotuzumab displayed a significant difference in the incidence of myelosuppression (RR: 0.39, 95% CI: 0.18–0.83, $P = 0.02$) (**Figure S8**).

Eight nonhaematological toxicities were reported, namely cardiac toxicity, gastrointestinal toxicity, dermatologic toxicity, pulmonary toxicity, renal toxicity, constitutional symptoms, infection, and laboratory findings. The results indicated statistically significant differences for dermatologic toxicity (RR: 9.83, 95% CI: 3.54–27.28, $P < 0.0001$) and laboratory findings (RR: 1.59, 95% CI: 1.05–2.42, $P = 0.03$) in the anti-EGFR arm, whereas no significant differences were observed for other nonhaematological toxicities (**Table S2**). A subgroup analysis confirmed that patients who received anti-EGFR treatment had higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia than those in the control group (**Table S2**). Higher incidences of diarrhoea, acne-like rashes, and hypomagnesaemia were observed in patients who received anti-EGFR agents in combination with CRT than in those who received



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CRT alone (**Figures S9–S11**). Increased risks of diarrhoea, acne-like rashes, and hypomagnesaemia were found among patients who received anti-EGFR mAbs, especially cetuximab (**Figures S9–S11**). Additionally, anti-EGFR TKIs, particularly erlotinib, increased the incidence of acne-like rash (**Figure S10**), whereas gefitinib was associated with an increased risk of diarrhoea (**Figure S9**).

**Sensitivity Analysis**

For sensitivity analysis, studies by Ruhstaller et al. 2018[14], Chen et al. 2014a,[25] Chen et al. 2014b,[20] Sun et al. 2014,[26] and Wu et al. 2014[27] were excluded. No significant changes in the pooled estimates of OS, PFS, ORR, and DCR indicated that the results were statistically stable and reliable (**Table S3**). No other sensitivity analysis had any effect on the results.

**Publication Bias**

A funnel plot of the included studies, which was used to assess the possibility of publication bias, showed an asymmetry, thereby indicating that the publication bias existed (**Figures 4, S12, and S13**).

## Discussion

OC, a malignant digestive system neoplasm with a poor prognosis, is regarded as a threat to public health. Currently, anti-EGFR agents are commonly used as valuable therapeutic options for OC. However, various studies have published contradictory results regarding these agents. Therefore, we conducted this meta-analysis of RCTs to evaluate the efficacy and safety of combination therapy with anti-EGFR agents and conventional treatments compared with those



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of conventional treatments alone. To the best of our knowledge, this is the first meta-analysis to evaluate the benefits and toxicities of anti-EGFR and non-anti-EGFR treatment strategies for OC. The findings of this meta-analysis indicated that anti-EGFR-including treatments might be more effective than non-anti-EGFR treatments in terms of prolonging the OS and increasing the ORR and DCR in patients with OC. However, anti-EGFR agents may be associated with higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia.

The addition of anti-EGFR agents to CT, CRT, or RT regimens significantly improved the OS of patients with OC. Because CRT is the standard treatment approach for OC, the combination of anti-EGFR agents with CT or RT was not considered in this meta-analysis, owing to limited availability of studies. According to this meta-analysis, the addition of anti-EGFR agents to CRT may be an important strategy to improve clinical outcomes in patients with OC. Most importantly, anti-EGFR mAbs and TKIs, especially nimotuzumab and erlotinib, significantly improved the OS. The clinical benefits of nimotuzumab may be explained by its ability to directly inhibit the EGFR signalling pathway, which results in an antiproliferative effect on tumour cells.[42] Furthermore, nimotuzumab enhances the sensitivity of tumour cells to RT and CT,[43] which may explain the positive results obtained in this study. The combination of erlotinib with conventional treatments for OC is still controversial because of the limited number of RCTs that tested this TKI. TKIs have been demonstrated to induce G1/S phase cell-cycle arrest in various cancers. Studies have also reported that TKIs inhibit the growth of OC cells,[44-47] which might explain the significant improvement of OS in the present study.



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However, cetuximab did not significantly increase the OS, which was consistent with the data from a previous report.[48] It has been suggested that the reduced efficacy of cetuximab in combination with CRT is due to proinflammatory and antitumour proliferative effects of this mAb.[49] Moreover, cetuximab may limit the free radical-induced DNA damage by platinum drugs,[50] which could also explain the nonsignificant outcomes in this meta-analysis. For gefitinib, no improvement in OS was observed either, which was consistent with the results of a previous meta-analysis.[51] Currently, there is no evidence that gefitinib can suppress EGFR, which indicates its limited clinical efficacy.

In this meta-analysis, there was no significant improvement in the PFS outcome in the anti-EGFR arm, which was consistent with the results of a previous meta-analysis.[52] A longer PFS was only confirmed for nimotuzumab combination therapy, as also reported by Jing et al.[53] The biological characteristics of nimotuzumab differ from those of other anti-EGFR agents, which might explain its different clinical effects.

With regard to the ORR and DCR, significant differences were noted between the anti-EGFR and non-anti-EGFR arms. Both anti-EGFR mAbs and TKIs, especially nimotuzumab, cetuximab, and erlotinib, caused significant differences in the ORR and DCR for OC. However, previous studies have reported different ORRs and DCRs.[54]

Regarding the ORR, our results were consistent with those of some studies that indicated a significant advantage of combining anti-EGFR agents with CT, CRT, or RT.[55 56] Furthermore, an ORR of 47.8% has been previously reported in the nimotuzumab group compared to 15.4% in the control group.[17] Additionally, anti-EGFR TKIs have been found to be superior to



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conventional therapy in terms of the ORR.[57] Our meta-analysis indicated that erlotinib significantly increased the ORR. However, no combined data on the ORR have been reported for OC. It remains unknown whether erlotinib therapy is better than other treatment strategies owing to insufficient and conflicting homogeneity studies.

The DCR was greatly improved when anti-EGFR agents were combined with CT or RT; however, this result could not be confirmed as only few studies reported this outcome. Erlotinib and cetuximab demonstrated notable beneficial effects on the DCR in OC, which was consistent with the results of previous studies.[48 57] Most meta-analyses have only reported results for mAb and TKI therapy, without specifying the individual drugs; therefore, it was impossible to verify the effectiveness of each drug.

This meta-analysis confirmed that patients with OC had higher incidences of myelosuppression, diarrhoea, acne-like rashes, and hypomagnesaemia in the anti-EGFR than in the non-anti-EGFR arm, which was consistent with the data of previous studies.[58 59] An increased risk of myelosuppression was confirmed for combination therapy with anti-EGFR agents and CT. In particular, this meta-analysis showed that nimotuzumab increased the risk of myelosuppression. This finding might be due to the ability of nimotuzumab to enhance the sensitivity of tumour cells to CT.[43] However, these findings should be interpreted with caution because of the lack of sufficient high-quality, multicentre, large-sample RCTs.

Nonhaematological toxicities, such as diarrhoea, acne-like rashes, and hypomagnesaemia, occurred more often in patients with OC who received anti-EGFR agents with CRT than in those who received CRT alone. Furthermore, the combination of cetuximab or gefitinib with



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conventional treatment resulted in a higher risk of diarrhoea than conventional treatment alone. The underlying mechanisms of diarrhoea-causing effects of anti-EGFR mAbs and TKIs remain poorly understood and might be due to multiple factors, including altered gut motility, colonic crypt damage, changes to the intestinal microbiota, and altered colonic transport.[60] A plausible explanation could be the inhibition of the EGFR pathway in enterocytes.[61] Thus, a better understanding of the underlying mechanisms may help in the identification of new approaches to the prophylaxis, risk reduction, and development of more effective management of diarrhoea.

Acne-like rash is the most common side effect of anti-EGFR agents,[62] and this meta-analysis confirmed that anti-EGFR mAbs and TKIs, such as cetuximab and erlotinib, could induce acne-like rashes. The mechanism leading to skin toxicity is not well known but undoubtedly involves the activation of associated signalling pathways, particularly the RAS/RAF/MEK/ERK pathway, which affects cell cycle regulation, including the proliferation and differentiation of epidermal cells.[63] Skin toxicity often leads to physical and psychological discomfort, which may result in treatment interruptions, dose modifications, or even nonadherence to anticancer therapies.[62] Therefore, it is imperative to understand skin toxicities and develop proper approaches to control or relieve their adverse effects in patients.

An increased incidence of hypomagnesaemia was confirmed for cetuximab, in accordance with the data of another report.[48] Hypomagnesaemia, a frequent side effect of cetuximab, is a serious electrolyte abnormality associated with an increased risk of seizures and cardiac events.[64] The exact mechanism of cetuximab-induced hypomagnesaemia has not been well elucidated. Some studies have indicated that EGFR is strongly expressed in the kidney,



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particularly in the ascending limb of the loop of Henle, where 70% of the filtered magnesium is reabsorbed, and cetuximab might interfere with the magnesium transport by blocking EGFR.[64 65] However, hypomagnesaemia is often overlooked because of its nonspecific symptoms, which are attributed to underlying tumours or previous CT regimens.[66] Consequently, serum magnesium should be monitored during cetuximab-based therapy.

Several limitations should be considered when interpreting the results of this meta-analysis. First, we lacked access to individual patient data, which might have caused publication, reporting, or selection bias. Second, because of the insufficient number of studies conducted, different anti-EGFR agents and conventional treatments were included in this meta-analysis, which increased clinical heterogeneity of the data. Subsequently, subgroup and sensitivity analyses were performed to improve the reliability of our results. Third, the methods were unclear, especially in the domain of 'allocation concealment and blinding', which might have affected the overall methodological quality of the included studies. High methodological quality of RCTs is needed for further research. Finally, considering all these limitations, the results of this meta-analysis should be interpreted with caution, and further research is necessary to verify the effects of anti-EGFR therapy in OC.

In conclusion, this meta-analysis indicated that the addition of anti-EGFR agents to multimodal therapy was beneficial for patients with OC in terms of their OS, ORR, and DCR. Therefore, anti-EGFR agents may be recommended as a treatment option for OC to improve treatment outcomes. The results of this study may help establish an optimum therapeutic regimen for OC. However, considering the higher incidences of myelosuppression, diarrhoea,



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acne-like rashes, and hypomagnesaemia when using the combination of anti-EGFR agents and multimodal therapy, careful monitoring of patients and effective management of their condition are recommended. Further high-quality, large-scale, multicentre studies are needed to confirm the efficacy and safety of anti-EGFR agents in patients with OC.

**Contributors** LJZ and RY designed the study; LJZ, YLS, and NJ performed the literature search, extracted the data, and evaluated the quality of included studies; LJZ, YQH, BD, WL, YZH, YC, and HBL conducted statistical analyses; LJZ, YLS, and NJ were involved in the interpretation of the results; LJZ, YLS, NJ, and RY were responsible for the writing and critical revision of the manuscript. All authors read and approved the final manuscript.

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## Figure legends

**Figure 1** Flow diagram of the selection process of relevant studies

**Figure 2** Forest plot of the overall survival (OS) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

**Figure 3** Forest plot of the progression-free survival (PFS) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

**Figure 4** Funnel plot of publication bias

**Figure S1** Subgroup analysis of overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S2** Subgroup analysis of progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S3** Forest plot of the objective response rate (ORR) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S5** Forest plot of the disease control rate (DCR) in patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment



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regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S7** Forest plot of grade 3–5 haematological toxicities in patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

**Figure S12** Funnel plot of publication bias for the objective response rate (ORR)

**Figure S13** Funnel plot of publication bias for the disease control rate (DCR)

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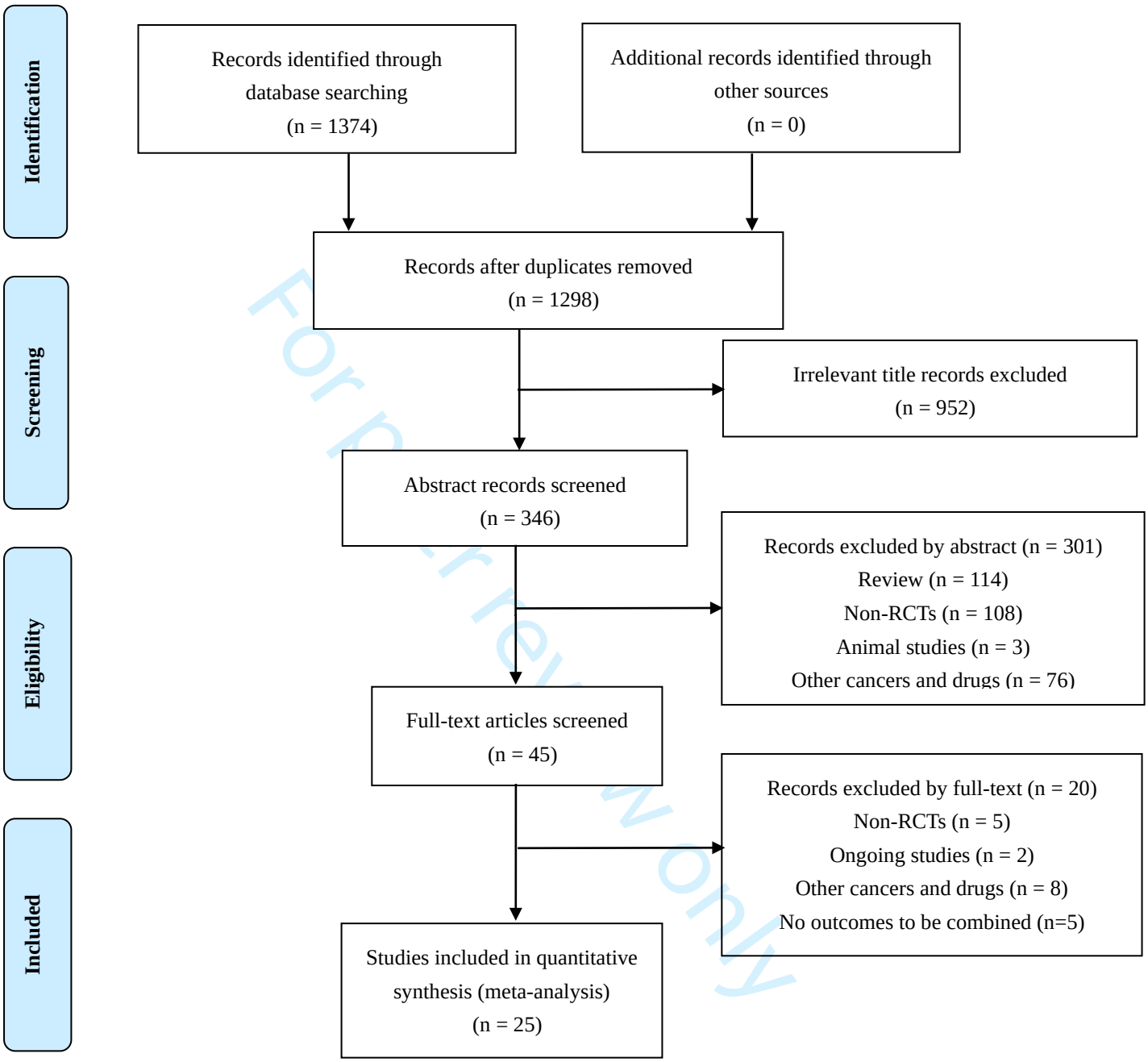



**Figure 1** Flow diagram of the selection process for relevant studies

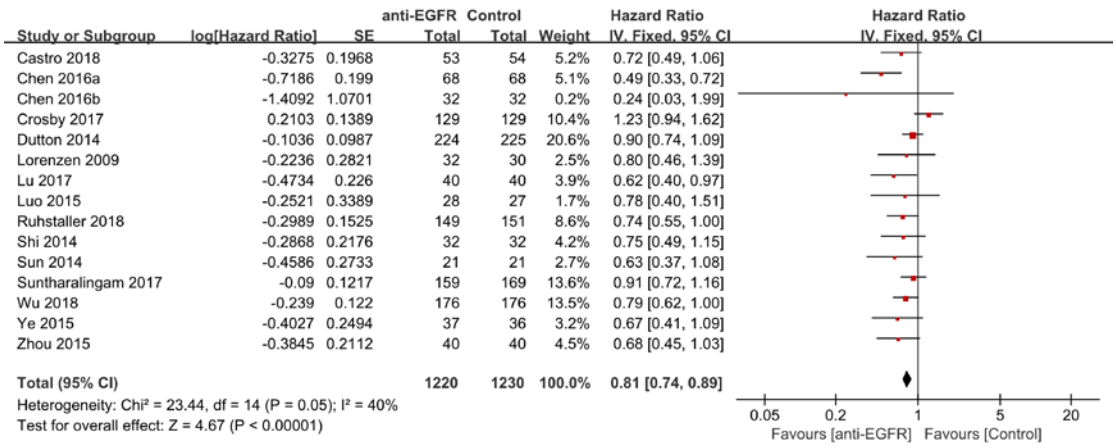



**Figure 2** Forest plot of the overall survival (OS) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

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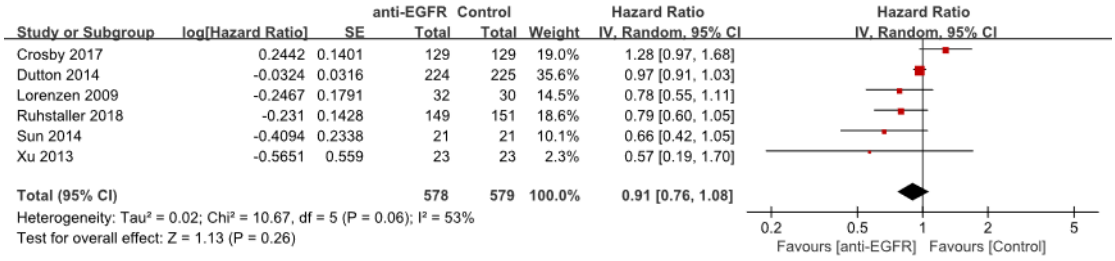



**Figure 3** Forest plot of the progression-free survival (PFS) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

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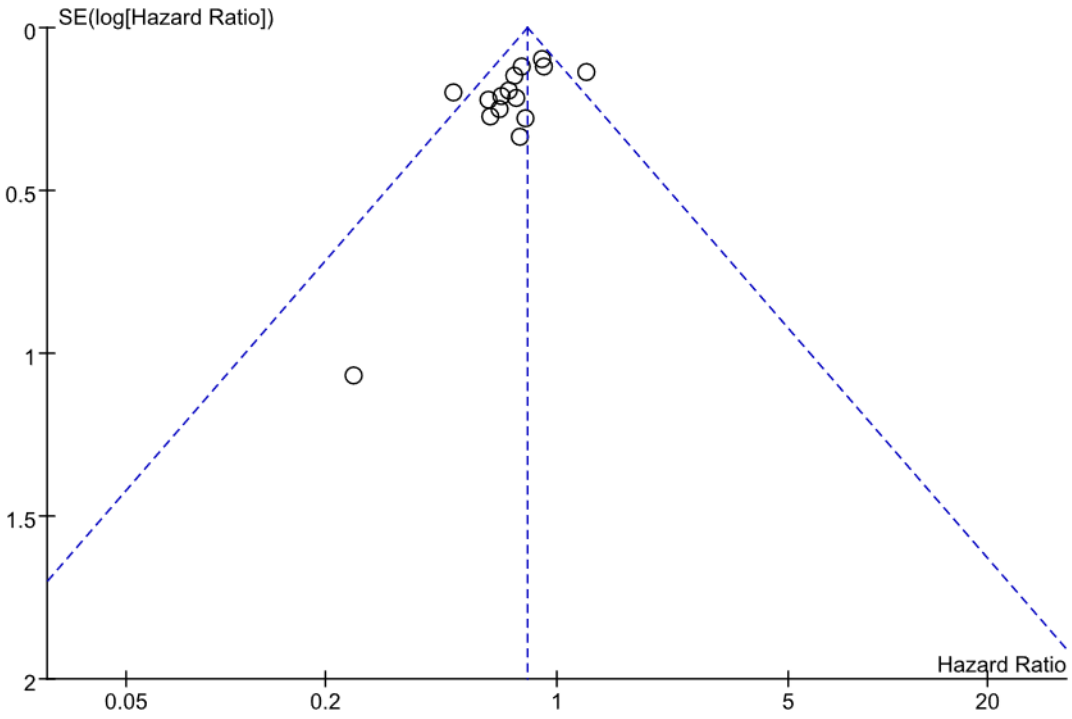



**Figure 4** Funnel plot of publication bias

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**Table S1** Basic characteristics of included studies in the meta-analysis

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Lorenzen et al. 2009[11] | Germany | RCT Multicenter | 2004.12-2006.12 | 32 | 30 | 61 (40-76) | 62 (40-74) | Cetuximab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | OS, PFS, ORR, DCR, Toxicities |
| Crosby et al. 2013[41] | UK | RCT Multicenter | 2008.02-2012.02 | 129 | 129 | 66.9 (44.9-84.1) | 66.6 (35.7-81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, Toxicities |
| Chen et al. 2014a[25] | China | RCT Single-institution | 2011.07-2012.01 | 20 | 20 | 57.3 (44-73) | | Surgery+Cetuximab+CT (Cisplatin+5-fluorouracil） | Surgery+CT (Cisplatin+5-fluorouracil) | OS, PFS, Toxicities |
| Crosby et al. 2017[40] | UK | RCT Multicenter | 2008.02-2012.02 | 129 | 129 | 66.9 (44.9-84.1) | 66.6 (35.7-81.9) | Cetuximab+CRT (Cisplatin+Capecitabine+RT) | CRT (Cisplatin+Capecitabine+RT) | OS, PFS, Toxicities |
| Suntharalingam et al. 2017[15] | USA | RCT Multicenter | 2008.06-2013.02 | 159 | 169 | 65 (39-87) | 63 (32-85) | Cetuximab+CRT (Cisplatin+Paclitaxel+RT) | CRT (Cisplatin+Paclitaxel+RT) | OS, Toxicities |
| Lu et al. 2017[37] | China | RCT Single-institution | 2013.01-2015.06 | 40 | 40 | 61.2 (45-80) | 61.3 (45-80) | Cetuximab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | OS, ORR, DCR, Toxicities |
| Ruhstaller et al. 2018[14] | Switzerland | RCT Multicenter | 2010.05-2016.10 | 149 | 151 | 61 (56-68) | 61 (53-68) | Cetuximab+IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+ RT) +Surgery | IC (Cisplatin+Docetaxel) +CRT (Cisplatin+Docetaxel+RT) + Surgery | OS, PFS, Toxicities |
| Ramos-Suzarte et al. 2012[17] | Cuba | RCT Multicenter | 2005.12-2008.12 | 33 | 30 | 58.5 | 58.9 | Nimotuzumab+CRT (Cisplatin+5-fluorouracil+RT) | CRT (Cisplatin+5-fluorouracil+RT) | ORR, DCR, Toxicities |
| Xu et al. 2013[36] | China | RCT Single-institution | 2009.01-2012.12 | 23 | 23 | 63 (40-74) | 62 (41-75) | Nimotuzumab+CT (Cisplatin+5-fluorouracil) | CT (Cisplatin+5-fluorouracil) | PFS, ORR, DCR, Toxicities |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Shi et al. 2014[35] | China | RCT Single-institution | 2009.10-2010.10 | 32 | 32 | 57 (42-68) | | Nimotuzumab+RT | RT | OS, ORR, DCR |
| Sun et al. 2014[26] | China | RCT Single-institution | 2008.01-2009.07 | 21 | 21 | 54 (36-78) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, PFS, ORR, DCR, Toxicities |
| Wu et al. 2014[27] | China | RCT Single-institution | 2009.10-2011.09 | 16 | 16 | 55 (38-68) | | Nimotuzumab+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | ORR |
| Luo et al. 2015[33] | China | RCT Single-institution | NR | 28 | 27 | 62 (32-68) | 63 (31-69) | Nimotuzumab+CRT (Docetaxel+RT] | CRT (Docetaxel+RT] | OS, ORR, DCR, Toxicities |
| Chen et al. 2016b[30] | China | RCT Single-institution | 2012.01-2013.01 | 32 | 32 | 53.43 (39-68) | 54.52 (41-68) | Nimotuzumab+CRT (Nadaplatin+RT] | CRT (Nadaplatin+RT] | OS, ORR, DCR |
| Fu et al. 2017[29] | China | RCT Single-institution | 2013.12-2016.12 | 40 | 40 | 60.55 (31-76) | 61.38 (31-75) | Nimotuzumab+CRT (Docetaxel+Cisplatin++RT) | CRT (Docetaxel+Cisplatin++RT) | ORR, DCR |
| Castro et al. 2018[16] | Brazil | RCT Multicenter | 2009.02-2011.08 | 53 | 54 | 60.0 (40-75) | 58.5 (36-82) | Nimotuzumab+CRT (Cisplatin+Fluorouracil+RT) | CRT (Cisplatin+Fluorouracil+RT) | OS, Toxicities |
| Chen et al. 2014b[20] | China | RCT Single-institution | 2010.01-2012.01 | 60 | 60 | 58.7 (49-75) | | Surgery+Erlotinib+CT (Cisplatin+5-fluorouracil) | Surgery+CT (Cisplatin+5-fluorouracil) | CR, ORR, DCR |
| Zhang et al. 2014[34] | China | RCT Single-institution | 2009.10-2012.10 | 42 | 42 | 59.1 (49-74) | 58.4 (47-72) | Erlotinib+RT | RT | CR, ORR, DCR |
| Ye et al. 2015[32] | China | RCT Single-institution | 2011.02-2012.04 | 37 | 36 | 53.8 (NR) | 55.2 (NR) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, DCR |

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**Table S1** Continued

| Study (year) | Country | Study design | Enrollment period | Sample Size (anti-EGFR/Control) | | Mean/Median age (range, years) (anti-EGFR/Control) | | Treatment Protocol (anti-EGFR/Control) | | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|
| Zhou et al. 2015[38] | China | RCT Single-institution | 2009.05-2011.11 | 40 | 40 | 55 (46-70) | 54 (45-68) | Erlotinib+RT | CT+RT (Docetaxel+Cisplatin)+RT | OS, ORR, DCR |
| Chen et al. 2016a[39] | China | RCT Single-institution | 2012.11-2014.11 | 68 | 68 | 47.3 (28-76) | 47.9 (26-78) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, ORR |
| Zhao et al. 2016[31] | China | RCT Single-institution | 2013.08-2014.04 | 38 | 38 | 64.5 (61-83) | 63.4 (59-81) | Erlotinib+RT | RT | CR, ORR, DCR |
| Wu et al. 2018[19] | China | RCT Multicenter | 2007.12-2015.06 | 176 | 176 | 61 (35-70) | 61.5 (40-70) | Erlotinib+CRT (Paclitaxel+Cisplatin+RT) | CRT (Paclitaxel+Cisplatin+RT) | OS, Toxicities |
| Zeng et al. 2014[21] | China | RCT Single-institution | 2018.03-2010.08 | 26 | 26 | NR (40-80) | | Gefitinib+CT (Oxaliplatin+5-fluorouracil) | CT (Oxaliplatin+5-fluorouracil) | ORR, DCR |
| Dutton et al. 2014[22] | UK | RCT Multicenter | 2009.03-2011.11 | 224 | 225 | 64.7 (58.0-70.1) | 64.9 (58.2-70.7) | Gefitinib+CRT (NR) | Placebo+CRT (NR) | OS, PFS, Toxicities |

*RCT randomized controlled trial, NR Not reported, OS Overall survival, PFS Progression-free survival, ORR Objective response rate, DCR Disease control rate, RT Radiotherapy, CRT Chemoradiotherapy, IC Induction chemotherapy, CT Chemotherapy*

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**Table S2** Non-haematological toxicities of grade 3–5

| Toxicities | Subgroups | No. of studies | Sample Size | | RR [95% CI] | *P* | Heterogeneity | |
|---|---|---|---|---|---|---|---|---|
| | | | anti-EGFR (events/total) | Control (events/total) | | | $I^2$ | *P* |
| Cardiac toxicity** | ALL | 4 | 19/664 | 9/676 | 2.13 [0.97, 4.67] | 0.06 | 0% | 0.72 |
| Gastrointestinal toxicity | ALL | 28 | 359/2927 | 330/2990 | 1.11 [0.97, 1.28] | 0.13 | 16% | 0.23 |
| | Nausea and Vomiting | 6 | 74/558 | 81/565 | 0.94 [0.70, 1.26] | 0.67 | 43% | 0.12 |
| | Diarrhoea | 7 | 55/782 | 24/790 | 2.26 [1.42, 3.60] | 0.0005* | 30% | 0.21 |
| | Dehydration | 2 | 37/206 | 28/220 | 1.42 [0.91, 2.23] | 0.12 | 40% | 0.20 |
| | Dysphagia | 4 | 100/493 | 104/505 | 0.97 [0.76, 1.24] | 0.83 | 0% | 0.79 |
| | Oesophagitis | 5 | 64/520 | 63/531 | 1.03 [0.75, 1.42] | 0.84 | 0% | 0.70 |
| | Anorexia | 4 | 29/368 | 30/379 | 1.00 [0.61, 1.63] | 1.00 | 0% | 0.99 |
| Dermatologic toxicity | Acne-like rash | 6 | 37/749 | 2/760 | 9.83 [3.54, 27.28] | <0.0001* | 0% | 0.90 |
| Pulmonary toxicity | Dyspnoea | 3 | 10/335 | 6/349 | 1.72 [0.63, 4.68] | 0.29 | 26% | 0.25 |
| Renal toxicity | Renal insufficiency | 4 | 7/588 | 6/601 | 1.17 [0.41, 3.34] | 0.77 | 0% | 0.54 |
| Constitutional symptoms | ALL | 9 | 120/1084 | 104/1109 | 1.17 [0.92, 1.49] | 0.20 | 0% | 0.67 |
| | Fatigue | 6 | 109/749 | 93/760 | 1.19 [0.92, 1.53] | 0.19 | 0% | 0.51 |
| | Weight loss | 3 | 11/335 | 11/349 | 1.05 [0.47, 2.35] | 0.90 | 0% | 0.47 |
| Infection | ALL | 9 | 15/817 | 7/839 | 2.21 [0.91, 5.36] | 0.08 | 0% | 0.97 |
| | Fever | 4 | 2/271 | 1/280 | 2.17 [0.20, 23.69] | 0.53 | Not applicable | Not applicable |
| | Pneumonitis | 3 | 6/364 | 2/376 | 3.14 [0.64, 15.34] | 0.16 | 0% | 0.70 |
| | Stomatitis | 2 | 7/182 | 4/183 | 1.77 [0.53, 5.92] | 0.36 | 0% | 0.83 |
| Laboratory findings | ALL | 9 | 52/1005 | 34/1047 | 1.59 [1.05, 2.42] | 0.03* | 0% | 0.70 |
| | Hypomagnesaemia | 3 | 15/335 | 6/349 | 2.61 [1.02, 6.68] | 0.04* | 5% | 0.31 |
| | Hypokalaemia | 3 | 20/335 | 19/349 | 1.10 [0.60, 2.01] | 0.75 | 0% | 0.70 |
| | Hyponatraemia | 3 | 17/335 | 9/349 | 1.97 [0.91, 4.25] | 0.08 | 0% | 0.80 |

*$P < 0.05$, **including cardiac disorder/ischaemia/infarction*



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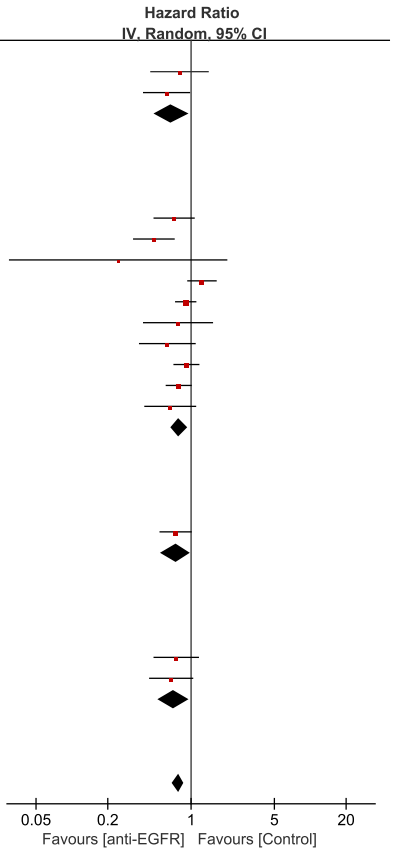
**Table S3** Sensitivity analysis by excluding the Ruhstaller et al. 2018[14], Chen et al. 2014a[25], Chen et al. 2014b[20], Sun et al. 2014[26] and Wu et al. 2014[27]

| Outcome | No. of studies | Ruhstaller et al. 2018[14] | | Chen et al. 2014a[25] | | Chen et al. 2014b[20] | | Sun et al. 2014[26] | | Wu et al. 2014[27] | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* | HR or RR [95% CI] | *P* |
| OS | 15 | 0.82 [0.75, 0.90] | <0.0001 | NR | NR | NR | NR | 0.82 [0.75, 0.89] | <0.00001 | NR | NR |
| PFS | 6 | 0.93 [0.76, 1.14] | 0.48 | NR | NR | NR | NR | 0.94 [0.79, 1.12] | 0.48 | NR | NR |
| ORR | 16 | NR | NR | NR | NR | 1.33 [1.14, 1.54] | 0.0002 | 1.32 [1.14, 1.52] | 0.0002 | 1.34 [1.19, 1.51] | <0.00001 |
| DCR | 15 | NR | NR | NR | NR | 1.24 [1.11, 1.38] | <0.0001 | 1.23 [1.13, 1.34] | <0.00001 | NR | NR |

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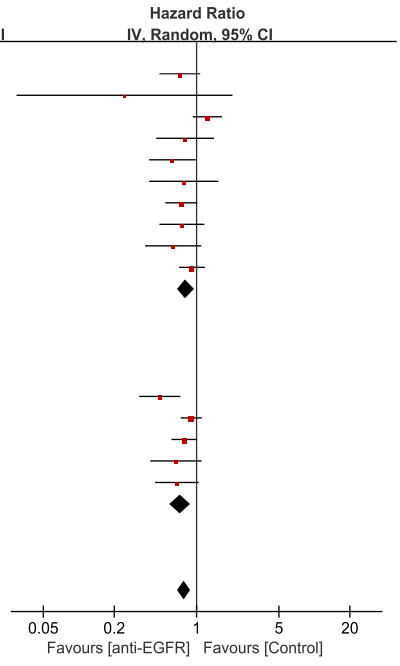
**A**

| Study or Subgroup | log[Hazard Ratio] | SE | anti-EGFR Total | Control Total | Weight | Hazard Ratio IV, Random, 95% CI |
|---|---|---|---|---|---|---|
| **2.1.1 EGFR+CT** | | | | | | |
| Lorenzen 2009 | -0.2236 | 0.2821 | 32 | 30 | 3.9% | 0.80 [0.46, 1.39] |
| Lu 2017 | -0.4734 | 0.226 | 40 | 40 | 5.5% | 0.62 [0.40, 0.97] |
| **Subtotal (95% CI)** | | | **72** | **70** | **9.4%** | **0.69 [0.49, 0.97]** |
| Heterogeneity: Tau² = 0.00; Chi² = 0.48, df = 1 (P = 0.49); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.13 (P = 0.03) | | | | | | |
| **2.1.2 EGFR+CRT** | | | | | | |
| Castro 2018 | -0.3275 | 0.1968 | 53 | 54 | 6.6% | 0.72 [0.49, 1.06] |
| Chen 2016a | -0.7186 | 0.199 | 68 | 68 | 6.5% | 0.49 [0.33, 0.72] |
| Chen 2016b | -1.4092 | 1.0701 | 32 | 32 | 0.3% | 0.24 [0.03, 1.99] |
| Crosby 2017 | 0.2103 | 0.1389 | 129 | 129 | 9.8% | 1.23 [0.94, 1.62] |
| Dutton 2014 | -0.1036 | 0.0987 | 224 | 225 | 12.8% | 0.90 [0.74, 1.09] |
| Luo 2015 | -0.2521 | 0.3389 | 28 | 27 | 2.9% | 0.78 [0.40, 1.51] |
| Sun 2014 | -0.4586 | 0.2733 | 21 | 21 | 4.1% | 0.63 [0.37, 1.08] |
| Suntharalingam 2017 | -0.09 | 0.1217 | 159 | 169 | 11.0% | 0.91 [0.72, 1.16] |
| Wu 2018 | -0.239 | 0.122 | 176 | 176 | 11.0% | 0.79 [0.62, 1.00] |
| Ye 2015 | -0.4027 | 0.2494 | 37 | 36 | 4.8% | 0.67 [0.41, 1.09] |
| **Subtotal (95% CI)** | | | **927** | **937** | **69.8%** | **0.80 [0.67, 0.95]** |
| Heterogeneity: Tau² = 0.04; Chi² = 20.33, df = 9 (P = 0.02); I² = 56% | | | | | | |
| Test for overall effect: Z = 2.60 (P = 0.009) | | | | | | |
| **2.1.3 EGFR+IC+CRT+S** | | | | | | |
| Ruhstaller 2018 | -0.2989 | 0.1525 | 149 | 151 | 8.9% | 0.74 [0.55, 1.00] |
| **Subtotal (95% CI)** | | | **149** | **151** | **8.9%** | **0.74 [0.55, 1.00]** |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.96 (P = 0.05) | | | | | | |
| **2.1.5 EGFR+RT** | | | | | | |
| Shi 2014 | -0.2868 | 0.2176 | 32 | 32 | 5.8% | 0.75 [0.49, 1.15] |
| Zhou 2015 | -0.3845 | 0.2112 | 40 | 40 | 6.0% | 0.68 [0.45, 1.03] |
| **Subtotal (95% CI)** | | | **72** | **72** | **11.8%** | **0.71 [0.53, 0.96]** |
| Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 1 (P = 0.75); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.22 (P = 0.03) | | | | | | |
| **Total (95% CI)** | | | **1220** | **1230** | **100.0%** | **0.78 [0.69, 0.88]** |
| Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40% | | | | | | |
| Test for overall effect: Z = 3.95 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.84, df = 3 (P = 0.84), I² = 0% | | | | | | |




**B**

| Study or Subgroup | log[Hazard Ratio] | SE | anti-EGFR Total | Control Total | Weight | Hazard Ratio IV, Random, 95% CI |
|---|---|---|---|---|---|---|
| **2.2.1 mAb** | | | | | | |
| Castro 2018 | -0.3275 | 0.1968 | 53 | 54 | 6.6% | 0.72 [0.49, 1.06] |
| Chen 2016b | -1.4092 | 1.0701 | 32 | 32 | 0.3% | 0.24 [0.03, 1.99] |
| Crosby 2017 | 0.2103 | 0.1389 | 129 | 129 | 9.8% | 1.23 [0.94, 1.62] |
| Lorenzen 2009 | -0.2236 | 0.2821 | 32 | 30 | 3.9% | 0.80 [0.46, 1.39] |
| Lu 2017 | -0.4734 | 0.226 | 40 | 40 | 5.5% | 0.62 [0.40, 0.97] |
| Luo 2015 | -0.2521 | 0.3389 | 28 | 27 | 2.9% | 0.78 [0.40, 1.51] |
| Ruhstaller 2018 | -0.2989 | 0.1525 | 149 | 151 | 8.9% | 0.74 [0.55, 1.00] |
| Shi 2014 | -0.2868 | 0.2176 | 32 | 32 | 5.8% | 0.75 [0.49, 1.15] |
| Sun 2014 | -0.4586 | 0.2733 | 21 | 21 | 4.1% | 0.63 [0.37, 1.08] |
| Suntharalingam 2017 | -0.09 | 0.1217 | 159 | 169 | 11.0% | 0.91 [0.72, 1.16] |
| **Subtotal (95% CI)** | | | **675** | **685** | **58.9%** | **0.81 [0.69, 0.96]** |
| Heterogeneity: Tau² = 0.02; Chi² = 13.92, df = 9 (P = 0.13); I² = 35% | | | | | | |
| Test for overall effect: Z = 2.47 (P = 0.01) | | | | | | |
| **2.2.2 TKI** | | | | | | |
| Chen 2016a | -0.7186 | 0.199 | 68 | 68 | 6.5% | 0.49 [0.33, 0.72] |
| Dutton 2014 | -0.1036 | 0.0987 | 224 | 225 | 12.8% | 0.90 [0.74, 1.09] |
| Wu 2018 | -0.239 | 0.122 | 176 | 176 | 11.0% | 0.79 [0.62, 1.00] |
| Ye 2015 | -0.4027 | 0.2494 | 37 | 36 | 4.8% | 0.67 [0.41, 1.09] |
| Zhou 2015 | -0.3845 | 0.2112 | 40 | 40 | 6.0% | 0.68 [0.45, 1.03] |
| **Subtotal (95% CI)** | | | **545** | **545** | **41.1%** | **0.73 [0.59, 0.89]** |
| Heterogeneity: Tau² = 0.03; Chi² = 8.52, df = 4 (P = 0.07); I² = 53% | | | | | | |
| Test for overall effect: Z = 3.04 (P = 0.002) | | | | | | |
| **Total (95% CI)** | | | **1220** | **1230** | **100.0%** | **0.78 [0.69, 0.88]** |
| Heterogeneity: Tau² = 0.02; Chi² = 23.44, df = 14 (P = 0.05); I² = 40% | | | | | | |
| Test for overall effect: Z = 3.95 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.75, df = 1 (P = 0.39), I² = 0% | | | | | | |



**C**

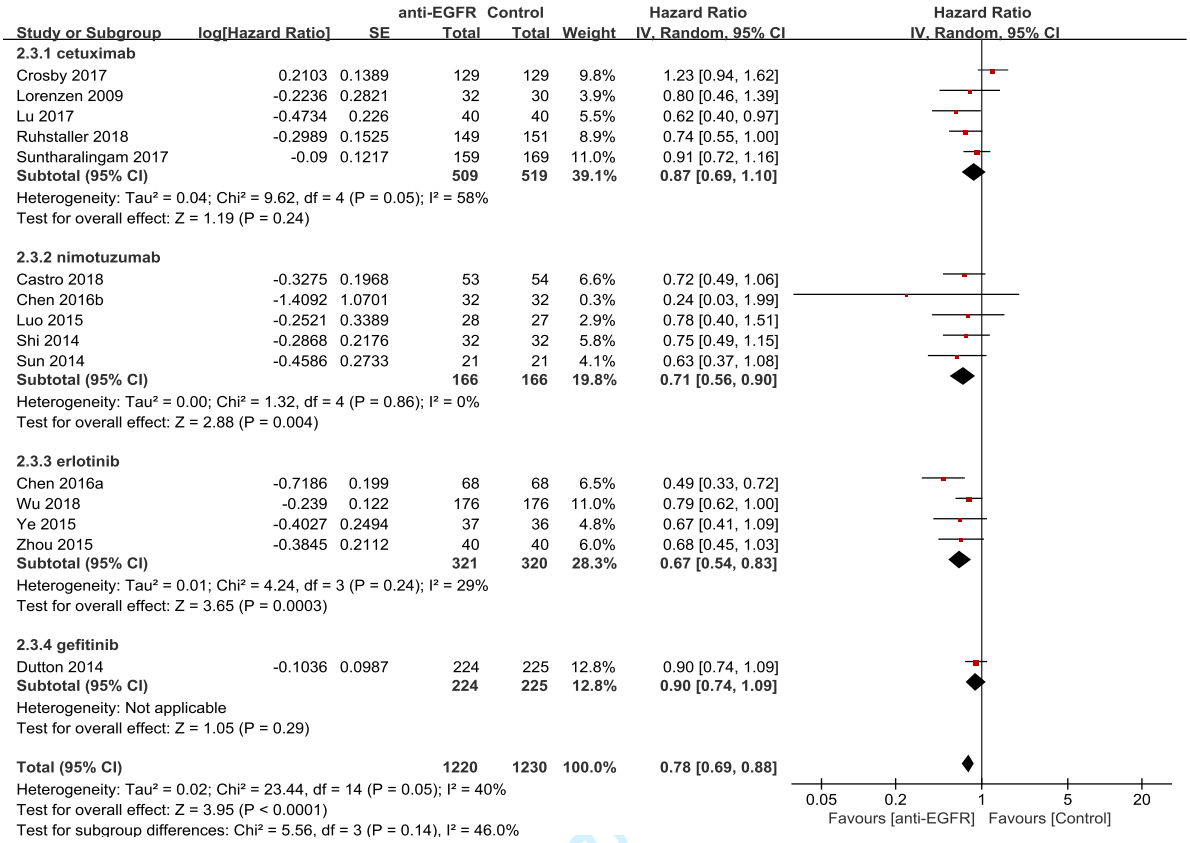
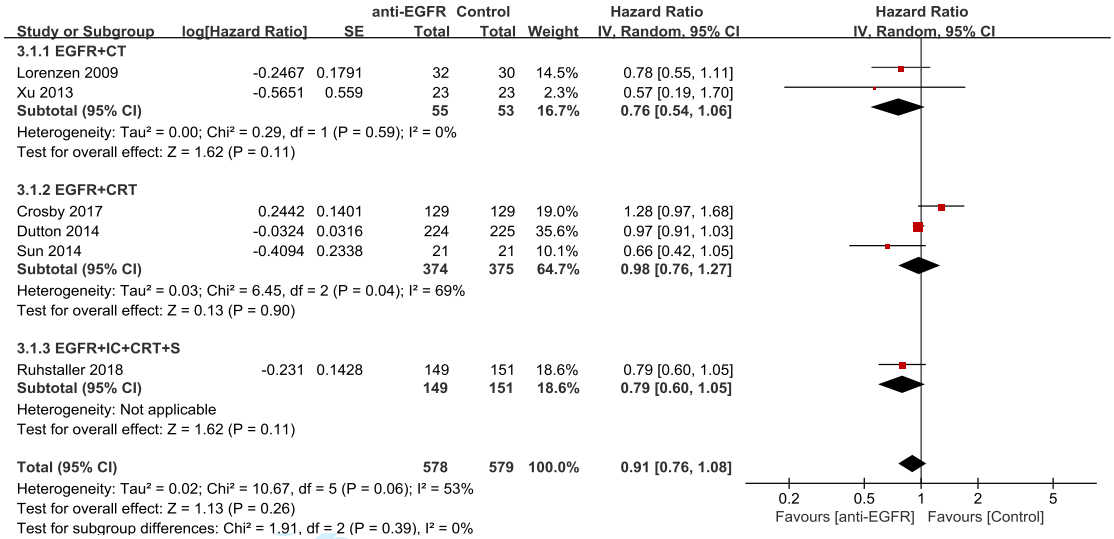


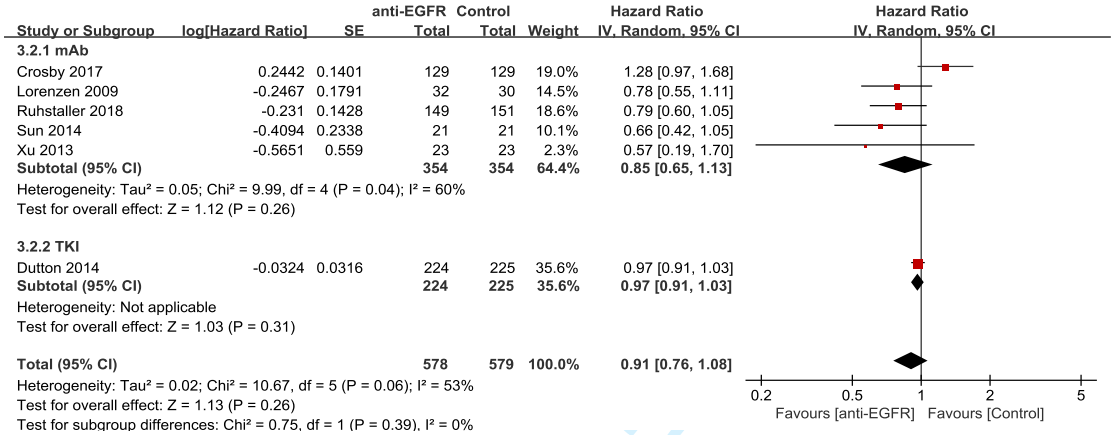

**Figure S1** Subgroup analysis of overall survival (OS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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**A**




**B**




**C**

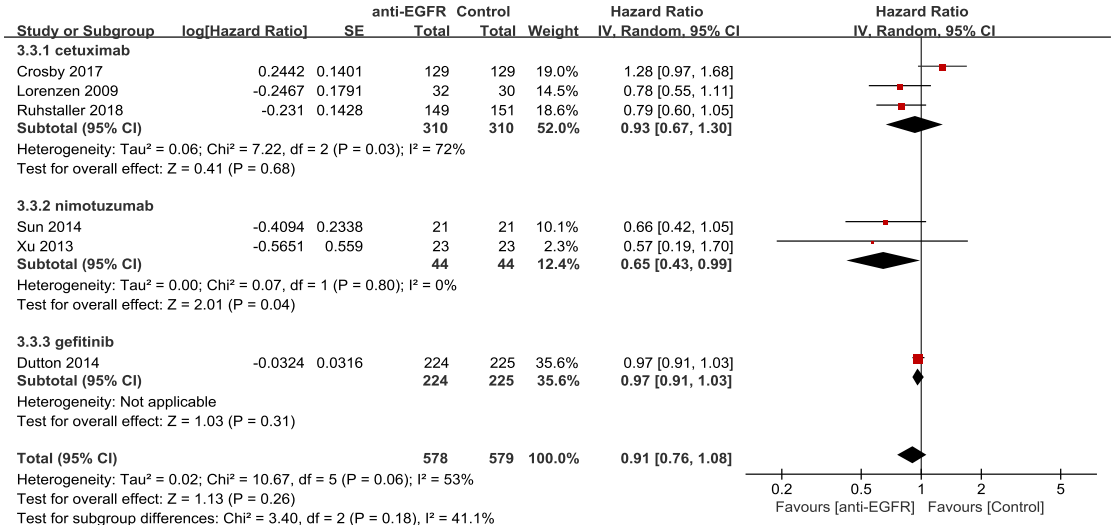



**Figure S2** Subgroup analysis of progression-free survival (PFS) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

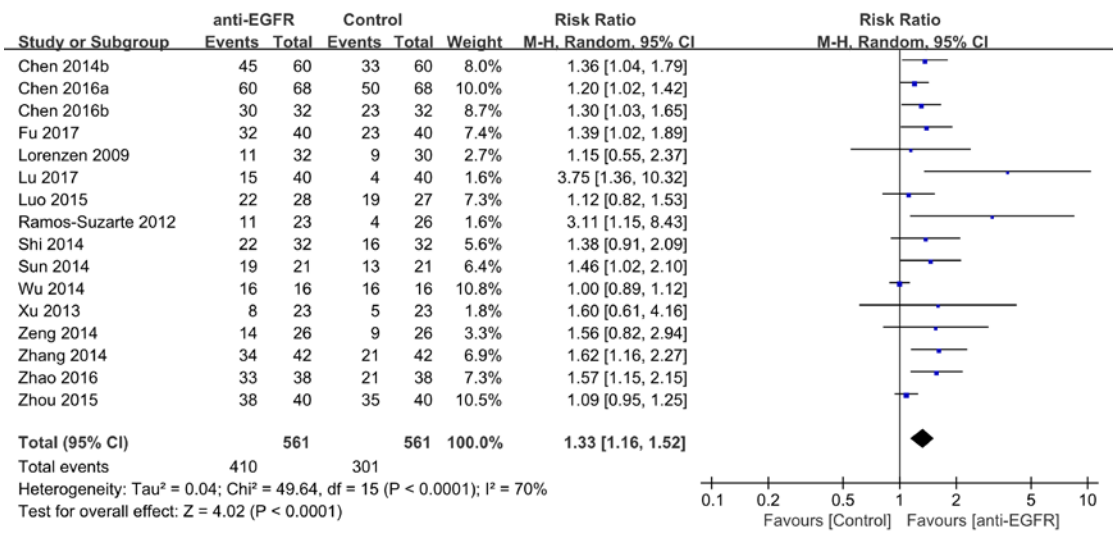
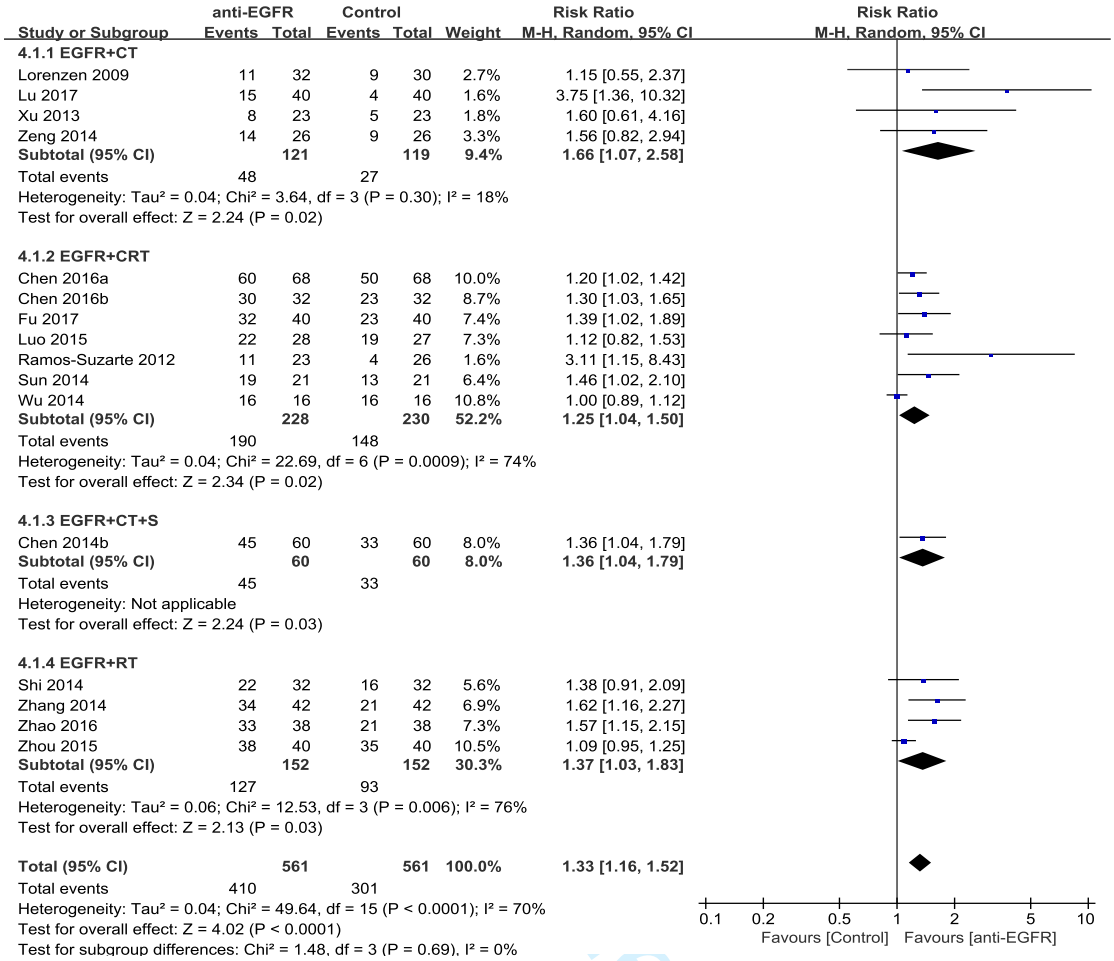


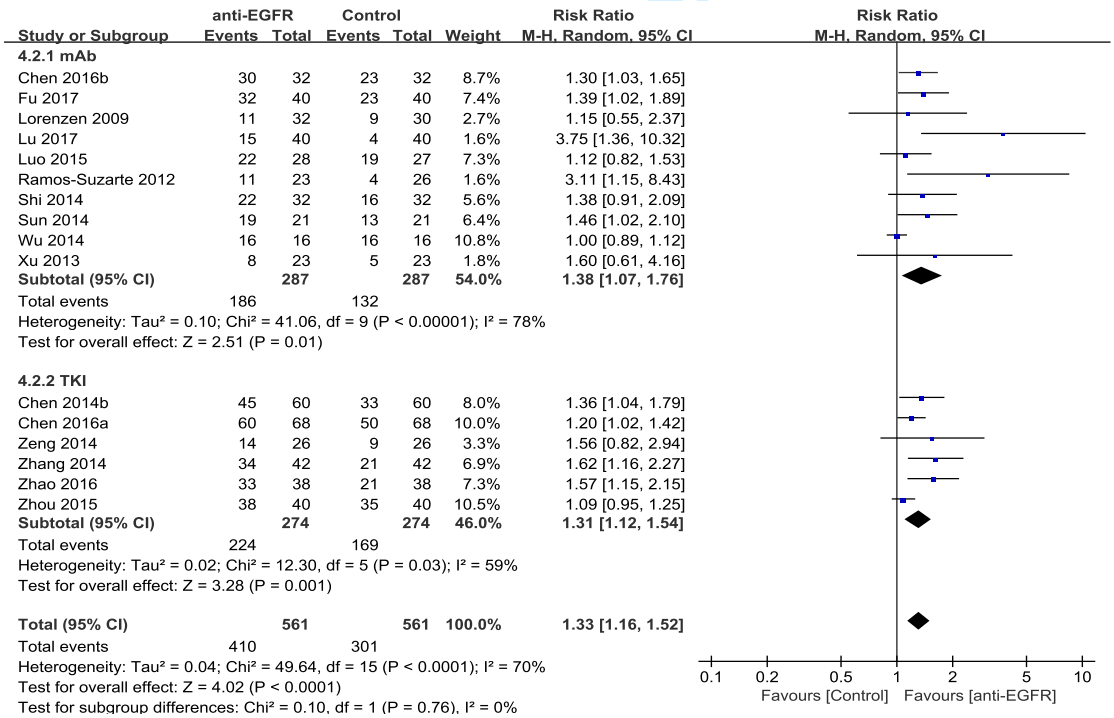

**Figure S3** Forest plot of the objective response rate (ORR) of patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

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**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.1.1 EGFR+CT
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Subtotal (95% CI) 121 119 9.4% 1.66 [1.07, 2.58]
Total events 48 27
Heterogeneity: Tau² = 0.04; Chi² = 3.64, df = 3 (P = 0.30); I² = 18%
Test for overall effect: Z = 2.24 (P = 0.02)
4.1.2 EGFR+CRT
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Subtotal (95% CI) 228 230 52.2% 1.25 [1.04, 1.50]
Total events 190 148
Heterogeneity: Tau² = 0.04; Chi² = 22.69, df = 6 (P = 0.0009); I² = 74%
Test for overall effect: Z = 2.34 (P = 0.02)
4.1.3 EGFR+CT+S
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Subtotal (95% CI) 60 60 8.0% 1.36 [1.04, 1.79]
Total events 45 33
Heterogeneity: Not applicable
Test for overall effect: Z = 2.24 (P = 0.03)
4.1.4 EGFR+RT
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 152 152 30.3% 1.37 [1.03, 1.83]
Total events 127 93
Heterogeneity: Tau² = 0.06; Chi² = 12.53, df = 3 (P = 0.006); I² = 76%
Test for overall effect: Z = 2.13 (P = 0.03)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 1.48, df = 3 (P = 0.69), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.2.1 mAb
Chen 2016b 30 32 23 32 8.7% 1.30 [1.03, 1.65]
Fu 2017 32 40 23 40 7.4% 1.39 [1.02, 1.89]
Lorenzen 2009 11 32 9 30 2.7% 1.15 [0.55, 2.37]
Lu 2017 15 40 4 40 1.6% 3.75 [1.36, 10.32]
Luo 2015 22 28 19 27 7.3% 1.12 [0.82, 1.53]
Ramos-Suzarte 2012 11 23 4 26 1.6% 3.11 [1.15, 8.43]
Shi 2014 22 32 16 32 5.6% 1.38 [0.91, 2.09]
Sun 2014 19 21 13 21 6.4% 1.46 [1.02, 2.10]
Wu 2014 16 16 16 16 10.8% 1.00 [0.89, 1.12]
Xu 2013 8 23 5 23 1.8% 1.60 [0.61, 4.16]
Subtotal (95% CI) 287 287 54.0% 1.38 [1.07, 1.76]
Total events 186 132
Heterogeneity: Tau² = 0.10; Chi² = 41.06, df = 9 (P < 0.00001); I² = 78%
Test for overall effect: Z = 2.51 (P = 0.01)
4.2.2 TKI
Chen 2014b 45 60 33 60 8.0% 1.36 [1.04, 1.79]
Chen 2016a 60 68 50 68 10.0% 1.20 [1.02, 1.42]
Zeng 2014 14 26 9 26 3.3% 1.56 [0.82, 2.94]
Zhang 2014 34 42 21 42 6.9% 1.62 [1.16, 2.27]
Zhao 2016 33 38 21 38 7.3% 1.57 [1.15, 2.15]
Zhou 2015 38 40 35 40 10.5% 1.09 [0.95, 1.25]
Subtotal (95% CI) 274 274 46.0% 1.31 [1.12, 1.54]
Total events 224 169
Heterogeneity: Tau² = 0.02; Chi² = 12.30, df = 5 (P = 0.03); I² = 59%
Test for overall effect: Z = 3.28 (P = 0.001)
Total (95% CI) 561 561 100.0% 1.33 [1.16, 1.52]
Total events 410 301
Heterogeneity: Tau² = 0.04; Chi² = 49.64, df = 15 (P < 0.0001); I² = 70%
Test for overall effect: Z = 4.02 (P < 0.0001)
Test for subgroup differences: Chi² = 0.10, df = 1 (P = 0.76), I² = 0%
0.1 0.2 0.5 1 2 5 10
Favours [Control] Favours [anti-EGFR]

**C**

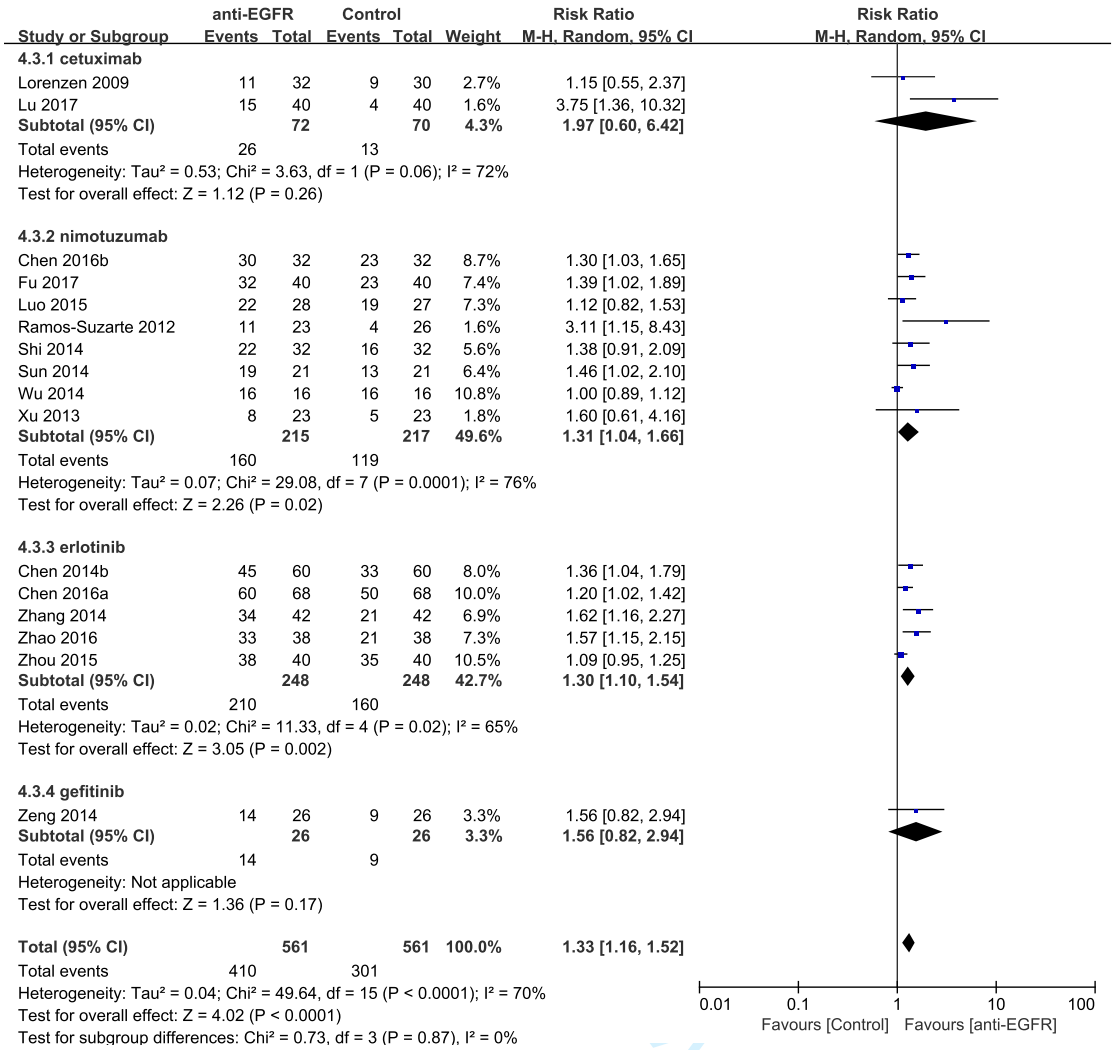



**Figure S4** Subgroup analysis of the objective response rate (ORR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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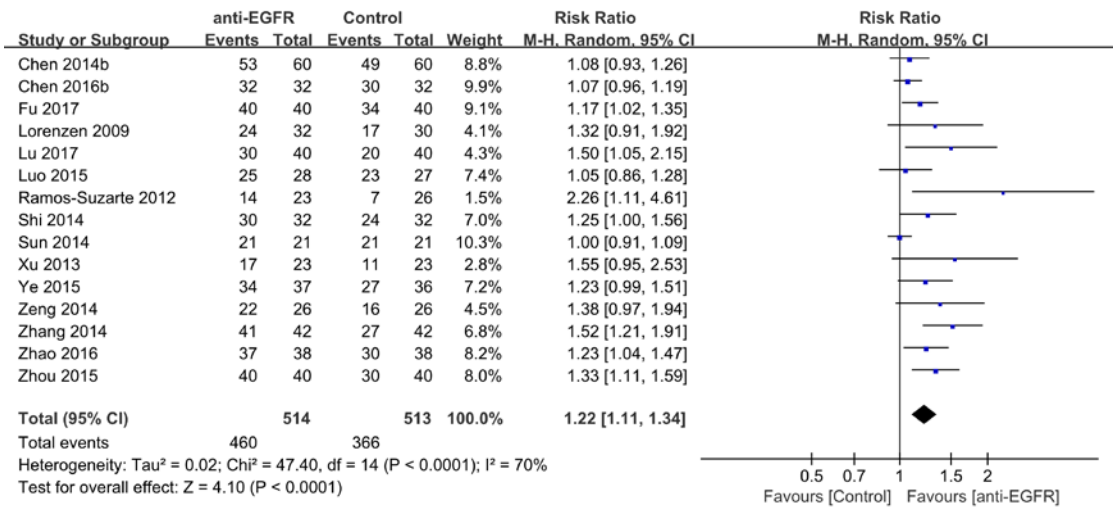
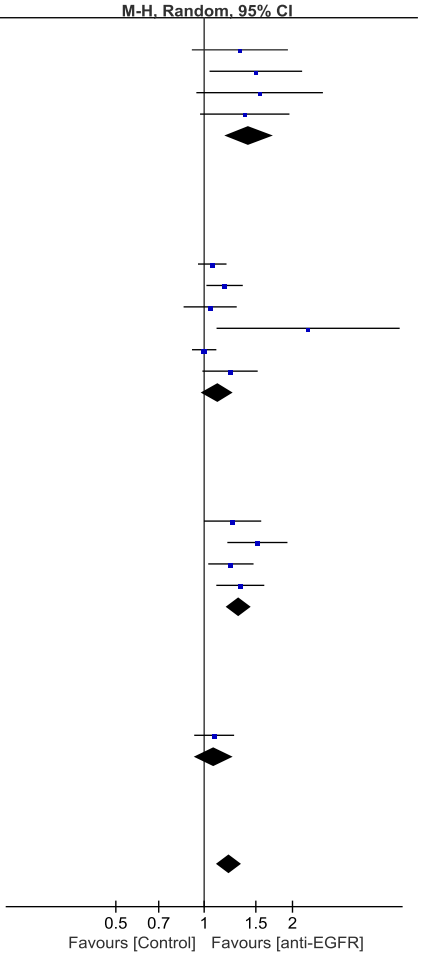



**Figure S5** Forest plot of the disease control rate (DCR) in patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm

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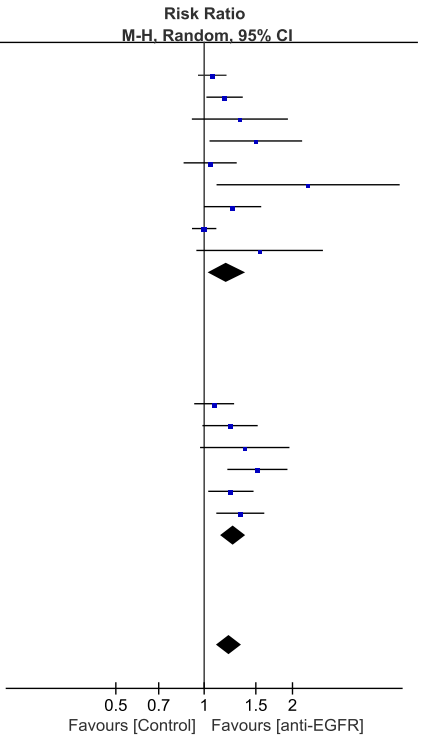
**A**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.1.1 EGFR+CT** | | | | | | |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| **Subtotal (95% CI)** | | **121** | | **119** | **15.7%** | **1.42 [1.17, 1.72]** |
| Total events | 93 | | 64 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 0.38, df = 3 (P = 0.94); I² = 0% | | | | | | |
| Test for overall effect: Z = 3.61 (P = 0.0003) | | | | | | |
| **5.1.2 EGFR+CRT** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| **Subtotal (95% CI)** | | **181** | | **182** | **45.5%** | **1.11 [0.98, 1.26]** |
| Total events | 166 | | 142 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 18.13, df = 5 (P = 0.003); I² = 72% | | | | | | |
| Test for overall effect: Z = 1.69 (P = 0.09) | | | | | | |
| **5.1.3 EGFR+RT** | | | | | | |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **152** | | **152** | **30.0%** | **1.31 [1.19, 1.45]** |
| Total events | 148 | | 111 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 2.33, df = 3 (P = 0.51); I² = 0% | | | | | | |
| Test for overall effect: Z = 5.41 (P < 0.00001) | | | | | | |
| **5.1.4 EGFR+CT+S** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| **Subtotal (95% CI)** | | **60** | | **60** | **8.8%** | **1.08 [0.93, 1.26]** |
| Total events | 53 | | 49 | | | |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.02 (P = 0.31) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 9.05, df = 3 (P = 0.03), I² = 66.9% | | | | | | |




**B**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **5.2.1 mAb** | | | | | | |
| Chen 2016b | 32 | 32 | 30 | 32 | 9.9% | 1.07 [0.96, 1.19] |
| Fu 2017 | 40 | 40 | 34 | 40 | 9.1% | 1.17 [1.02, 1.35] |
| Lorenzen 2009 | 24 | 32 | 17 | 30 | 4.1% | 1.32 [0.91, 1.92] |
| Lu 2017 | 30 | 40 | 20 | 40 | 4.3% | 1.50 [1.05, 2.15] |
| Luo 2015 | 25 | 28 | 23 | 27 | 7.4% | 1.05 [0.86, 1.28] |
| Ramos-Suzarte 2012 | 14 | 23 | 7 | 26 | 1.5% | 2.26 [1.11, 4.61] |
| Shi 2014 | 30 | 32 | 24 | 32 | 7.0% | 1.25 [1.00, 1.56] |
| Sun 2014 | 21 | 21 | 21 | 21 | 10.3% | 1.00 [0.91, 1.09] |
| Xu 2013 | 17 | 23 | 11 | 23 | 2.8% | 1.55 [0.95, 2.53] |
| **Subtotal (95% CI)** | | **271** | | **271** | **56.5%** | **1.20 [1.04, 1.38]** |
| Total events | 233 | | 187 | | | |
| Heterogeneity: Tau² = 0.03; Chi² = 37.25, df = 8 (P < 0.0001); I² = 79% | | | | | | |
| Test for overall effect: Z = 2.43 (P = 0.01) | | | | | | |
| **5.2.2 TKI** | | | | | | |
| Chen 2014b | 53 | 60 | 49 | 60 | 8.8% | 1.08 [0.93, 1.26] |
| Ye 2015 | 34 | 37 | 27 | 36 | 7.2% | 1.23 [0.99, 1.51] |
| Zeng 2014 | 22 | 26 | 16 | 26 | 4.5% | 1.38 [0.97, 1.94] |
| Zhang 2014 | 41 | 42 | 27 | 42 | 6.8% | 1.52 [1.21, 1.91] |
| Zhao 2016 | 37 | 38 | 30 | 38 | 8.2% | 1.23 [1.04, 1.47] |
| Zhou 2015 | 40 | 40 | 30 | 40 | 8.0% | 1.33 [1.11, 1.59] |
| **Subtotal (95% CI)** | | **243** | | **242** | **43.5%** | **1.26 [1.14, 1.38]** |
| Total events | 227 | | 179 | | | |
| Heterogeneity: Tau² = 0.00; Chi² = 7.16, df = 5 (P = 0.21); I² = 30% | | | | | | |
| Test for overall effect: Z = 4.54 (P < 0.00001) | | | | | | |
| **Total (95% CI)** | | **514** | | **513** | **100.0%** | **1.22 [1.11, 1.34]** |
| Total events | 460 | | 366 | | | |
| Heterogeneity: Tau² = 0.02; Chi² = 47.40, df = 14 (P < 0.0001); I² = 70% | | | | | | |
| Test for overall effect: Z = 4.10 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.29, df = 1 (P = 0.59), I² = 0% | | | | | | |



**C**

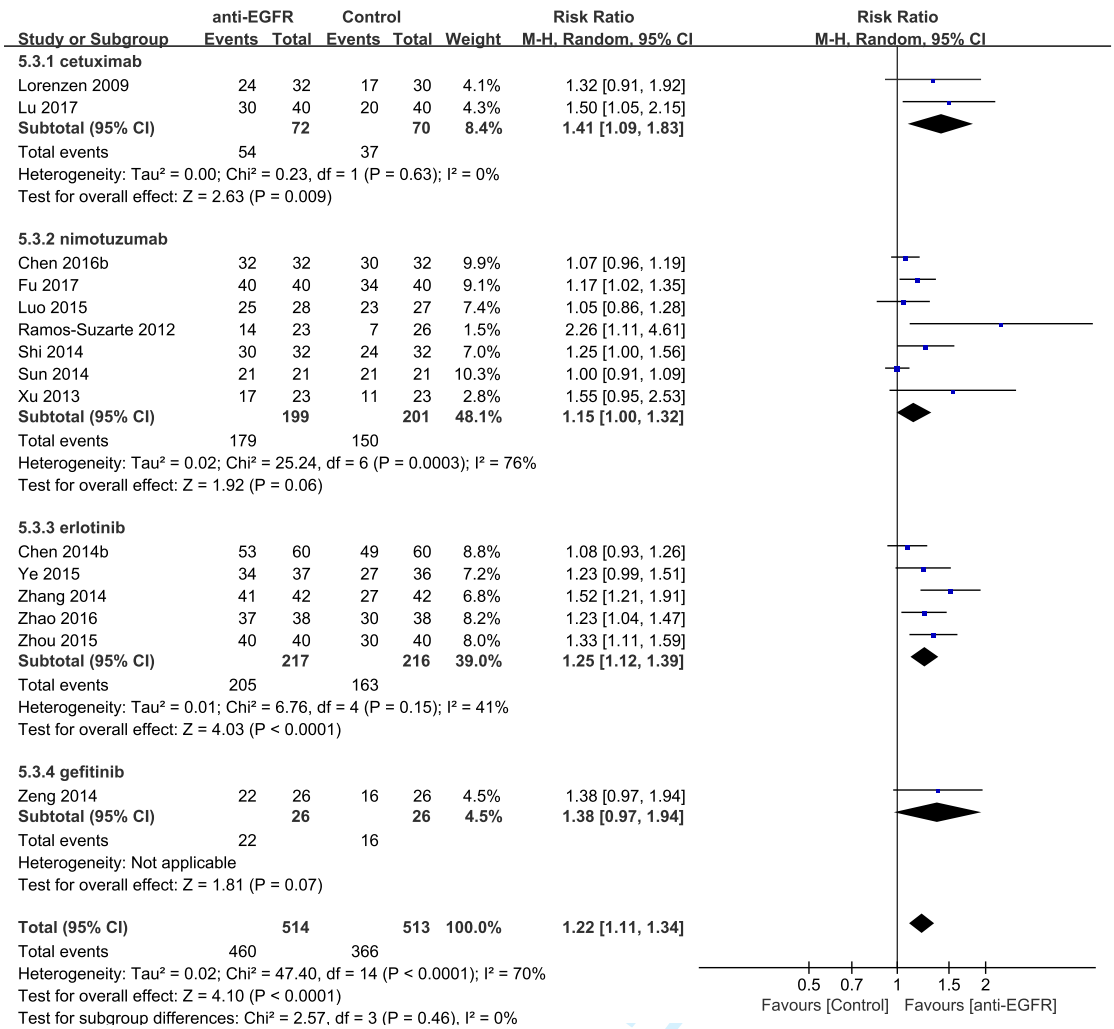



**Figure S6** Subgroup analysis of the disease control rate (DCR) according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

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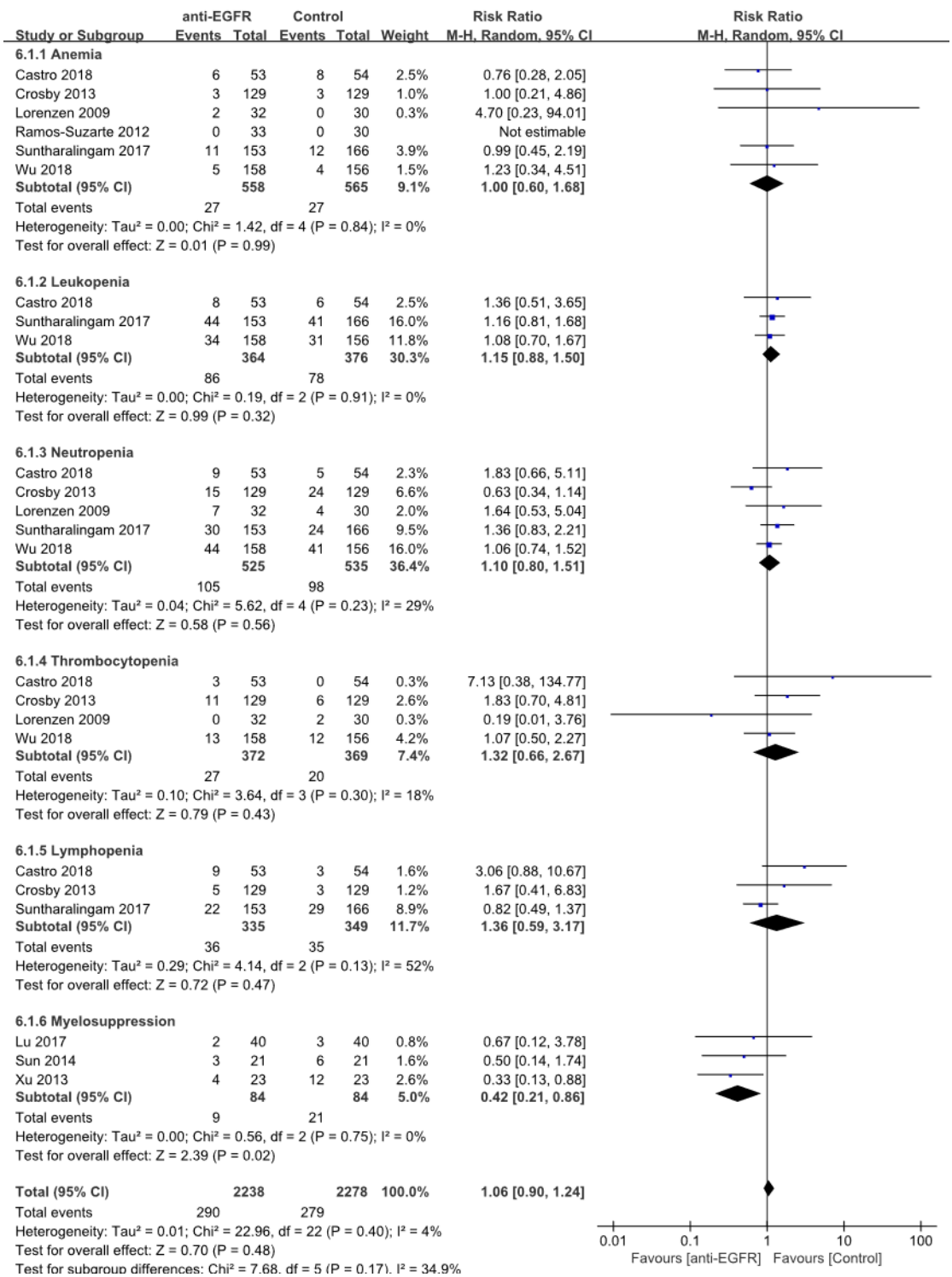
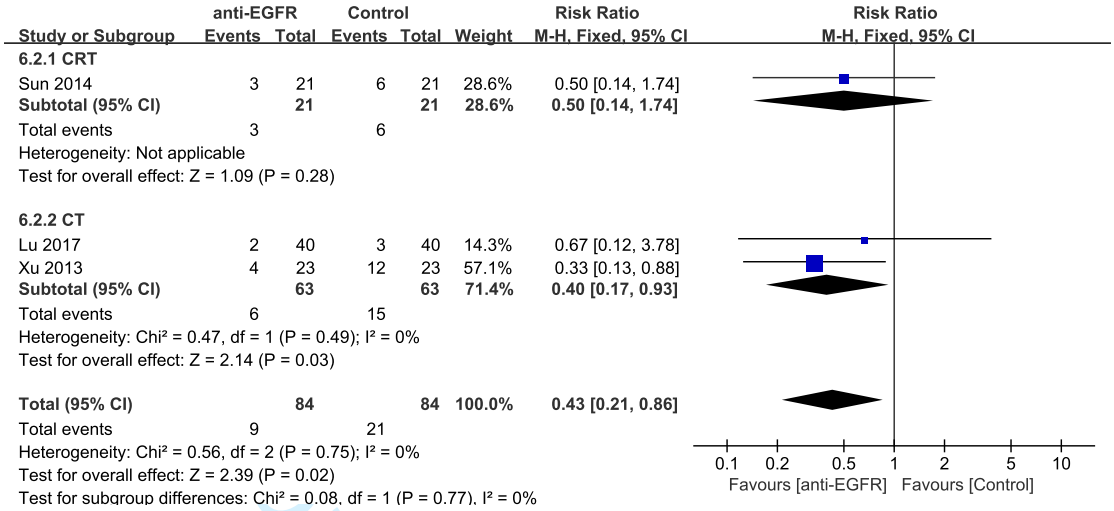



**Figure S7** Forest plot of grade 3–5 haematological toxicities in patients with oesophageal cancer in the anti-EGFR arm versus the non-anti-EGFR arm



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**A**




**B**

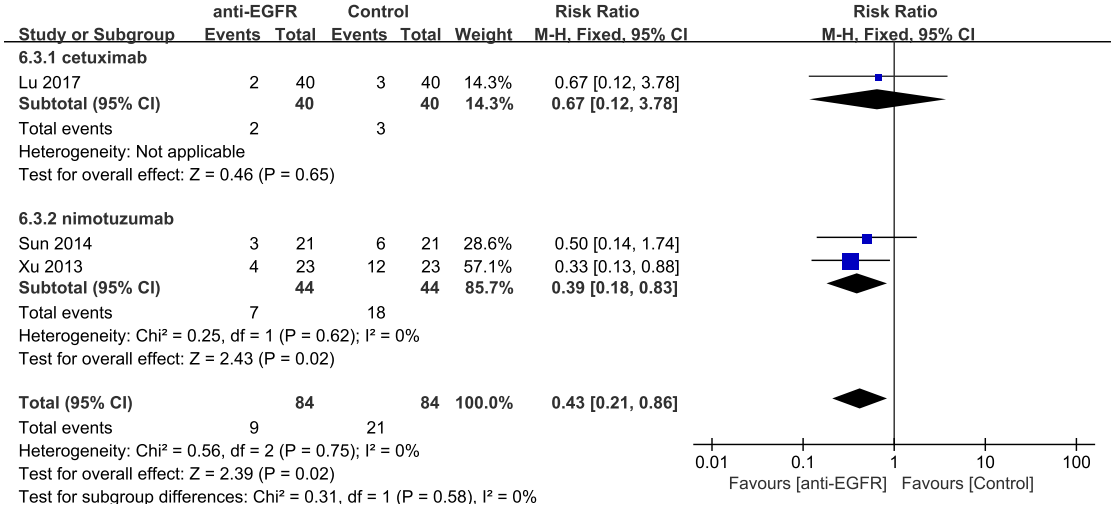
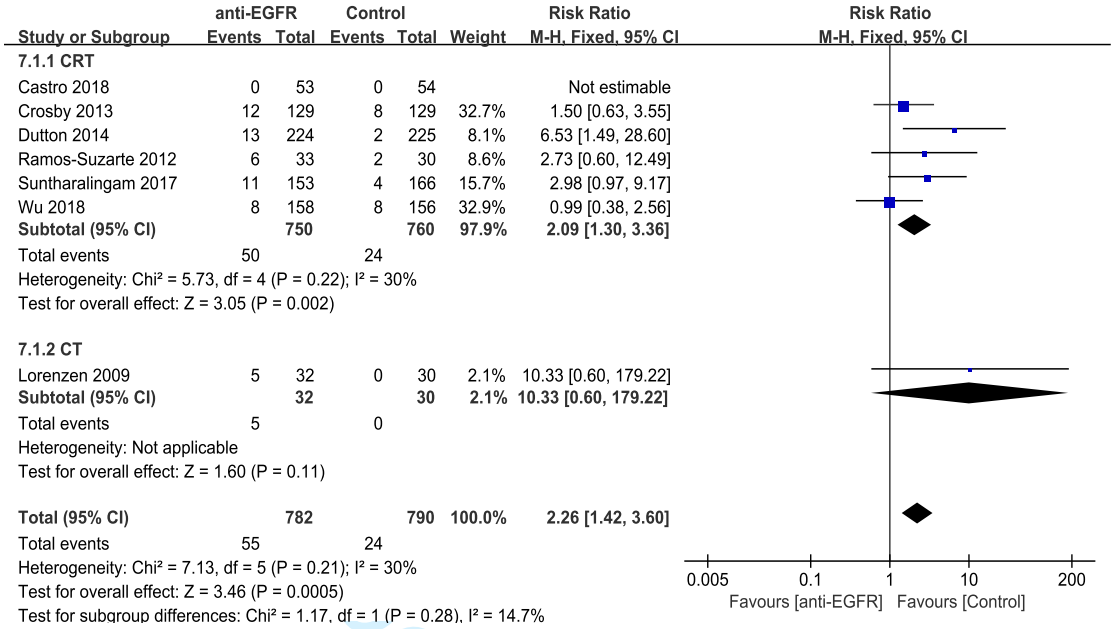
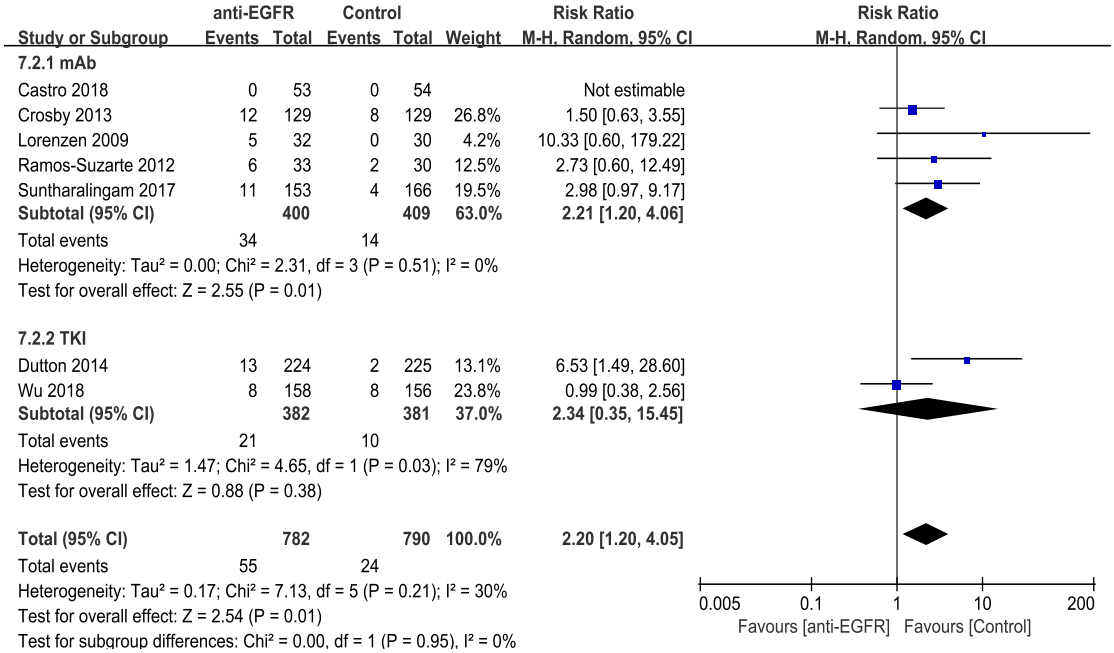



**Figure S8** Subgroup analysis of grade 3–5 myelosuppression according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR agents

**A**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
7.1.1 CRT
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 32.7% 1.50 [0.63, 3.55]
Dutton 2014 13 224 2 225 8.1% 6.53 [1.49, 28.60]
Ramos-Suzarte 2012 6 33 2 30 8.6% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 15.7% 2.98 [0.97, 9.17]
Wu 2018 8 158 8 156 32.9% 0.99 [0.38, 2.56]
Subtotal (95% CI) 750 760 97.9% 2.09 [1.30, 3.36]
Total events 50 24
Heterogeneity: Chi² = 5.73, df = 4 (P = 0.22); I² = 30%
Test for overall effect: Z = 3.05 (P = 0.002)
7.1.2 CT
Lorenzen 2009 5 32 0 30 2.1% 10.33 [0.60, 179.22]
Subtotal (95% CI) 32 30 2.1% 10.33 [0.60, 179.22]
Total events 5 0
Heterogeneity: Not applicable
Test for overall effect: Z = 1.60 (P = 0.11)
Total (95% CI) 782 790 100.0% 2.26 [1.42, 3.60]
Total events 55 24
Heterogeneity: Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 3.46 (P = 0.0005)
Test for subgroup differences: Chi² = 1.17, df = 1 (P = 0.28), I² = 14.7%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**B**


anti-EGFR
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
7.2.1 mAb
Castro 2018 0 53 0 54 Not estimable
Crosby 2013 12 129 8 129 26.8% 1.50 [0.63, 3.55]
Lorenzen 2009 5 32 0 30 4.2% 10.33 [0.60, 179.22]
Ramos-Suzarte 2012 6 33 2 30 12.5% 2.73 [0.60, 12.49]
Suntharalingam 2017 11 153 4 166 19.5% 2.98 [0.97, 9.17]
Subtotal (95% CI) 400 409 63.0% 2.21 [1.20, 4.06]
Total events 34 14
Heterogeneity: Tau² = 0.00; Chi² = 2.31, df = 3 (P = 0.51); I² = 0%
Test for overall effect: Z = 2.55 (P = 0.01)
7.2.2 TKI
Dutton 2014 13 224 2 225 13.1% 6.53 [1.49, 28.60]
Wu 2018 8 158 8 156 23.8% 0.99 [0.38, 2.56]
Subtotal (95% CI) 382 381 37.0% 2.34 [0.35, 15.45]
Total events 21 10
Heterogeneity: Tau² = 1.47; Chi² = 4.65, df = 1 (P = 0.03); I² = 79%
Test for overall effect: Z = 0.88 (P = 0.38)
Total (95% CI) 782 790 100.0% 2.20 [1.20, 4.05]
Total events 55 24
Heterogeneity: Tau² = 0.17; Chi² = 7.13, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 2.54 (P = 0.01)
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.95), I² = 0%
0.005 0.1 1 10 200
Favours [anti-EGFR] Favours [Control]

**C**

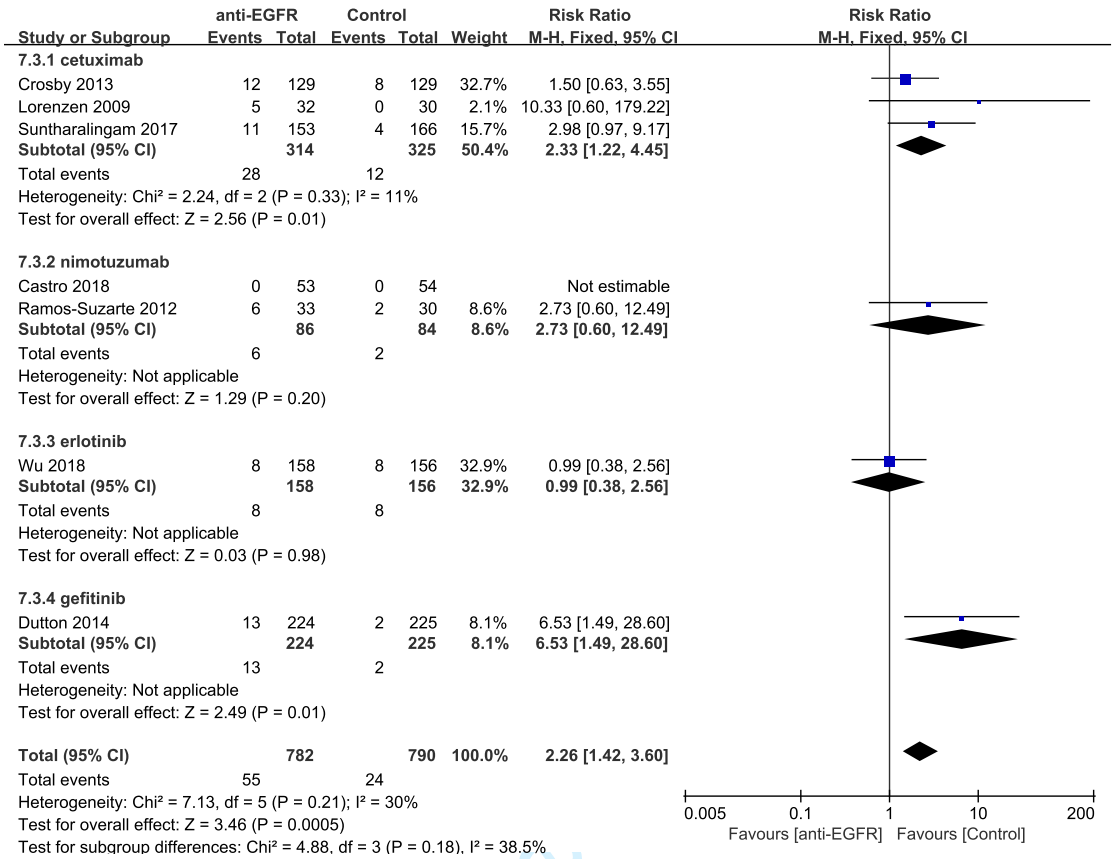
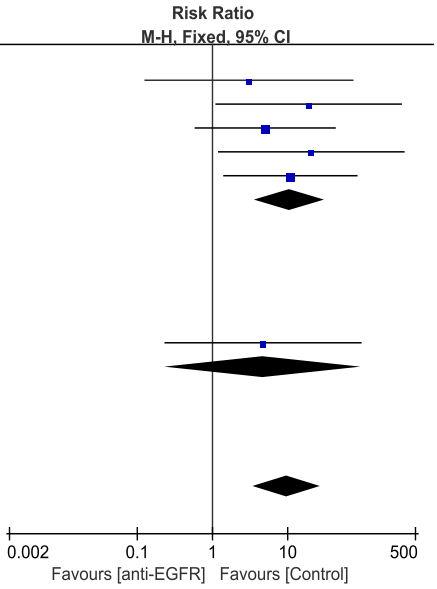



**Figure S9** Subgroup analysis of grade 3–5 diarrhoea according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

eview only

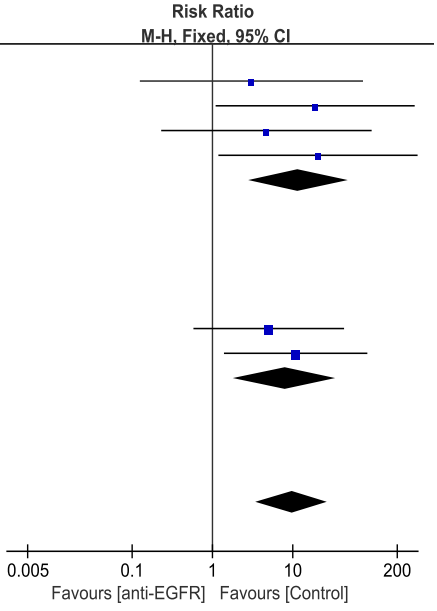
**A**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **8.1.1 CRT** | | | | | | |
| Castro 2018 | 1 | 53 | 0 | 54 | 12.4% | 3.06 [0.13, 73.37] |
| Crosby 2013 | 9 | 129 | 0 | 129 | 12.5% | 19.00 [1.12, 323.06] |
| Dutton 2014 | 5 | 224 | 1 | 225 | 25.0% | 5.02 [0.59, 42.65] |
| Suntharalingam 2017 | 9 | 153 | 0 | 166 | 12.0% | 20.60 [1.21, 351.02] |
| Wu 2018 | 11 | 158 | 1 | 156 | 25.2% | 10.86 [1.42, 83.12] |
| **Subtotal (95% CI)** | | **717** | | **730** | **87.1%** | **10.59 [3.56, 31.49]** |
| Total events | 35 | | 2 | | | |
| Heterogeneity: Chi² = 1.43, df = 4 (P = 0.84); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.24 (P < 0.0001) | | | | | | |
| **8.1.2 CT** | | | | | | |
| Lorenzen 2009 | 2 | 32 | 0 | 30 | 12.9% | 4.70 [0.23, 94.01] |
| **Subtotal (95% CI)** | | **32** | | **30** | **12.9%** | **4.70 [0.23, 94.01]** |
| Total events | 2 | | 0 | | | |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z = 1.01 (P = 0.31) | | | | | | |
| **Total (95% CI)** | | **749** | | **760** | **100.0%** | **9.83 [3.54, 27.28]** |
| Total events | 37 | | 2 | | | |
| Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.39 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.25, df = 1 (P = 0.62), I² = 0% | | | | | | |




**B**

| Study or Subgroup | anti-EGFR Events | anti-EGFR Total | Control Events | Control Total | Weight | Risk Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **8.2.1 mAb** | | | | | | |
| Castro 2018 | 1 | 53 | 0 | 54 | 12.4% | 3.06 [0.13, 73.37] |
| Crosby 2013 | 9 | 129 | 0 | 129 | 12.5% | 19.00 [1.12, 323.06] |
| Lorenzen 2009 | 2 | 32 | 0 | 30 | 12.9% | 4.70 [0.23, 94.01] |
| Suntharalingam 2017 | 9 | 153 | 0 | 166 | 12.0% | 20.60 [1.21, 351.02] |
| **Subtotal (95% CI)** | | **367** | | **379** | **49.8%** | **11.71 [2.80, 48.99]** |
| Total events | 21 | | 0 | | | |
| Heterogeneity: Chi² = 1.31, df = 3 (P = 0.73); I² = 0% | | | | | | |
| Test for overall effect: Z = 3.37 (P = 0.0007) | | | | | | |
| **8.2.2 TKI** | | | | | | |
| Dutton 2014 | 5 | 224 | 1 | 225 | 25.0% | 5.02 [0.59, 42.65] |
| Wu 2018 | 11 | 158 | 1 | 156 | 25.2% | 10.86 [1.42, 83.12] |
| **Subtotal (95% CI)** | | **382** | | **381** | **50.2%** | **7.95 [1.84, 34.37]** |
| Total events | 16 | | 2 | | | |
| Heterogeneity: Chi² = 0.27, df = 1 (P = 0.61); I² = 0% | | | | | | |
| Test for overall effect: Z = 2.78 (P = 0.005) | | | | | | |
| **Total (95% CI)** | | **749** | | **760** | **100.0%** | **9.83 [3.54, 27.28]** |
| Total events | 37 | | 2 | | | |
| Heterogeneity: Chi² = 1.61, df = 5 (P = 0.90); I² = 0% | | | | | | |
| Test for overall effect: Z = 4.39 (P < 0.0001) | | | | | | |
| Test for subgroup differences: Chi² = 0.14, df = 1 (P = 0.71), I² = 0% | | | | | | |



**C**

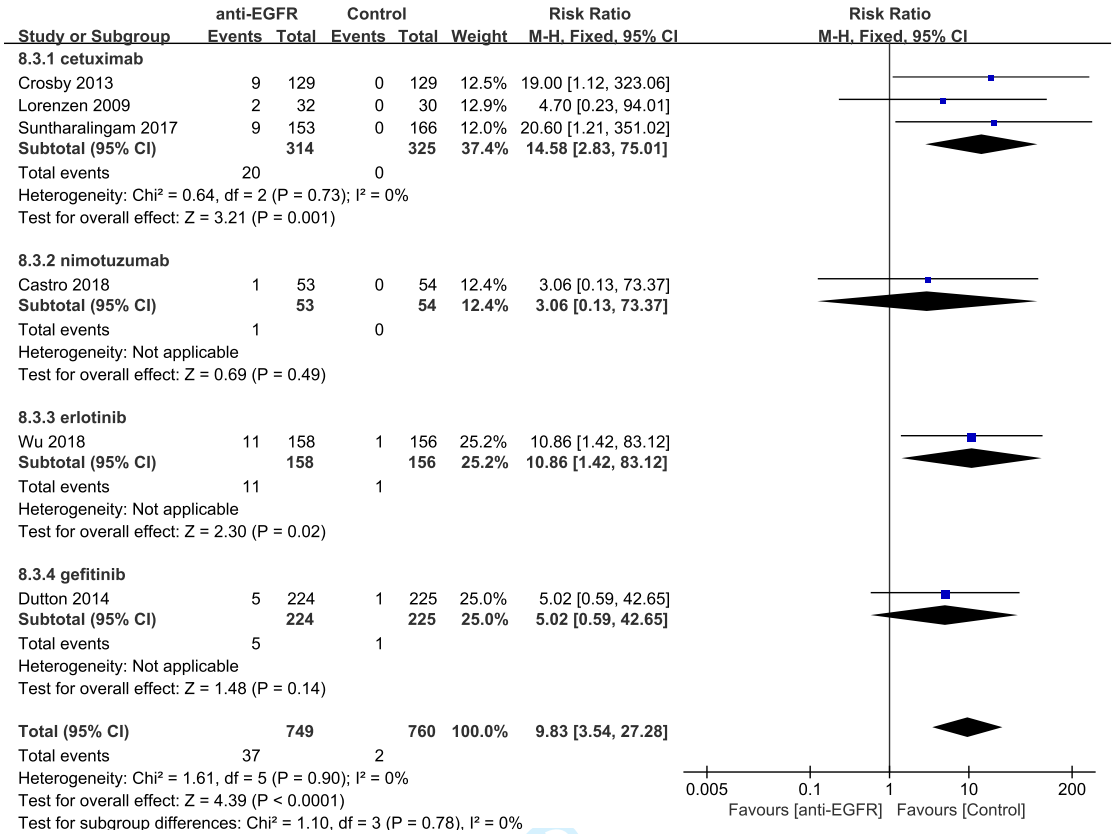



**Figure S10** Subgroup analysis of grade 3–5 acne-like rashes according to different treatment regimens: (A) different treatment modalities; (B) different anti-EGFR groups; (C) different anti-EGFR agents

eview only

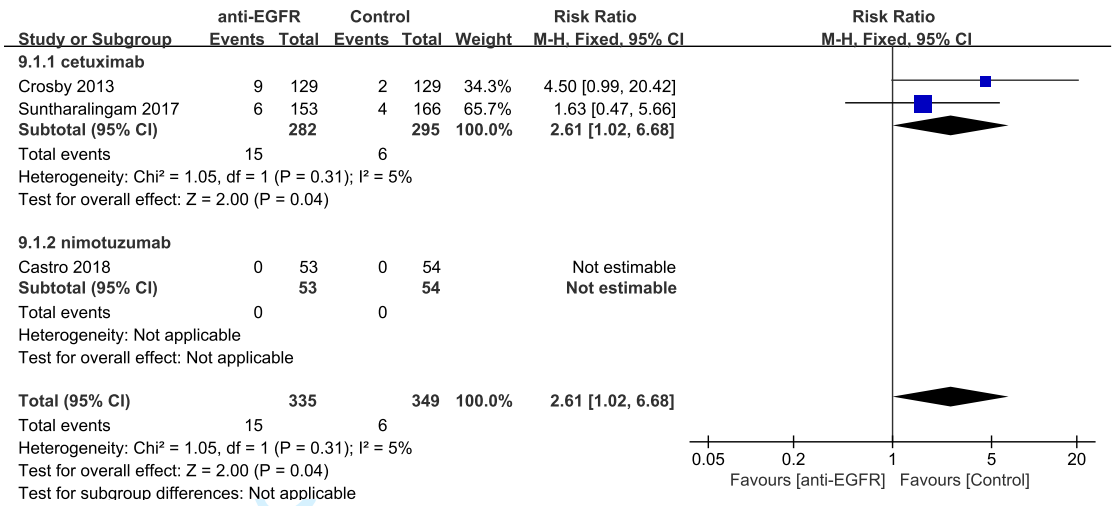



**Figure S11** Subgroup analysis of grade 3–5 hypomagnesaemia according to different anti-EGFR agents

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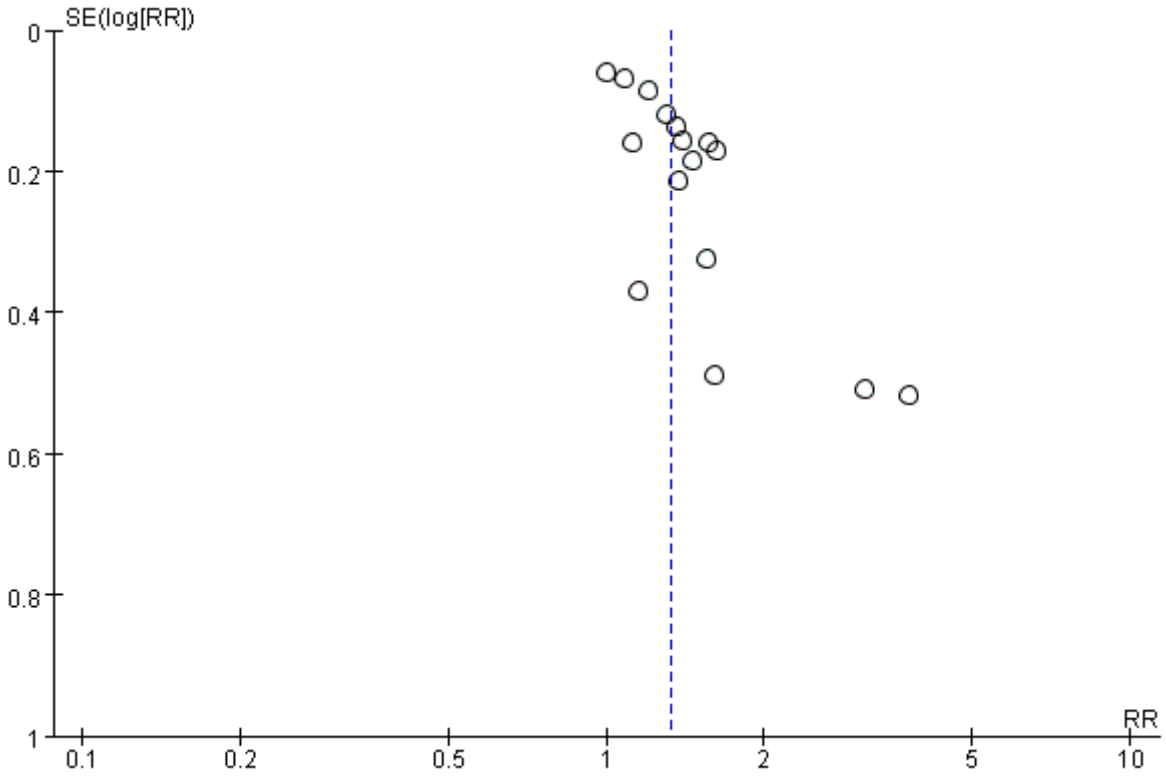



**Figure S12** Funnel plot of publication bias for the objective response rate (ORR)

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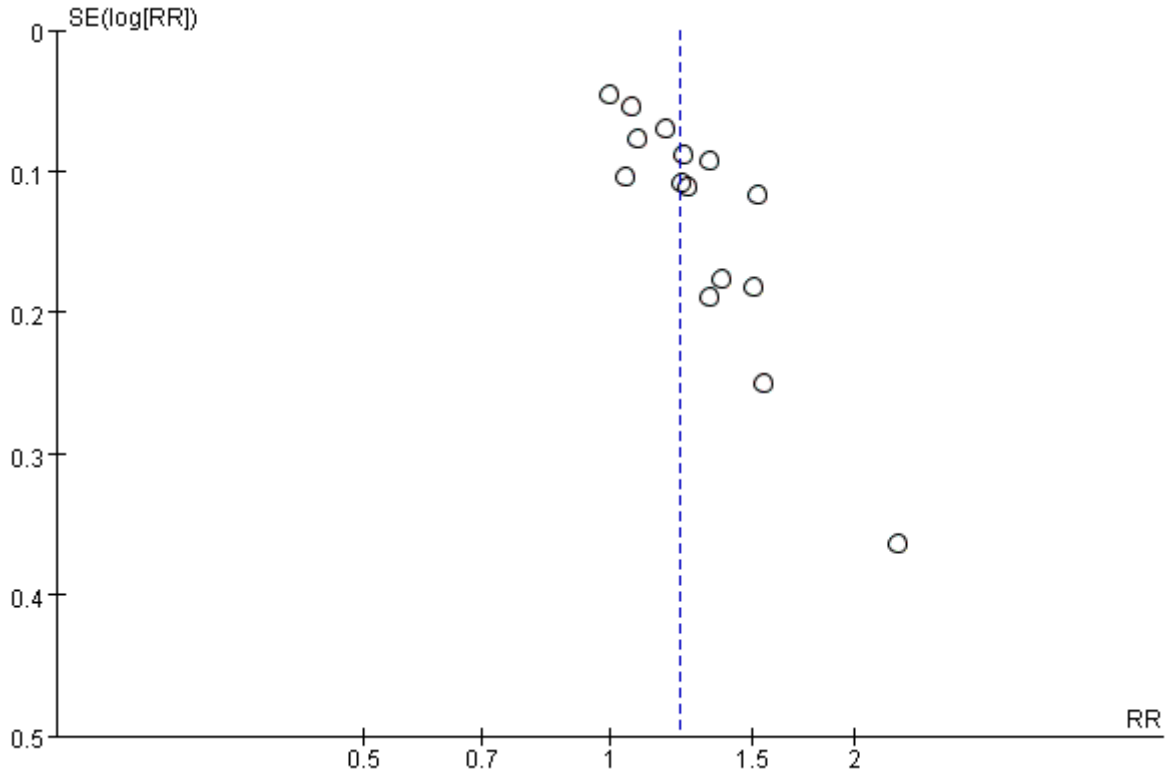



**Figure S13** Funnel plot of publication bias for the disease control rate (DCR)

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**Supplementary Data S1**

**Appendix 1. PubMed search strategy (N = 86)**

#1 “esophageal neoplasms”.[Mesh]

#2 “esophageal neoplasm*”.ti,ab.

#3 “esophagus neoplasm*”.ti,ab.

#4 “oesophageal neoplasm*”.ti,ab.

#5 “esophageal cancer*”.ti,ab.

#6 “esophagus cancer*”.ti,ab.

#7 “oesophageal cancer*”.ti,ab.

#8 “cancer of esophagus”.ti,ab.

#9 “cancer of the esophagus”.ti,ab.

#10 “esophageal tumor*”.ti,ab.

#11 “esophagus tumor*”.ti,ab.

#12 “oesophageal tumor*”.ti,ab.

#13 “esophageal carcinoma*”.ti,ab.

#14 “esophagus carcinoma*”.ti,ab.

#15 “oesophageal carcinoma*”.ti,ab.

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 “EGFR inhibitor*”.ti,ab.

#18 “epidermal growth factor receptor inhibitor*”.ti,ab.

#19 “anti-human epidermal growth factor receptor”.ti,ab.

#20 #17 OR #18 OR #19

#21 “cetuximab”.[Mesh]

#22 “erbitux”.ti,ab.

#23 “IMC C225”.ti,ab.

#24 “IMC-C225”.ti,ab.

#25 “MAb C225”.ti,ab.

#26 “C225”.ti,ab.

#27 “C-225”.ti,ab.



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#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 “panitumumab”.[Mesh]

#30 “ABX-EGF monoclonal antibody”.ti,ab.

#31 “ABX-EGF MAb”.ti,ab.

#32 “vectibix”.ti,ab.

#33 #29 OR #30 OR #31 OR #32

#34 “nimotuzumab”.ti,ab.

#35 “theraloc”.ti,ab.

#36 #34 OR #35

#37 “erlotinib hydrochloride”.[Mesh]

#38 “erlotinib HCl”.ti,ab.

#39 “erlotinib”.ti,ab.

#40 “tarceva”.ti,ab.

#41 “11C-erlotinib”.ti,ab.

#42 “11C erlotinib”.ti,ab.

#43 “OSI-774”.ti,ab.

#44 “OSI 774”.ti,ab.

#45 “OSI774”.ti,ab.

#46 “CP 358774”.ti,ab.

#47 “CP-358774”.ti,ab.

#48 “CP358774”.ti,ab.

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 “gefitinib”.[Mesh]

#51 “iressa”.ti,ab.

#52 “ZD1839”.ti,ab.

#53 “ZD-1839”.ti,ab.

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 “randomized controlled trials as topic”.[Mesh]

#57 “randomized controlled trial[publication type]”



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#58 “random*”.ti,ab.

#59 “clinical trials as topic”.[Mesh]

#60 “clinical trial[publication type]”

#61 “controlled clinical trials as topic”.[Mesh]

#62 “controlled clinical trial[publication type]”

#63 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62

#64 #16 AND #55 AND #63

**Appendix 2. EMBASE search strategy (N = 831)**

#1 'esophagus tumor'/exp

#2 'esophageal neoplasm*':ab,ti

#3 'esophagus neoplasm*':ab,ti

#4 'oesophageal neoplasm*':ab,ti

#5 'esophageal cancer*':ab,ti

#6 'esophagus cancer*':ab,ti

#7 'oesophageal cancer*':ab,ti

#8 'cancer of esophagus':ab,ti

#9 'cancer of the esophagus':ab,ti

#10 'esophageal tumor*':ab,ti

#11 'esophagus tumor*':ab,ti

#12 'oesophageal tumor*':ab,ti

#13 'esophageal carcinoma*':ab,ti

#14 'esophagus carcinoma*':ab,ti

#15 'oesophageal carcinoma*':ab,ti

#16 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15

#17 'egfr inhibitor*':ab,ti

#18 'epidermal growth factor receptor inhibitor*':ab,ti

#19 'anti-human epidermal growth factor receptor':ab,ti

#20 #17 OR #18 OR #19



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#21 'cetuximab'/exp

#22 'erbitux':ab,ti

#23 'imc c225':ab,ti

#24 'imc-c225':ab,ti

#25 'mab c225':ab,ti

#26 'c225':ab,ti

#27 'c-225':ab,ti

#28 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27

#29 'panitumumab'/exp

#30 'abx-egf monoclonal antibody':ab,ti

#31 'abx-egf mab':ab,ti

#32 'vectibix':ab,ti

#33 #29 OR #30 OR #31 OR #32

#34 'nimotuzumab'/exp

#35 'theraloc':ab,ti

#36 #34 OR #35

#37 'erlotinib'/exp

#38 'erlotinib hcl':ab,ti

#39 'erlotinib':ab,ti

#40 'tarceva':ab,ti

#41 '11c-erlotinib':ab,ti

#42 '11c erlotinib':ab,ti

#43 'osi-774':ab,ti

#44 'osi 774':ab,ti

#45 'osi774':ab,ti

#46 'cp 358774':ab,ti

#47 'cp-358774':ab,ti

#48 'cp358774':ab,ti

#49 #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48

#50 'gefitinib'/exp



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#51 'iressa':ab,ti

#52 'zd1839':ab,ti

#53 'zd-1839':ab,ti

#54 #50 OR #51 OR #52 OR #53

#55 #20 OR #28 OR #33 OR #36 OR #49 OR #54

#56 'randomized controlled trial'/exp

#57 'randomized controlled trial (topic)'/exp

#58 'random*':ab,ti

#59 'clinical trial (topic)'/exp

#60 'clinical trial'/exp

#61 'clinical trial*':ab,ti

#62 'controlled clinical trial'/exp

#63 'controlled clinical trial (topic)'/exp

#64 'controlled clinical trial*':ab,ti

#65 #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62 OR #63 OR #64

#66 #16 AND #55 AND #65

**Appendix 3. CENTRAL search strategy (N = 94)**

#1 MeSH descriptor: [esophageal neoplasms] explode all trees

#2 (esophageal neoplasm*):ti,ab,kw OR (esophagus neoplasm*):ti,ab,kw OR (oesophageal neoplasm*):ti,ab,kw OR (esophageal cancer*):ti,ab,kw OR (esophagus cancer*):ti,ab,kw. (Word variations have been searched)

#3 (oesophageal cancer*):ti,ab,kw OR (cancer of esophagus):ti,ab,kw OR (cancer of the esophagus):ti,ab,kw OR (esophageal tumor*):ti,ab,kw OR (esophagus tumor*):ti,ab,kw. (Word variations have been searched)

#4 (oesophageal tumor*):ti,ab,kw OR (esophageal carcinoma*):ti,ab,kw OR (esophagus carcinoma*):ti,ab,kw OR (oesophageal carcinoma*):ti,ab,kw. (Word variations have been searched)

#5 #1 OR #2 OR #3 OR #4

#6 (EGFR inhibitor*):ti,ab,kw OR (epidermal growth factor receptor inhibitor*):ti,ab,kw OR (anti-human epidermal growth factor receptor):ti,ab,kw. (Word variations have been searched)



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#7 MeSH descriptor: [cetuximab] explode all trees

#8 (erbitux):ti,ab,kw OR (IMC C225):ti,ab,kw OR (IMC-C225):ti,ab,kw. (Word variations have been searched)

#9 (MAb C225):ti,ab,kw OR (C225):ti,ab,kw OR (C-225):ti,ab,kw. (Word variations have been searched)

#10 #7 OR #8 OR #9

#11 MeSH descriptor: [panitumumab] explode all trees

#12 (ABX-EGF monoclonal antibody):ti,ab,kw OR (ABX-EGF MAb):ti,ab,kw OR (vectibix):ti,ab,kw. (Word variations have been searched)

#13 #11 OR #12

#14 (nimotuzumab):ti,ab,kw OR (theraloc):ti,ab,kw. (Word variations have been searched)

#15 MeSH descriptor: [erlotinib hydrochloride] explode all trees

#16 (erlotinib HCl):ti,ab,kw OR (erlotinib):ti,ab,kw OR (tarceva):ti,ab,kw OR (11C-erlotinib):ti,ab,kw OR (11C erlotinib):ti,ab,kw. (Word variations have been searched)

#17 (OSI-774):ti,ab,kw OR (OSI 774):ti,ab,kw OR (OSI774):ti,ab,kw. (Word variations have been searched)

#18 (CP 358774):ti,ab,kw OR (CP-358774):ti,ab,kw OR (CP358774):ti,ab,kw. (Word variations have been searched)

#19 #15 OR #16 #17 OR #18

#20 MeSH descriptor: [gefitinib] explode all trees

#21 (iressa):ti,ab,kw OR (ZD1839):ti,ab,kw OR (ZD-1839):ti,ab,kw. (Word variations have been searched)

#22 #20 OR #21

#23 #6 OR #10 OR #13 OR #14 OR #19 OR #22

#24 MeSH descriptor: [randomized controlled trials as topic] explode all trees

#25 MeSH descriptor: [randomized controlled trial] explode all trees

#26 (random*):ti,ab,kw. (Word variations have been searched)

#27 MeSH descriptor: [clinical trials as topic] explode all trees

#28 MeSH descriptor: [clinical trial] explode all trees

#29 (clinical trial*):ti,ab,kw. (Word variations have been searched)



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#30 MeSH descriptor: [controlled clinical trial] explode all trees

#31 MeSH descriptor: [controlled clinical trials as topic] explode all trees

#32 (controlled clinical trial*):ti,ab,kw. (Word variations have been searched)

#33 #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32

#34 #5 AND #23 AND #33

**Appendix 4. Web of Science search strategy (N = 191)**

#1 TS=(esophageal neoplasms OR esophageal neoplasm* OR esophagus neoplasm* OR oesophageal neoplasm* OR esophageal cancer* OR esophagus cancer* OR oesophageal cancer* OR cancer of esophagus OR cancer of the esophagus OR esophageal tumor* OR esophagus tumor* OR oesophageal tumor* OR esophageal carcinoma* OR esophagus carcinoma* OR oesophageal carcinoma*)

#2 TS=(EGFR inhibitor* OR epidermal growth factor receptor inhibitor* OR anti-human epidermal growth factor receptor)

#3 TS=(cetuximab OR erbitux OR IMC C225 OR IMC-C225 OR MAb C225 OR C225 OR C-225) #4 TS=(panitumumab OR ABX-EGF monoclonal antibody OR ABX-EGF MAb OR vectibix)

#5 TS=(nimotuzumab OR theraloc)

#6 TS=(Erlotinib Hydrochloride OR Erlotinib HCl OR Erlotinib OR Tarceva OR 11C-erlotinib OR 11C erlotinib OR OSI-774 OR OSI 774 OR OSI774 OR CP 358774 OR CP-358774 OR CP358774)

#7 TS=(Gefitinib OR Iressa OR ZD1839 OR ZD-1839)

#8 #7 OR #6 OR #5 OR #4 OR #3 OR #2

#9 TS=(randomized controlled trials as topic OR randomized controlled trial OR clinical trials as topic OR clinical trial OR controlled clinical trial OR controlled clinical trials as topic OR random*)

#10 #9 AND #8 AND #1

**Appendix 5. ClinicalTrials.gov search strategy (N = 24)**

#1 Condition or disease: esophageal neoplasms

#2 Intervention/treatment: cetuximab OR panitumumab OR nimotuzumab OR erlotinib hydrochloride OR gefitinib

#3 Study Results: studies with results

#4 #1 AND #2 AND #3



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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 4-5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 1 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 6-7 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5-6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 5-6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 7 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 7-8 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 7 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 7-8 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 7-8 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 7-8 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 7-8 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 9 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 9-10 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 10-11 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 9-14 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 9-14 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 9-14 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 9-14 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 14-20 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 14-20 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 19-20 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 20 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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